

## Brief Device Description

The Newport HT70 family of ventilators are state of the art ventilators that combine ruggedness, ease of use and clinical proficiency with exceptional mobility to provide ventilatory support for infant, pediatric and adult patients in emergency care, transport, subacute care and home care applications. It is also ideal for emergency preparedness applications.

The compact, lightweight HT70 ventilator is built for hard work with a durable polymer exterior and robust overall design that stands up to harsh environments.

The HT70 Ventilator defines ease of use with all essential controls at your fingertips using a simple membrane button and touch screen combination. There are no complicated menus or difficult sequences to follow in order to make necessary adjustments for common operations.

A three-tiered management domain system makes it very easy for caregivers to manage all controls while providing quick access to the more essential elements in transport situations and significantly enhanced safety and simplicity in the homecare environment.

## Sophisticated Clinical Capabilities

In addition to its durability and ease of use, the HT70 ventilator offers the complete array of clinical capabilities needed for managing patients.

The twin micro-piston pump's ability to deliver a variable flow enables the HT70 to provide a full range of operating modes and breath types with servo-controlled, leak-compensated PEEP. Leak compensation helps to improve triggering and avoid auto-triggering when a leak is present. The HT70 may be used with an endotracheal tube, tracheal tube, face mask, nasal mask or prongs, or mouthpiece.

There are 3 models for the HT70 series of ventilators:

|        |                                                                                              |
|--------|----------------------------------------------------------------------------------------------|
| HT70S  | HT70 Basic for use when Pressure Support is not needed.                                      |
| HT70   | HT70 Classic, adds Pressure Support and related parameters and Trends screen                 |
| HT70PM | HT70 Plus, adds on-airway flow sensor option with graphics, flow trigger and exhaled volumes |

The HT70 Basic and Classic models provide monitoring of inspiratory tidal volume (every breath), inspiratory minute volume, total respiratory rate, peak pressure, mean pressure and baseline (PEEP) pressure. Real-time patient circuit pressure is displayed at all times

# 1 Introduction

on the airway pressure gauge on the face panel. A comprehensive alarm system is built-in to alert the user to violations of user-set or ventilator safety limits. An optional built-in oxygen sensor allows monitoring of O<sub>2</sub> with high and low O<sub>2</sub> alarms.

The HT70 Plus model adds an on-airway flow sensor with onscreen graphics, exhaled tidal and minute volume monitoring/alarms, and flow trigger. This manual describes the HT70 Plus model and will denote features that are not available on the HT70 and HT70S models.

Gas delivery to the patient may be enriched with oxygen (0.21-1.00) using either the optional Air Oxygen Entrainment (50 psi) Mixer or optional Low Flow Oxygen Reservoir.

## **Exceptional Mobility**

The ventilator's unique design provides maximum mobility and safety for short or long distance transport of critically ill patients and also for patients who are going about their normal activities of daily life. This exceptional mobility is derived from two sources: Newport's patented, power conserving dual-micro-piston technology which eliminates the need for an external compressed gas source, and the Internal Dual Battery System which allows virtually continuous use from battery power through hot-swappable technology.

The HT70's micro-pistons use a fraction of the power that is consumed by turbines and blowers. This enables longer battery use time. Our patented system also uses considerably less supplemental oxygen than turbine or blower systems, again improving mobility for transport or homecare use. The superior technology of our micro-piston system over the turbine and blower systems allow the HT70 to ventilate safely over a wide range of environmental conditions and altitudes.

The HT70's twin micro-piston internal pump is made of mechanically moving components. As with any other gas delivery system made of moving components, it may emit a minor level of noise during operation. This is not a malfunction and does not affect the performance of the ventilator.

The Internal Dual Battery System consists of two independent but coordinated lithium ion batteries, the Power Pac battery, located on the back of the ventilator and the Backup Battery inside the ventilator. The Internal Dual Battery System can provide up to 10 hours of operation at standard settings when new and fully charged. This system assures continued support during transport, daily activities or power outages.

The detachable Power Pac is 'hot-swappable'. That is, if more battery time is needed, a depleted Power Pac can easily be removed from the back of the HT70 and replaced with a recharged Power Pac without interrupting ventilation. No tools are needed. The secondary Backup Battery maintains operation without interruption when the Power Pac is swapped out and also provides a minimum of 30 minutes of full operation when all other power sources are depleted. The Power Pac weighs two pounds and is charged anytime the ventilator is connected to an external power source (AC or DC). It can also be charged separately.

The HT70 may be operated from a variety of AC (100-240 VAC @ 50 / 60 Hz) or DC (12-24 VDC) external power sources or from the Internal Dual Battery System. The optional DC Auto Lighter Power Adapter accessory enables connection to an automobile-type DC outlet. Any time the ventilator is connected to external power, both batteries in the Internal Dual Battery System are charging, whether or not the ventilator is in use.

### **Travel Certified**

The HT70 has been tested for and meets requirements for use in helicopter and fixed wing transport and for use on commercial airlines. Before traveling, be sure to speak with your airline representative about their particular concerns and clear all of your equipment with them well before your departure. The labeling that the FAA requires to be on the ventilator is located on the bottom of the HT70.

### **Intended Use**

Newport HT70 family of ventilators is intended to provide continuous or intermittent positive pressure mechanical ventilatory support for the care of individuals who require mechanical ventilation through invasive or noninvasive interfaces.

Specifically, the Newport HT70 family of ventilators is applicable for infant, pediatric and adult patients greater than or equal to 5 kg (11 lbs) in hospital, sub-acute, emergency department, and home care environments as well as for transport and emergency response applications.

**NOTE:** Federal law (US) restricts sale by or on the order of a physician.

# 1 Introduction

## Ventilator Configurations

Newport Medical offers five configurations for the 3 models in the HT70 family of ventilators. See Table 1. In addition, the front control panel labeling is available in various languages and regional power cords, i.e. North American, European, etc., can be specified. See your Newport Medical Representative for details.

**Table 1**

| Part Number   | Description                    | Distinguishing Features                                                                                                                                                                        |
|---------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HT70PM</b> | HT70 Plus                      | Full featured with Flow sensor, graphics and built-in oxygen monitor with alarms. Compatible with optional plug and play pulse oximeter                                                        |
| <b>HT70</b>   | HT70, w/o Oxygen Sensor        | Classic features<br>No built-in oxygen monitor                                                                                                                                                 |
| <b>HT70M</b>  | HT70, with Oxygen Sensor       | Classic features<br>Includes built-in oxygen monitor with alarms                                                                                                                               |
| <b>HT70S</b>  | HT70, Basic w/o Oxygen Sensor  | Classic features except: <ul style="list-style-type: none"><li>• No Pressure Support or Pressure Support parameters</li><li>• No Trends</li><li>• No built-in oxygen monitor</li></ul>         |
| <b>HT70SM</b> | HT70, Basic with Oxygen Sensor | Classic features except: <ul style="list-style-type: none"><li>• No Pressure Support or Pressure Support parameters</li><li>• No Trends</li></ul> Includes built-in oxygen monitor with alarms |

## Warnings, Cautions, and Notes

Please review all WARNINGS and Cautions outlined in this manual before operating the ventilator.

Use of the product requires full understanding and strict observation of all sections of these instructions. The equipment is only to be used for the purposes specified under Intended Use and in conjunction with appropriate patient observation and monitoring. Observe all WARNINGS and Cautions that appear in this manual and on equipment labels.

**WARNING** A warning describes a condition that can cause injury.

**Caution:** A caution describes a condition that can cause damage to equipment.

**NOTE:** A note emphasizes information that is important or convenient.

## General Notes

The Newport HT70 has been designed to accommodate connectivity with nurse call/monitoring systems. Because it is not possible to anticipate every configuration of hardware and software associated with nurse call/monitoring system, it is the user's responsibility to confirm proper functionality of the system when used in conjunction with the HT70. Verification of alarms, alerts and patient data transmissions is required. If the system performance is not as expected, contact Newport Medical Technical Support for assistance troubleshooting the set-up. Do not use the HT70 ventilator with a nurse call/monitoring system until the functionality of the ventilator/system combination has been confirmed.

## General Cautions

Do not place liquids on or near the ventilator.

Damage can occur if the HT70 is exposed to extreme temperatures. Do not store the HT70 in areas where it may be exposed to temperatures below -40° C (-40° F) or above 65° C (149° F).

To avoid the risk of electric shock, the ventilator should not be opened by anyone other than an approved service provider.

## General Warnings

The design of the HT70 ventilator, the Operating and Service manuals, and the labeling on the ventilator take into consideration that the purchase and use of the equipment is restricted to trained professionals, and that certain inherent characteristics of the ventilator are known to the operator. Instructions, warnings and caution statements are therefore limited to the specifics of the HT70.

This manual excludes references to various hazards which are obvious to medical professionals and operators of this equipment including consequences of product misuse, and potential adverse effects in patients with abnormal conditions.

Transport of patients with the HT70 requires that medical staff have a good working knowledge of the ventilator's use and

# 1 Introduction

problem resolution. Proper emergency backup equipment must be immediately available during transport.

Product modification or misuse can be dangerous. Newport Medical Instruments, Inc. disclaims all liability for the consequences of product alterations or modifications, as well as for the consequences which might result from the combination of this ventilator with other products, whether supplied by Newport or by other manufacturers, unless such a combination has been specifically endorsed by Newport Medical. There is a risk of explosion if used in the presence of flammable anesthetics.

A patient connected to a ventilator requires the constant attention of trained caregivers to the patient's condition.

Ventilator alarms are a critical element in the safety net of patient care. It is extremely important for patient safety that caregivers immediately identify and correct alarm violations.

Always have an alternate power source and means of ventilation available when the ventilator is in use so that they are easy to access in case of a mechanical or system problem.

If a fault is detected in the ventilator and its life support functions are in doubt, immediately discontinue use; use an alternative method of ventilation until the fault has been corrected. Contact your service provider immediately.

Do not block the Emergency Gas Intake (on the bottom panel) or the Fresh Gas Intake Port (on the right side panel).

Always use appropriate monitors to ensure sufficient oxygenation and ventilation (such as a pulse oximeter and/or a capnograph) when the HT70 Ventilator is in use on a patient.

The optional Air/Oxygen Entrainment Mixer and Low Flow Oxygen Reservoir are designed to operate with medical grade oxygen.

Ensure that the oxygen source is not empty before and during the use of the optional Air/Oxygen Entrainment Mixer or Low Flow Oxygen Reservoir.

When the optional Air/Oxygen Entrainment Mixer is secured in place, ensure that the oxygen supply is enabled prior to powering the HT70 on to avoid putting stress on the internal pump and compromising gas delivery to the patient.

Calibrated oxygen monitoring at clinically appropriate levels is required for patient safety when supplemental oxygen is in use. The optional built-in oxygen sensor on the HT70 allows High and Low O<sub>2</sub> alarms to be enabled which can be used to assure proper oxygen delivery.

Always plug the HT70 into an external power supply source whenever it is available, even when HT70 is not in use, to keep the Internal Dual Battery System fully charged and to ensure best battery performance. Check battery capacity on the front panel before detaching from external power.

When installing a replacement Power Pac during battery operation, always ensure that the charge level LED on the replacement pack is green, indicating 90% or higher charge level.

Always ensure that the green External Power LED lights after connecting the ventilator to an external AC or DC power source.

To maintain grounding integrity when using AC power, only connect to properly grounded receptacles.

Use only the Newport supplied AC Power Supply (p/n PWR3204P) with the HT70 ventilator and HT70 Power Pac (p/n BAT3271A).

Always disconnect the external power supply prior to servicing.

After servicing the HT70, it must pass the Operational Verification Procedure (OVP) before it is returned to patient use. See section 6 of the HT70 Service Manual.

Do not use electrically conductive breathing circuits. Always use clean and dry breathing circuits.

Always use a clean, dry filter in the following locations: a standard bacteria filter on the gas output, a prox line (bacteria) filter on the proximal pressure tubing and an intake (bacteria) filter behind the filter cover.

Adding attachments or other components or sub-assemblies to the ventilator breathing circuit system can increase the patient's work of breathing and/or add resistance to patient exhalation.

Always ensure that the audible alarm loudness level is set at a volume that can be heard by the caregiver. Do not use the ventilator in an environment where audible alarms cannot be heard by the caregivers.

# 1 Introduction

The functioning of this machine may be adversely affected by the operation of other medical equipment, such as high frequency surgical(diathermy) equipment, defibrillators or short-wave therapy equipment in the vicinity.

This device has undergone EMC testing and found to be in conformance with IEC 60601-1-2:2001 and meets the requirement of CISPR11:2004 (Class B), IEC 61000-3-2:2006, and IEC 61000-3-3:1955 + A1:2001 + A2:2005. These requirements are designed to provide reasonable protection against harmful interference in a typical medical installation, as well as in homecare environments. The equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with these instructions, may cause harmful interference to other devices in the vicinity. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference with other devices, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures;

- Reorient or relocate the receiving device.
- Increase the separation between the equipment.
- Connect the equipment into an outlet on a circuit different from that to which the device(s) is connected.
- Consult the manufacturer or field service technician for help.

## Copyright Information

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| Front Panel Buttons - Symbols Version                                             |                           |
|-----------------------------------------------------------------------------------|---------------------------|
|  | Accept                    |
|  | Cancel                    |
|  | Alarm Silence Reset / LED |
|  | Breath Delivery Indicator |
|  | Brightness                |
|  | Device Alert LED          |
|  | External Power LED        |
|  | Manual Inflation          |
|  | Up/Down Arrow             |

## 2 Specifications

| Miscellaneous Reference Symbols                                                   |                                                                                                                                       |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|  | Manufacturer Name and Address                                                                                                         |
|  | Authorized Representative in the European community                                                                                   |
|  | Main Power Off / On (momentary switch)                                                                                                |
|  | Low (Paw or Min Vol) alarm                                                                                                            |
|  | High (Paw, Min Vol or RR) alarm                                                                                                       |
|  | Attention, see instructions for use                                                                                                   |
|  | Equipotentiality                                                                                                                      |
|  | Applied Parts Type BF                                                                                                                 |
|  | Brightness control                                                                                                                    |
|  | Alarm Silence                                                                                                                         |
|  | Up / Down arrow                                                                                                                       |
|  | Federal law (US) restricts sale by or on the order of a physician                                                                     |
|  | Meets FAA requirements in RTCA standard, DO160, sec 21 category M for use in all stages of air travel, including takeoff and landing. |
|  | Single patient use                                                                                                                    |

| Controls/Monitors                 |                                                                                                                                                                    |                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Controls / Monitors               | Range / Selection                                                                                                                                                  | Resolution                  |
| MODE (Pressure or Volume Control) | A/CMV                                                                                                                                                              |                             |
|                                   | SIMV                                                                                                                                                               |                             |
|                                   | SPONT                                                                                                                                                              |                             |
| Breath type (mandatory)           | Pressure Control or Volume Control                                                                                                                                 |                             |
| NIV (Non Invasive ventilation)    | On or Off, When On, allows ↓Min Vol Alarm to be set to Off, ↓P Alarm to be set 1 cmH <sub>2</sub> O/mbar above PEEP and allows adjustment of bias flow during PEEP |                             |
| VT (Tidal Volume)                 | 50 to 2,200 mL, ATPS, ± 10%                                                                                                                                        | 1.0 mL                      |
| PC (Pressure Control)             | 5 to 60 cmH <sub>2</sub> O / mbar                                                                                                                                  | 1.0 cmH <sub>2</sub> O/mbar |
| Flow                              | 6 to 100 L/min                                                                                                                                                     | 1.0 L/min                   |
| i time (Inspiratory Time)         | 0.1 to 3.0 sec                                                                                                                                                     | 0.1 sec                     |
| RR (Respiratory Rate)             | 1 to 99 b/min                                                                                                                                                      | 1.0 b/min                   |
| P trig (Sensitivity)              | -9.9 to 0 cmH <sub>2</sub> O / mbar, pressure triggering                                                                                                           | 0.1 cmH <sub>2</sub> O/mbar |
| Flow trig (Sensitivity)           | 0.1 to 10 L/min                                                                                                                                                    | 0.1 L/min                   |
| PEEP/CPAP                         | 0 to 30 cmH <sub>2</sub> O / mbar                                                                                                                                  | 1.0 cmH <sub>2</sub> O/mbar |
| PS (Pressure Support)*            | 0 to 60 cmH <sub>2</sub> O / mbar above baseline pressure, limited to PEEP + PS ≤ 60 cmH <sub>2</sub> O / mbar                                                     | 1.0 cmH <sub>2</sub> O/mbar |

\* Not available on the HT70S models

| Controls / Monitors            | Range / Selection                                                                                                                                                                                              | Resolution                                   |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| I:E Ratio                      | 1:99 to 3:1                                                                                                                                                                                                    | 0.1 for 9.9:1 to 1:99 and 1 for 99:1 to 1:99 |
| Airway Pressure Gauge          | -10 to 100 cmH <sub>2</sub> O / -10 to 98 mbar<br>includes indicator bars to show low and high PAW alarm limits                                                                                                |                                              |
| O <sub>2</sub> Sensor          | Enabled or Disabled<br>When Enabled, high and low O <sub>2</sub> alarms are active and O <sub>2</sub> Cylinder Time Calculator is available                                                                    |                                              |
| PS Max i time*                 | 0.1 - 3.0 sec                                                                                                                                                                                                  | 0.1 sec                                      |
| PS % Exp Threshold*            | 5 - 85%                                                                                                                                                                                                        | 5                                            |
| Slope Rise                     | 1 -10 (1 is slowest)                                                                                                                                                                                           | 1                                            |
| Flow Wave Pattern              | Square or Descending                                                                                                                                                                                           |                                              |
| Bias Flow                      | 0 L/min - PEEP Off<br>7 L/min - PEEP On<br>3-30 L/min - PEEP + NIV On                                                                                                                                          | 1                                            |
| AutoLock function              | Enabled/ Disabled                                                                                                                                                                                              |                                              |
| AutoLock icon                  | Touch for 3 seconds to unlock buttons if AutoLock is Enabled in <i>Utility</i> screen.<br>When Lock is on screen, all controls are locked except Alarm Silence Reset, manual inflation and Brightness control. |                                              |
| O <sub>2</sub> Cylinder Data   | Size: D, E, H, M, K, 100 L & 150 L<br>Cylinder Pressure: 300 to 2450 psi or 25 to 175 ATM or 2,000 to 17,000 kPa<br>Units: psi or ATM or kPa<br>O <sub>2</sub> Cylinder Monitor; Enabled/Disabled.             |                                              |
| Altitude                       | -1,000 to 10,000 feet, -1,000 to 10,000 meters (with Flow Sensor use)                                                                                                                                          |                                              |
| BUV Settings                   | Minimum RR: 8 to 30 b/min<br>Rate Factor: 1.1 to 1.8<br>SPONT Delta P: 5 to 20 cmH <sub>2</sub> O/mbar<br>SPONT i-time: 0.4 to 2.0 s                                                                           |                                              |
| <b>Monitor Data Selections</b> |                                                                                                                                                                                                                |                                              |
| Minute Volume                  | 0.01 L to 80.0 L                                                                                                                                                                                               | 0.01 L                                       |
| Insp./Exp. Tidal Volume        | 0.01 L to 3.0 L                                                                                                                                                                                                | 0.01 L                                       |
| RR total                       | 0 to 200 b/min                                                                                                                                                                                                 | 1 b/min                                      |

\* Not available on the HT70S models

## 2 Specifications

| Monitor Data Selections      |                                   |                                            |
|------------------------------|-----------------------------------|--------------------------------------------|
| P Peak                       | 0 to 100 cmH <sub>2</sub> O /mbar | 1 cmH <sub>2</sub> O/mbar                  |
| P Mean                       | 0 to 100 cmH <sub>2</sub> O /mbar | 1 cmH <sub>2</sub> O/mbar                  |
| P Base (PEEP)                | 0 to 100 cmH <sub>2</sub> O /mbar | 1 cmH <sub>2</sub> O/mbar                  |
| (Peak) Flow                  | 5 to 150 L/min                    | 0.1 L/min                                  |
| O <sub>2</sub> Cylinder time | hours / min                       | 15 minutes                                 |
| Battery time                 | hours / min                       | 15 minutes                                 |
| O <sub>2</sub> % (optional)  | 21 to 100                         | 1%                                         |
| I:E Ratio                    | 1:99.0 to 3.0:1                   | 0.1 (9.9:1 to 1:9.9)<br>1.0 (99:1 to 1:99) |

| Front Panel Membrane Buttons and Indicators |                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Cancel                                      | Press front panel button to cancel any touchscreen settings changes that have not been accepted                                                  |
| Accept                                      | Press to confirm or accept any touchscreen setting changes                                                                                       |
| Up / Down Arrows<br>▲ / ▼                   | Arrow buttons allow adjustment of settable parameters. Use up arrow to increase and down arrow to decrease.                                      |
| Brightness control                          | Press to select brightness levels, maximum, medium high, medium and low.                                                                         |
| Manual Inflation                            | 3 second maximum. While button is pressed, the ventilator closes the exhalation valve and delivers an operator controlled breath to the patient. |
| Breath Indicator LED                        | Lights to indicate the ventilator is delivering a breath.                                                                                        |
| External Power LED                          | Lights to indicate the ventilator is powered by external power                                                                                   |
| Device Alert LED                            | Lights to indicate a ventilator malfunction.<br><b>WARNING Use alternate ventilation source until malfunction is identified and corrected.</b>   |

| Alarms                           |                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handle LED                       | Alarm Indicators flash red or yellow for violated alarms.                                                                                                                                    |
| Alarm Silence/ Reset button      | Silences audible alarm for 60 sec; resets latched alarm messages. Press repeatedly (or push and hold for 3 seconds) to clear all latched messages.                                           |
| Alarm Silence LED                | LED remains lit during the alarm silence period                                                                                                                                              |
| Message Display area             | Alpha numeric display, turns color during an alarm violation and shows alarm message. Multiple alarm messages display in order of priority (red is high, amber is medium and yellow is low). |
| Alarm Loudness (in Alarm Screen) | 1 - 10 (10 is loudest)                                                                                                                                                                       |

| User Adjustable alarms                    | Alarm Priority | Range / Description                                                                                                                                                                 |
|-------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ↑P (High Pressure)                        | High           | 4 to 99 cmH <sub>2</sub> O / 4 to 99 mbar                                                                                                                                           |
| ↓P (Low Pressure)                         | High           | NIV Off: 3 to 98 cmH <sub>2</sub> O / 3 to 98 mbar (limited by PEEP + 3); 2 breath delay<br>NIV On: 1 to 98 cmH <sub>2</sub> O / 1 to 98 mbar (limited by PEEP + 1); 3 breath delay |
| ↑Min Vol (High Insp./ Exp. Minute Volume) | High           | NIV Off: 1.1 to 50 L/min<br>NIV On: 1.1 to 80 L/min                                                                                                                                 |
| ↓Min Vol (Low Insp./ Exp. Minute Volume)  | High           | NIV Off: 0.01 to 49.0 L/min<br>NIV On: Off, 0.01 to 49.0 L/min                                                                                                                      |
| ↑RR (High Respiratory Rate)               | Med            | Off, 30-100 b/min                                                                                                                                                                   |
| Apnea                                     | High           | 5 - 70 seconds                                                                                                                                                                      |
| ↑O <sub>2</sub>                           | Med            | Off, 24 - 100, only available when O <sub>2</sub> Sensor is enabled                                                                                                                 |
| ↓O <sub>2</sub>                           | Med            | Off, 22- 98, only available when O <sub>2</sub> Sensor is enabled                                                                                                                   |
| ↑VTE                                      | Med            | OFF, 0.06 to 2.2 Liters                                                                                                                                                             |
| Backup Ventilation (BUV)                  | Med            | Can be set to be activated by either the Low Minute Volume alarm or the Apnea alarm or both via the <i>More/Utilities/Custom Settings</i> /BUV Settings. Functional in all modes    |

| Automatic alarms                     | Alarm Priority | Description                                                                     |
|--------------------------------------|----------------|---------------------------------------------------------------------------------|
| Low Baseline Pressure                | High           | Paw < PEEP minus 2 cmH <sub>2</sub> O/mbar for 3 sec                            |
| High Baseline Pressure               | High           | Paw 5 above set PEEP at onset of a time-triggered breath                        |
| Pressure Control Setting Not Reached | High           | P Peak < 50% of PCV setting for 2 consecutive breaths                           |
| Check Circuit                        | High           | Circuit may be disconnected or proximal pressure line may be pinched or blocked |
| No External Power                    | Low            | Loss of External power, automatic switchover to Internal Dual Battery System    |
| Occlusion                            | High           | An occlusion or restriction in the circuit that interferes with exhalation      |
| Occlusion - Sustained                | High           | Occlusion continues for 10 sec or 2 breath periods, whichever is shorter        |
| Device Alert                         | High           | Ventilator malfunction, Device Alert LED lights red                             |

## 2 Specifications

| Automatic alarms                 | Alarm Priority | Description                                                                                                                         |
|----------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Shut Down Alert                  | High           | Silence by pressing Alarm Silence/Reset button                                                                                      |
| Motor Fault                      | High           | Hardware detected fault in the motor drive circuit has occurred                                                                     |
| Internal Temperature             | Low            | Internal temperature is > 60° C                                                                                                     |
| Backup Battery Temperature       | Low            | Backup Battery temperature is > 60° C                                                                                               |
| Power Pac Battery Temperature    | Low            | Power Pac Battery temperature is > 60° C                                                                                            |
| Power Pac Battery Pack Low       | Med            | Less than 2 Ah of charge is left on Power Pac battery pack                                                                          |
| Switching to Backup Battery      | Med            | Indicates that the Power Pac battery pack is not available or useable. Ventilator is switching battery operation to Backup Battery. |
| Running on Backup Battery        | Med            | The Ventilator is operating on Backup Battery for > 15 minutes. Audible alarm will sound every 5 minutes thereafter                 |
| Backup Battery Low               | High           | Backup Battery has insufficient charge, less than 1Ah                                                                               |
| Backup Battery Shutdown Imminent | High           | Backup Battery is extremely low and will lose power very soon. Connect to external power or insert new Power Pac battery pack       |
| Backup Battery Failure           | High           | Indicates a failure in Backup Battery due to communication failure with host processor or capacity is below 1 Ah                    |
| Flow Sensor disconnect           | Med            | Flow sensor is no longer detected or is faulty                                                                                      |

| Hardware Requirements        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Patient outlet               | 22 mm OD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| AC Power input               | 100 to 240 VAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DC Power input               | 12 to 24 VDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Power switch                 | momentary switch to power on and off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| RS-232C Interface            | 9 pin standard RS232 connector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Nurse Call / Remote Alarm    | RJ435 connector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| USB Ports                    | Two USB ports for connecting to central monitoring systems, uploading software upgrades or download data files.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Electrical                   | Applied parts type BF<br>Class I Protection Against Electric Shock<br>100-240 VAC, max. 2 A, 50 / 60 Hz<br>12-24 VDC, max. 5 A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Internal Dual Battery System | <p>Power Pac Battery Pack: 14.4 VDC, 6.5 amp hours<br/>Recharge: minimum 3 hours for 100% charge<br/>When new and fully charged, the primary Lithium Ion battery pack supplies power for up to 10 hours of operation at these settings: A/CMV mode, RR=15, Tidal Volume=500 mL, i time=1.0 sec, PEEP=0, max. airway pressure 30 cmH<sub>2</sub>O /mbar, Power Save On, Bias flow Off.</p> <p><b>NOTE:</b> The Power Pac and Backup batteries are charged whenever the HT70 is connected to an external power source. Battery charge level is best maintained by keeping the HT70 continuously connected to external power.</p> <p>Backup Battery: 14.4 VDC, 2 amp hour<br/>The secondary Lithium Ion backup battery will supply power for a minimum of 30 minutes.</p> |
| Pneumatics                   | Dual micro-piston system requires no external air compressor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 2 Specifications

| Hardware Requirements                      | Description                                                                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emergency Intake                           | Maximum inspiratory and expiratory pressure drop at single fault conditions: 10 cmH <sub>2</sub> O/ L/sec (measured at patient connection port)                                                                                    |
| Maximum Limited Pressure (Pressure Relief) | 100 cmH <sub>2</sub> O/mbar                                                                                                                                                                                                        |
| Environment                                |                                                                                                                                                                                                                                    |
| Description                                |                                                                                                                                                                                                                                    |
| Operating Temperature                      | -18°C to 40°C<br><b>NOTE:</b> For proper operation at low range temperatures (-18°C), the HT70 must be started in a normal room temperature environment and allowed to run for 30 minutes prior to transfer to colder environment. |
| Water Ingress Protection                   | IEC 60529 IPX4                                                                                                                                                                                                                     |
| Operating Humidity                         | 15 to 95% non-condensing                                                                                                                                                                                                           |
| Operating Altitude                         | Sea level to 15,000 ft (0 to 4,572 m)<br>There is no altitude limitation when HT70 is operated in a pressurized environment.                                                                                                       |
| Operating Pressure                         | 600 to 1,100 mbar                                                                                                                                                                                                                  |
| Storage and Shipping Temperature           | -40° C to 65° C                                                                                                                                                                                                                    |
| Storage and Shipping Humidity              | 0 to 95% non-condensing                                                                                                                                                                                                            |
| Size and Weight                            |                                                                                                                                                                                                                                    |
| Height (includes handle)                   | 10.25 inches (26.04 cm)                                                                                                                                                                                                            |
| Width                                      | 9.75 inches (24.77 cm)                                                                                                                                                                                                             |
| Depth                                      | 11 inches (27.94 cm)                                                                                                                                                                                                               |
| Weight                                     | 15.4 lbs. (6.9 kg)                                                                                                                                                                                                                 |

| Factory Default Parameters |                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient Settings:          |                                                                                                                                                                                                                                                                                                                                                              |
| MODE                       | A/CMV                                                                                                                                                                                                                                                                                                                                                        |
| VT (Volume Control)        | 500 mL                                                                                                                                                                                                                                                                                                                                                       |
| i time                     | 1.0 sec                                                                                                                                                                                                                                                                                                                                                      |
| RR                         | 12 b/min                                                                                                                                                                                                                                                                                                                                                     |
| P <sub>trig</sub>          | 2 cmH <sub>2</sub> O                                                                                                                                                                                                                                                                                                                                         |
| ↓Paw Alarm                 | 5 cmH <sub>2</sub> O                                                                                                                                                                                                                                                                                                                                         |
| ↑Paw Alarm                 | 40 cmH <sub>2</sub> O                                                                                                                                                                                                                                                                                                                                        |
| ↑MV Alarm                  | 9 L/min                                                                                                                                                                                                                                                                                                                                                      |
| ↓MV Alarm                  | 3 L/min                                                                                                                                                                                                                                                                                                                                                      |
| ↑O <sub>2</sub>            | Off                                                                                                                                                                                                                                                                                                                                                          |
| ↓O <sub>2</sub>            | Off                                                                                                                                                                                                                                                                                                                                                          |
| PEEP/CPAP                  | Off                                                                                                                                                                                                                                                                                                                                                          |
| PS                         | Off                                                                                                                                                                                                                                                                                                                                                          |
| Alarm Loudness Level       | 6                                                                                                                                                                                                                                                                                                                                                            |
| Miscellaneous              |                                                                                                                                                                                                                                                                                                                                                              |
| Patient Circuit            | <p>Reusable or disposable 22 mm I.D. adult or 15 mm I.D. pediatric circuit with 3/16 inch (4.8 mm) I.D. proximal pressure sensing line, 1/8 inch (3.2 mm) I.D. exhalation valve control drive tubing, and exhalation valve.</p> <p><b>NOTE:</b> Newport Medical cannot guarantee the safe use of breathing circuits that are not recommended by Newport.</p> |

### (Optional) Air / Oxygen Entrainment Mixer (MX70A) Specifications

|                          |                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Pneumatic Requirements:  |                                                                                                                         |
| Oxygen                   | 35 to 65 psig (2.4 to 4.5 Bar) full operating range, maximum accuracy 40 to 50 psig (2.7 to 3.4 Bar) accuracy $\pm .08$ |
| Air                      | Atmospheric pressure                                                                                                    |
| FiO <sub>2</sub> Control | adjusted continuously from 0.21 to 1.00                                                                                 |

**NOTE:** Oxygen source gas must be medical grade, 100% oxygen.

### (Optional) Low Flow Oxygen Reservoir (RSV3215A)

|                          |                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------|
| Pneumatic Requirements:  |                                                                                     |
| Oxygen                   | 0-10 L/min                                                                          |
| Air                      | Atmospheric pressure                                                                |
| FiO <sub>2</sub> Control | FiO <sub>2</sub> , indirectly adjusted from 0.21 up to 1.00 via oxygen flow (L/min) |

**WARNING** Appropriate oxygen monitoring is required for patient safety.

## 2 Specifications

| <b>Regulatory and Agency Standards</b>                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Testing and evaluation of the HT70 Ventilator has been conducted in compliance with the following voluntary standards:                                                                           |
| ASTM F 1246-91:2005 Standard Specifications for Electrically Powered Home Care Ventilators – Part 1: Positive-Pressure Ventilators and Ventilator Circuits                                       |
| IEC 60068-2-6:2008 Environmental Testing- Part 2: Tests- Test FE: Vibration (Sinusoidal)                                                                                                         |
| IEC 60068-2-27:2008 Environmental Testing- Part 2: Tests- Test EA and Guidance: Shock                                                                                                            |
| IEC 68-2-34:1973 Environmental Testing- Part 2: Tests- Test Fd: Random Vibration Wide Band – General Requirements                                                                                |
| IEC 60601-1:1988 (+A1:1991 +A2:1995) Medical Electrical Equipment – Part 1: General Requirements for Safety                                                                                      |
| IEC 60601-1-1:2000 Medical Electrical Equipment, Collateral Standard: Safety Requirements for Medical Electrical Systems                                                                         |
| IEC 60601-1-2:2001 Medical Electrical Equipment, Collateral Standard: Electromagnetic Compatibility- Requirements and Tests                                                                      |
| IEC 60601-1-4:2000 Medical Electrical Equipment, Collateral Standard: Programmable Electrical Medical Systems                                                                                    |
| IEC 60601-1-6:2004 Medical Electrical Equipment, Collateral Standard: Usability                                                                                                                  |
| IEC 60601-1-8:2006 Medical Electrical Equipment, Collateral Standard: General Requirements, Tests, and Guidance for Alarm Systems in Medical Electrical Equipment and Medical Electrical Systems |
| MIL-STD-810E Test Method Standard for Environmental Engineering Considerations and Laboratory Tests                                                                                              |

## Device Description

The Newport HT70 ventilator is a state of the art ventilator that combines ruggedness, exceptional portability and ease of use with clinical proficiency to provide ventilatory support for infant, pediatric and adult patients. The HT70 is built for hard work with a durable polymer exterior and robust overall design that stands up to the harsh environments found in the world of emergency response, transport and homecare ventilation.

The unique design of the HT70 provides maximum mobility and safety for short or long distance transport of critically ill patients and for patients to enjoy their normal activities of daily life. This exceptional mobility is derived from two sources: the patented, power conserving dual-micro-piston technology which eliminates the need for an external compressed gas source, and the Integrated Battery System which allows virtually continuous use from battery power through hot-swappable technology.

The Integrated Battery System consists of two independent but coordinated Lithium ion batteries – the Power Pac battery, located on the back of the ventilator, and, the Backup Battery, located inside the HT70. When new and fully charged, the Integrated Battery System can provide up to 10 hours of operation at standard settings.

The Backup Battery maintains operation without interruption when the Power Pac is swapped out and also provides a minimum of 30 minutes operation when all other power sources are depleted.

The HT70 may be operated from the following power sources: A.C. (100-240 VAC @ 50 / 60), D.C. (12-24 VDC) external power source, HT70 Integrated Battery System, or the optional D.C. Auto Lighter Power Adapter accessory which enables connection to an automobile-type D.C. outlet.

All controls are easily found on the front touch screen and panel. There are no complicated menus or difficult sequences to follow in order to make necessary adjustments for common operations.

The compact lightweight HT70 ventilator incorporates powerful clinical capability for both invasive and non-invasive applications. The dual micro-piston ability to deliver variable flow enables the HT70 to provide a full range of operating modes and breath types with servo-controlled, leak-compensated PEEP.

# 3 Theory of Operation

The HT70 provides monitoring of inspiratory tidal volume (every breath), inspiratory minute volume, total respiratory rate, peak pressure, mean pressure and baseline (PEEP) pressure.

A comprehensive alarm system is built-in to alert the user to violations of user-set or ventilator set safety limits. An optional built-in oxygen monitor with high and low FiO<sub>2</sub> alarms may be added.

Gas delivery to the patient may be enriched with oxygen (.21-1.00) using either the optional Low Flow Oxygen Reservoir or high pressure entrainment mixer.

The HT70 has been tested for and meets the requirements for use in a medical evacuation helicopter and fixed wing transport.

## Functional Subsystems Overview

### Pneumatics System

The Newport HT70 pneumatics system consists of three subsystems: gas delivery, exhalation control, and safety. The gas delivery subsystem drives the movement of air into the lungs of the patient by converting electrical energy into positive air pressure. The exhalation control subsystem changes the breathing cycle from inspiration to expiration and provides positive end expiratory pressure (PEEP). The safety subsystem protects the patient by preventing high pressure and providing a capability to breathe in ambient air in the case of a potential failure of the ventilator. In addition, the safety subsystem is used to measure the air pressure requirements for ventilator safety, control, and monitoring. Refer to the section 8: Diagram 8.1 – Pneumatics System Diagram.

### Gas Delivery Subsystem

The gas delivery subsystem draws in air through the cover and inlet filter on the right-hand side of the ventilator. The semi-transparent cover allows the user to directly view the condition of the filter cover to allow for periodic replacement of the inlet filter. The air continues into the muffler, to reach the inlet side of the pump.

The HT70 does not require a gas source unless oxygen (O<sub>2</sub>) enrichment is desired. For oxygen enrichment, an optional oxygen

mixer or low flow oxygen adapter kit may be used. Both items connect to the air intake filter cover and allow the manual adjustment of oxygen enrichment from 21% to 100%. The low flow oxygen adapter provides a low flow of oxygen up to 10 L/min.

### Exhalation Valve Control Subsystem

The exhalation valve control subsystem connects within the manifold at the outlet port of the air delivery section. During inspiration, the expiratory valve control circuit actuates the exhalation control On/Off valve to pressurize the exhalation valve drive line. When Positive End-Expiratory Pressure (PEEP) is required, the exhalation control proportional solenoid valve applies pressure to the exhalation valve diaphragm, provides resistance to the expiratory flow, and maintains the patient airway pressure at a set level.

### Pneumatics Safety Subsystem

The mechanical relief valve provides the primary high pressure relief function which opens at a pressure of 100 cmH<sub>2</sub>O. The safety valve provides secondary high pressure relief function by actuating whenever the pressure exceeds 120 cmH<sub>2</sub>O. In the event of a pump failure, the safety system allows the patient to draw ambient air into the breathing circuit through the emergency intake valve and exhale through the exhalation valve. If an occlusion blocks the circuit for greater than 10 seconds, the safety valve opens and relieves pressure in the patient circuit.

## **Electronics System**

The microprocessor-based design incorporates a Graphical User Interface (GUI), a Single Board Computer (SBC), and the electronics to control the motor/pump, power management, and safety functions of the HT70.

Refer to section 8: Diagram 8.2 – Electronics System Diagram.

### Graphical User Interface (GUI) Subsystem

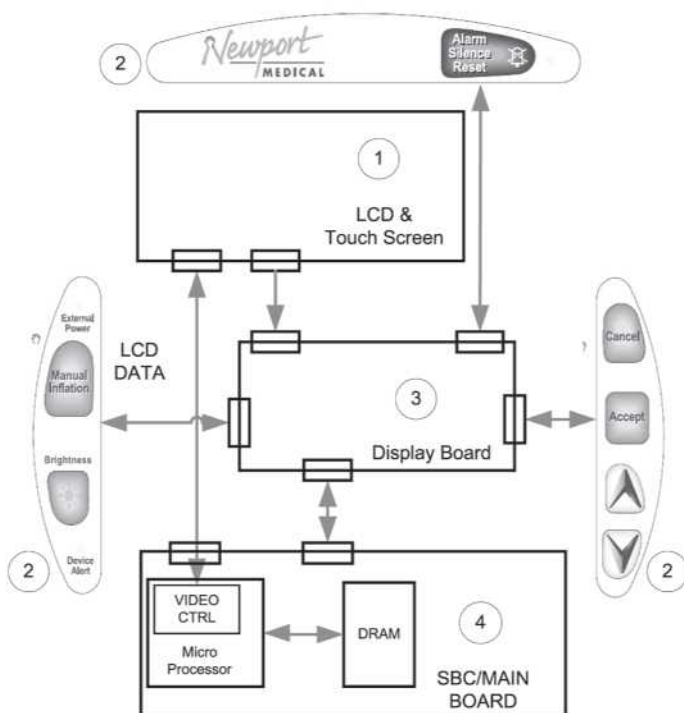
The Graphical User Interface (GUI) consists of a Touch Screen LCD display, membrane panels with embedded discrete Light Emitting Diodes (LED), Display Board, Single Board Computer (SBC) and Main Control Board, and supporting electronics. Refer to the Figure 3-1. Graphical User Interface Subsystem.

# 3

The touch screen overlays the LCD display and the modes and functions are shown on the LCD display. The user can select or change these parameters by pressing the soft keys shown on the LCD display.

The HT70 powers up into a standby condition, in which, there is no patient ventilation occurring. Upon completion of programming the patient settings while in the standby mode, the user can begin patient ventilation by pressing the 'Start Ventilation' soft key displayed on the LCD.

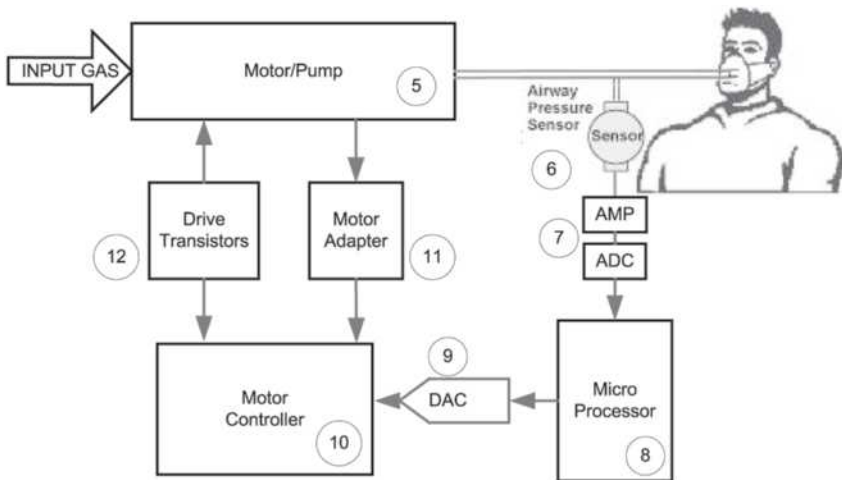
The membrane keys and the LEDs are controlled by the microcontroller on the SBC board.



**Figure 3-1 – Graphical User Interface Subsystem**

### Servo-controlled Motor and Pump Subsystem

The servo-controlled motor and pump subsystem utilizes the software algorithm to execute the servo controls for patient ventilation based on the patient settings. The software establishes the flow by controlling the motor speed which drives the piston-based pump. Refer to Figure 3-2. Motor and Pump Electronics Diagram.



**Figure 3-2 – Motor and Pump Electronics**

### Electronics Safety Subsystem

The HT70 can be brought to a safe state by the SBC microprocessor or independently by discrete hardware. The alarm system is comprised of two separate alarms for clinical alerts and hardware faults. The clinical alarm uses dual speakers and is controlled by the SBC microprocessor. The hardware fault alarm uses a piezo buzzer to indicate a 'Device Alert' alarm to force the system into a safe state.

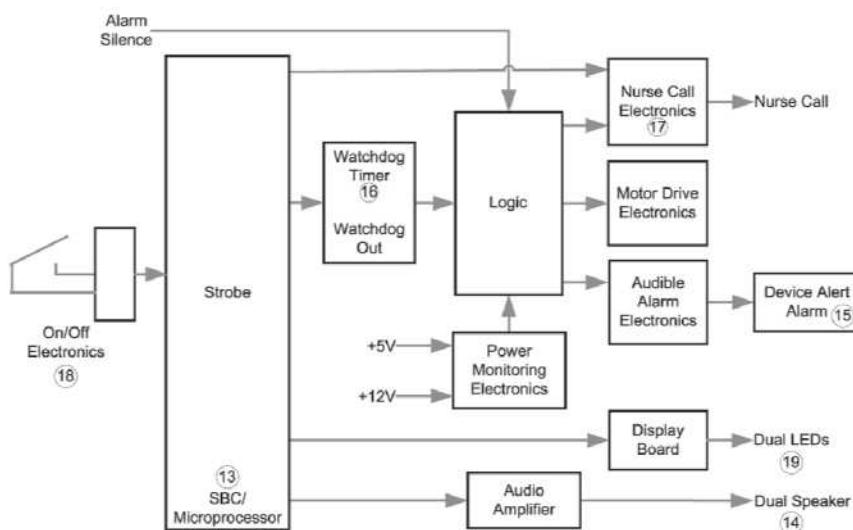
The remote alarm circuit is implemented with a software configurable set/reset relay which configures the nurse call for either a normally open or normally closed system. The relay is energized to the no alarm condition and de-energized to the alarm condition to ensure that nurse call is activated on loss of power.

Upon alarm activation, the software monitors the status of the signal to ensure the speaker is active. The device alert alarm uses a high reliability piezo buzzer powered by a super capacitor to provide an alarm for a minimum of two minutes at the detection of power loss. In addition to the audible alarms, the HT70 has dual LED indicators on the handle to visually display the alarms. When a high priority alarm is active, both LED indicators flash red. When a medium priority alarm is active, the red and yellow LED indicators both flash. When a low priority alarm is active, the LED indicators flash yellow.

To prevent the accidental shut down of the HT70, the safety subsystem uses an intelligent On/Off momentary contact switch. Upon confirmation from the user by pressing the 'Accept' membrane button, software shuts down the system and triggers an alarm. Press

### 3 Theory of Operation

the 'Alarm Silence' membrane button to silence the alarm. Refer to Figure 3-3. Electronics Safety Subsystem.



**Figure 3-3 – Electronics Safety Subsystem**

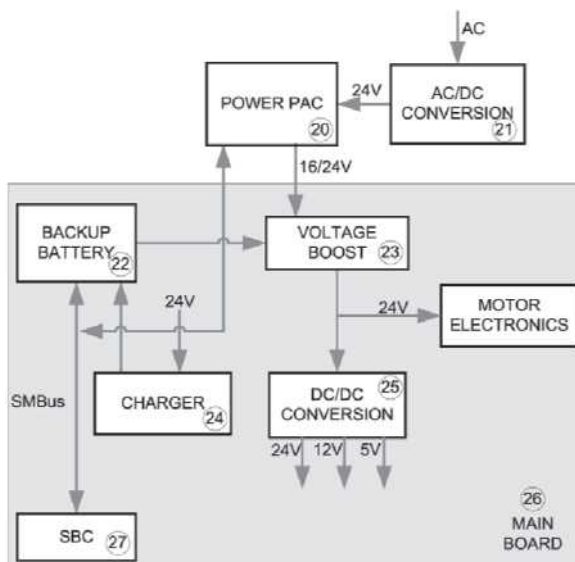
#### Power Management Subsystem

The HT70 draws power from three sources: external AC or DC power, Power Pac, and Backup Battery. External power to the HT70 is supplied through the Power Pac, The Power Pac electronics arbitrates the power source and sends the data to the Main Control board. The established power precedence is external power, Power Pac, and backup battery.

The Power Pac battery assembly slides and latches onto the HT70 and allows for easy removal when swapping the Power Pac is required. When swapping the Power Pac, the HT70 is powered from the Backup Battery.

The HT70 Backup Battery is mounted internally and cannot be swapped by the user. The Backup Battery contains the battery gas gauge and battery cells. The battery charge electronics is located on the Main Control Board. Refer to Figure 3-4. Power Management Subsystem.

The HT70 Internal Dual Battery System is charged when connected to an external AC or DC power source even while the ventilator is in use.



**Figure 3-4 – Power Management Subsystem**

## Ventilation Functions Overview

### Modes of Ventilation

#### A/CMV Mode

(Assist/Control Mandatory Ventilation)

In A/CMV mode, time activated (mandatory) breaths are delivered in accordance with the RR setting. Patients can trigger mandatory breaths in addition to, or in place of, time activated (mandatory) breaths if the effort they generate causes airway pressure to meet the Ptrig or Flow trigger setting. Every such patient effort results in a mandatory breath. The breath can be volume or pressure controlled. PEEP/CPAP may be added. In Pressure Control, tidal volume is determined by the target pressure, i-Time, and patient respiratory mechanics and in Volume Control, by the tidal volume setting.

#### SIMV Mode

(Synchronized Intermittent Mandatory Ventilation)

In SIMV mode, patients receive a fixed number of volume or pressure controlled mandatory breaths (time or patient activated) and may breathe spontaneously between mandatory breaths, with or without pressure support (Psupport). PEEP/CPAP may be added.

The first patient triggered breath in any mandatory breath interval will be a patient triggered mandatory breath. The patient has the rest of the interval to breathe spontaneously. If the patient does not trigger

# 3 Theory of Operation

the ventilator, and one complete mandatory breath interval has elapsed, a time triggered mandatory breath is delivered.

A mandatory breath lockout interval is activated whenever the patient triggers a mandatory breath. This limits the number of mandatory breaths (time triggered or patient triggered) the patient receives in 60 seconds to the RR (b/min) setting.

## **SPONT Mode**

(Spontaneous Ventilation)

In SPONT mode, mandatory breaths are not delivered but the user can adjust both PEEP/CPAP and pressure support (Psupport) levels. The patient has control over each breath.

When PEEP/CPAP is set above 0, the ventilator mode is CPAP (without Psupport) or Bi-level Positive Airway Pressure (with Psupport). Ensure that Ptrig is set so the HT70 detects all spontaneous patient efforts.

Entries for tidal volume, respiratory rate, i time and Low Paw alarm limit are all inactive in SPONT mode. However, users can preset these parameters for future A/CMV or SIMV operation. As with all HT70 operating modes, Backup Ventilation is activated if the BUV linked alarm is violated.

## **NIV**

(Non Invasive Ventilation)

The HT70 can be used for noninvasive ventilation in all modes. Press the *NIV* button on the left side of the touch screen to toggle ON noninvasive.

When NIV is on the following features are activated to assist with noninvasive ventilation:

- When PEEP is in use, bias flow becomes adjustable from 3 to 30 L/min to help manage leaks / minimize auto-triggering.
- The Low Minute Volume alarm can be turned off in the *Alarms* screen
- The Low Pressure alarm can be set closer to the base pressure (within 1 cmH<sub>2</sub>O /mbar above baseline)

**BUV****Backup Ventilation**

Backup Ventilation activates when the currently linked alarm occurs. This function can be linked with the Low Minute Volume (MVI) alarm, the Apnea alarm, or both alarms. During Backup Ventilation the linked alarm(s) will sound and the message window will indicate that Backup Ventilation is in use. There are default Backup Ventilation parameters, but the user may adjust these in the *Advanced Screen*.

Backup Ventilation is functional in all modes.

Backup Ventilation is not active for 60 seconds after the user adjusts ventilator controls, changes modes or starts ventilation from the Setting condition.

During Backup Ventilation, the Alarm Silence/Reset button can be pressed to silence the audible alarm. This will not cancel Backup Ventilation.

When linked with the Low Minute Volume alarm, Backup Ventilation is based on the exhaled or delivered inspiratory minute volume, depending on whether or not the on-airway flow sensor is in use. The inspiratory minute volume may be different from the expiratory minute volume in some conditions, such as in the case of a patient breathing circuit or airway leak.

**Backup Ventilation in A/CMV and SIMV Modes:**

The factory default setting for Backup Ventilation in these two modes will increase the respiratory rate by 1.5 times the set rate, up to a maximum of 99 b/min. The minimum breath rate delivered is 15 b/min.

The respiratory rate (RR) will only increase up to a rate that produces a 1:1 I:E ratio even if the calculated Backup Ventilation rate is higher.

**Backup Ventilation in Spont Mode:**

The factory default setting for Backup Ventilation in the SPONT mode will implement these changes:

SPONT = SIMV mode

Rate = 15 b/min

Pressure Control breath type = 15 cmH<sub>2</sub>O above set PEEP

i-time = 1.0 sec

# 3 Theory of Operation

## Cancellation of Backup Ventilation

### User Cancelled

If during Backup Ventilation, the user adjusts any ventilation parameter, Backup Ventilation is suspended for one minute and all user selected ventilation parameters are employed.

Sixty (60) seconds must pass after parameter adjustments before a linked alarm violation will result in Backup Ventilation.

### Patient Cancelled

If linked to low minute volume, when minute volume exceeds the Low MV alarm setting by 10%, Backup Ventilation is cancelled. If linked to Apnea Alarm, after 2 minutes of Backup Ventilation it is cancelled. At that time the audible alarm stops, the alarm indicator latches and the HT70 resumes ventilation at the user-selected parameters.

## Breath Types

### PC

(Pressure Control Ventilation)

The HT70 targets and maintains patient airway pressure at the set pressure control level throughout inspiration. Breath termination occurs when (1) the set i time elapses, or (2) Paw exceeds the Pressure Control setting by 8 cmH<sub>2</sub>O (mbar). Maximum airway pressure never exceeds the user set High P alarm setting.

The target airway pressure for pressure controlled mandatory breaths in A/CMV and SIMV is the display setting above ambient pressure, not above PEEP.

Both time and patient triggered mandatory breaths can be delivered in A/CMV and SIMV Pressure Control operation. During SIMV Pressure Control operation, patients can breathe spontaneously between mandatory breaths with or without pressure support.

When disconnecting the patient circuit during PC/PS ventilation, i.e. for suctioning, the flow may increase in order to compensate for the low pressure. After reconnecting the patient circuit, the flow will automatically readjust to meet the patient's demand.

**VC**

(Volume Control Ventilation)

During Volume Control ventilation, tidal volume can be set for mandatory breaths. If a volume setting is changed while the ventilator is operating, the change takes place in increments over a series of breaths.

When tidal volume is adjusted, inspiratory time remains constant and mandatory flow changes.

If an attempted tidal volume setting results in a flow rate in excess of 100 L/min or less than 6 L/min, flow adjustment ceases, the user is alerted by an audible beep, and a message appears in the Message Display Window. To allow further volume adjustment, change the inspiratory time to set the Flow to meet the need of the patient.

## Cleaning and Disinfecting

Use the information in this section in conjunction with hospital policy, physician prescription, Homecare Dealer or accessory manufacturer instructions.

### Definitions

**Cleaning:** A process that uses a medical detergent or alcohol based cleaning solution to remove blood, tissue and other residue. Rinse thoroughly with sterile, distilled water and allow to air dry.

**Disinfection:** A liquid chemical process that kills microbial organisms.

**Sterilization:** A process that uses steam autoclave or ethylene oxide (EtO) this is designed to render a product free of viable microorganisms.

**Caution:** When using liquid chemical agents, closely follow the manufacturer's recommendations. Prior to use, verify that the agent is compatible with plastics.

**WARNING** Ethylene Oxide (EtO) is toxic. All accessories **MUST** be completely dry prior to packaging for ethylene oxide sterilizing. After sterilizing, they must be properly aerated to dissipate residual gas absorbed by the material. Follow the EtO manufacturer's recommendations for the specific aeration periods required.

**Caution:** Ethylene Oxide (EtO) may cause superficial crazing of plastic components and will accelerate the aging of rubber components.

**Caution:** Always inspect breathing circuits and accessories after cleaning, disinfecting or sterilizing to check for deterioration. If the any part is damaged or shows excessive wear, replace with a new part. Do not use cracked or damaged parts.

## Ventilator

Wipe clean between patients and as needed while in use. The exterior of the ventilator should be wiped clean with a cloth dampened with a medical detergent, disinfectant or alcohol based cleaning solution.

**Caution:** Do not use agents that contain acetone, toluene, halogenated hydrocarbons, or strong alkalines on the face panel or ventilator housing.

## 4 Cleaning and Maintenance

**Caution:** Never autoclave or EtO sterilize the HT70 ventilator. These processes will damage the HT70, rendering it unusable.

### Accessories

#### Low Flow Oxygen Reservoir

Clean and disinfect between patients and as needed while in use, refer to the instructions provided with the oxygen reservoir.

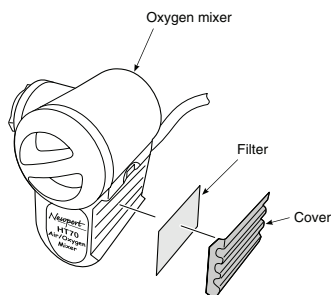
Disassembly from the HT70 ventilator: Remove the Oxygen Reservoir from the HT70 Fresh Gas Intake port. Disconnect the oxygen tubing.

General Cleaning Instructions: Hold the Low Flow Oxygen Reservoir in both hands and twist the top counterclockwise to disassemble. Separate all the parts and clean with soap and water, rinse thoroughly and then air dry.

**Caution:** Never mount the Low Flow Oxygen Reservoir onto the ventilator when wet.

#### Air/Oxygen Entrainment Mixer

Between patients and as needed while in use, the exterior of the mixer and attached hose should be wiped clean with a cloth dampened with a medical detergent, disinfectant or alcohol based cleaning solution.



Check the mixer intake filter (p/n FLT3209P) at setup and at least weekly and replace when dirty.

**WARNING** Always use a Mixer Intake Filter in the mixer to protect the internal mechanisms from contaminants and preserve the lifespan of your mixer.

**WARNING** Never reverse the Mixer Filter.

**Caution:** Do not wash or sterilize the Mixer Filter.

## Reusable Breathing Circuits and Exhalation Valves

The HT70 ventilator may be used with a standard single limb or “J” style breathing circuit with a quality exhalation valve. Reusable breathing circuits and exhalation valves are generally provided in clean, but not sterile, condition. Follow the manufacturer’s instructions to clean and/or disinfect prior to use.

**WARNING** Do not use electrically conductive breathing circuits. Always use clean and dry breathing circuits.

Reusable circuits should be cleaned and disinfected between patients and as needed while in use. Always use a clean, disinfected exhalation valve (and humidifier/probe assembly if appropriate) when a breathing circuit is reassembled for patient use. Clean and disinfect in accordance with the instructions provided by the manufacturer.

**Caution:** To avoid damage to a reusable circuit, attach and detach the circuit by grasping the cuffs at the end of the circuit tubing. Do not pull or twist the circuit tubing.

**General Cleaning Instructions:** Use a low flow of running water or low flow of air to clear tubings and passages of organic matter. Wash all components of the breathing circuit and exhalation valve with a soft brush in a mild medical detergent. Rinse thoroughly with sterile, distilled water. Shake off excess water and place all parts on a clean towel to air dry. (Do not heat or blow dry.) Always follow the instructions provided by the manufacturer.

**Disinfect:** Refer to the instructions provided by the breathing circuit and exhalation valve manufacturer.

**Sterilize:** Refer to the instructions provided by the breathing circuit and exhalation valve manufacturer.

### Air Intake Filter (p/n HT460300)

The Air Intake Filter, located on the right side of the ventilator behind the Filter Cover, keeps dirt and particles out of the ventilator’s piston system and patient gas pathway. As the filter becomes dirty it can reduce the volume of air drawn into the ventilator and add stress to the pump. Check the Intake Filter weekly. Replace with a new filter when the majority of the filter surface area is no longer white. Intake Filters are not reusable.

## 4 Cleaning and Maintenance

**WARNING** NEVER operate the HT70 without a clean Air Intake Filter in place. NEVER reverse the Air Intake Filter when dirty.

### **Proximal Inline Filter** (p/n HT6004701 or equivalent)

Check the Proximal (Prox) Inline Filter weekly and replace it at least every 3 months. Discard it and replace with a new filter if it appears to have gotten wet or come in contact with a contaminant. Proximal Inline filters are not reusable. If the filter becomes occluded, replace the filter. The primary indication for this would be a Check Circuit or Prox Line Alarm.

Newport Medical strongly recommends that extra Prox Inline Filters be available at all times when using the HT70 ventilator.

**WARNING** Always use a Proximal Inline Filter (p/n HT6004701 or equivalent) at the Prox. Line Connector to protect the internal pressure transducers from moisture or other contaminants.

**WARNING** Never reverse the Proximal Inline Filter.

**Caution:** Do not wash or sterilize the Prox Inline Filter.

## **Maintenance Guidelines**

### **Routine Maintenance**

- ✓ Perform the Circuit Check each time a fresh circuit/exhalation valve is installed.
- ✓ Check the Air Intake Filter (located behind the Filter Cover) at setup and at least weekly while in use. In some environments, it may need to be checked more often. Replace when the majority of the filter surface area is no longer white. Air Intake Filters are not reusable.

**WARNING** NEVER reverse the Air Intake Filter when dirty.

- ✓ Check the Prox Inline Filter weekly. Replace with a new filter if it appears to have gotten wet or come in contact with a contaminant. Inline filters are not reusable.
- ✓ Check the Mixer Intake Filter (located behind the Mixer Cover) at setup and at least weekly while in use. In some environments, it may need to be checked more often. Replace when the majority of the filter surface area is no longer white. Mixer Intake Filters are not reusable.

- ✓ Inspect the AC Power Adapter on a regular basis for signs of broken or frayed cord or connectors.
- ✓ Inspect the exhalation valve after each cleaning to verify that there are no cracks or damaged surfaces.
- ✓ Wipe down the surface of the ventilator housing regularly to remove any dust that might accumulate.
- ✓ Inspect and when necessary, replace accessories.
- ✓ If service is required, contact Newport Medical or your local equipment provider..
- ✓ To preserve the Internal Dual Battery System life:
  1. Whenever possible, plug into external power source to charge the batteries.
  2. Use the optional DC Auto Lighter Cable accessory to power the HT70 when traveling by automobile or to connect to an external battery.

See Section 7 of the Operations Manual, Battery Operation for more information on the proper operation of the HT70 Internal Dual Battery System.

### **6 Month Maintenance**

- ✓ Routine maintenance as described above
- ✓ Perform the Quick Check Procedure (described in Section 5 of the Operations Manual)

### **12 Month Maintenance**

- ✓ Routine maintenance as described above
- ✓ Perform the Quick Check Procedure (described in Section 5 of the Operations Manual)

### **24 Month Maintenance**

- ✓ Replace air intake and prox inline filter (HT46300 and HT6004701)
- ✓ Replace the primary integrated battery (Power Pac) (BAT3271A)
- ✓ Replace the secondary internal back up battery (BAT3205A)
- ✓ Replace the oxygen sensor (if installed) (SEN2103P)

## 4 Cleaning and Maintenance

- ✓ Replace the Coin Battery (BAT3207P)
- ✓ Replace the cooling fan filter (FCT3202P)
- ✓ Calibration and OVP performed by Authorized Service Provider

### 15,000 Hour Maintenance

- ✓ A comprehensive maintenance should be performed after 15,000 hours of operation. Refer to the HT70 Service Manual, or contact the Newport Medical Technical Service Department for detailed information on the 15,000 Hour Maintenance.

Do not attempt to open or perform any service procedures on the HT70. Only Newport Medical trained technicians are authorized to service the ventilator.

Newport Medical Technical Service Department:  
Telephone: 800-255-6774 option #4 & option #2  
Email: [venttechsupport@covidien.com](mailto:venttechsupport@covidien.com)

### General Warnings

Preventive maintenance work, repairs and service may only be performed by Newport Medical trained or factory-authorized personnel.

Always follow accepted hospital procedures or physician instructions for handling equipment contaminated with body fluids.

The ventilator and its accessories must be thoroughly cleaned and disinfected after each patient use. Perform all cleaning and sterilization of external parts and accessories in accordance with established hospital procedures and manufacturer's instructions.

Certain components of the ventilator, such as the exhalation valve and the front panel, consist of materials that are sensitive to some organic solvents used for cleaning and disinfection (e.g. phenols, halogen releasing compounds, oxygen releasing compounds, and strong organic acids). Exposure to such substances may cause damage that is not immediately recognizable.

The reusable exhalation valve, reusable breathing circuit and other parts that come in direct contact with the patient should be disinfected or sterilized between uses according to hospital policy.

## Factory Maintenance or Repair

An authorized Newport Medical Instruments factory-trained technician must do all service or repairs performed on the HT70.

**Caution:** Always disconnect the external power supply prior to servicing.

Scheduled maintenance or repair services are available from the Newport Technical Service Department. To send your ventilator in for service, see Repackaging /Return Information that follows in this section.

Current pricing for scheduled maintenance and labor rates can be found in Newport Medical Instruments Annual Price List. To obtain a copy, please contact your local Newport Sales Representative or contact our Customer Service Department.

## Repacking / Return Information

Use the original packing carton and material to ship the ventilator back to Newport Medical. Or you can contact Newport Medical Customer Service department to order replacement packing material.

Prior to returning your ventilator for service or repair you must obtain a returned goods authorization number (RGA) from our Technical Service Department. Refer to the HT70 Service Manual or contact Technical Service department for complete instructions.

See Contact Information page at the front of this manual for address, phone and website details.

## General Information

This chapter describes how to repair the HT70 ventilator including instructions on how to remove and replace the major assemblies, subassemblies, and components, if necessary. Unless otherwise instructed, reinstall the assemblies, subassemblies, and components by reversing the disassembly procedure.

Prior to servicing the HT70, be sure to do the following:

1. Disconnect from external power.
2. Disconnect oxygen (O<sub>2</sub>), if applicable.
3. Put on ESD protective equipment and ensure the service area is an ESD safe environment.

### WARNING

These instructions are intended for use only by a Newport Medical Instruments factory-trained technician. Do not perform any unauthorized modifications or repairs to the ventilator or its components.

### Caution

To avoid damaging equipment, always use standard electrostatic discharge (ESD) precautions, including an ESD wrist strap, when servicing the ventilator.

## Tools Required

#1 Phillips Screwdriver (short)

#1 Phillips Screwdriver

#1 Flat Screwdriver

5/64" Hex Driver

3/32" Hex Driver

3/16" Nut Driver

Large Pliers

Anti-static Wrist Strap

Anti-static Mat or Grounded Workstation

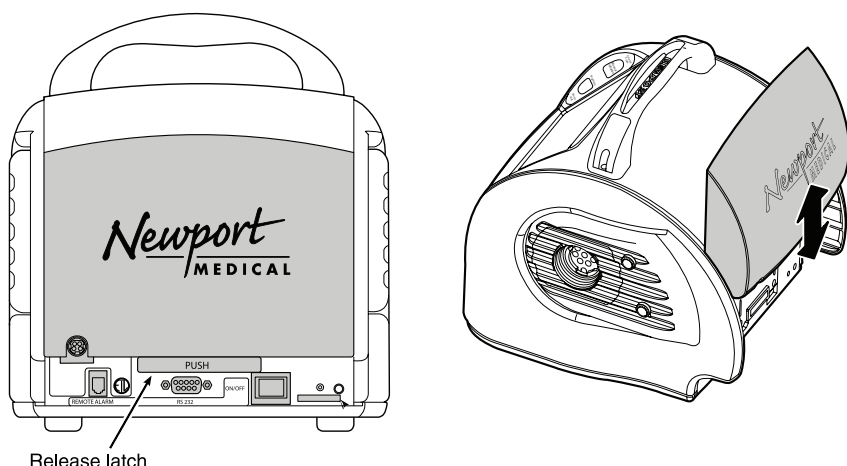
# 5 Service and Repair

## Removal and Replacement of the Rear Panel Assembly

Including associated assemblies

### Rear Panel Assembly

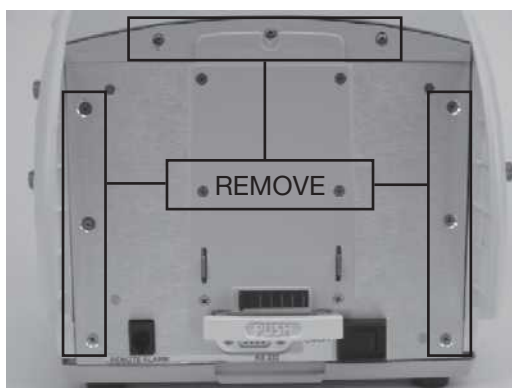
1. Disconnect the power cord PWR3204P from the HT70.
2. Press the release latch and remove the Power Pac BAT3271A.



**Figure 5.1 Power Pac Removal**

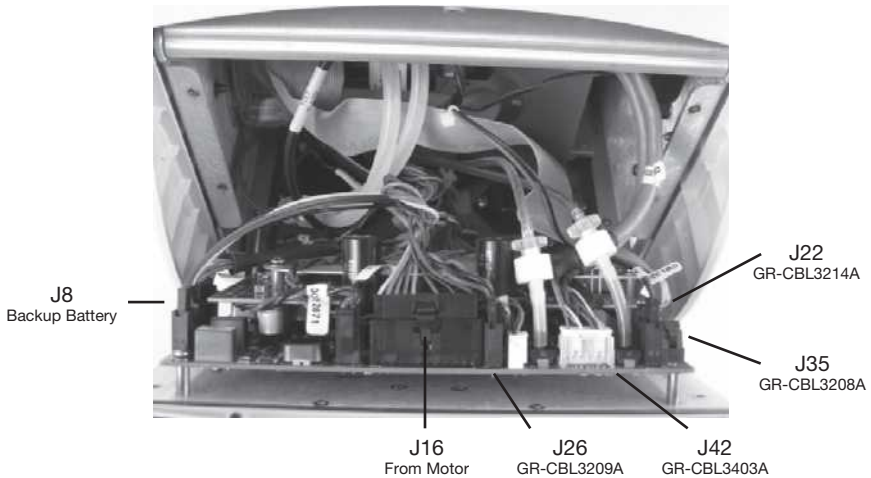
3. Using a #1 Phillips screwdriver, remove the nine (9) outside screws on the rear panel.

**Note:** The two (2) outside top screws are longer and must be used to secure the top portion only.



**Figure 5.2 – Rear Panel Screws**

4. Carefully slide the rear panel back to expose the cable harnesses and pneumatic tubes connected to the Main Control Board (MCB).
5. Disconnect the following cables from the MCB: J8, J16, J22, J26 J35 and J42.



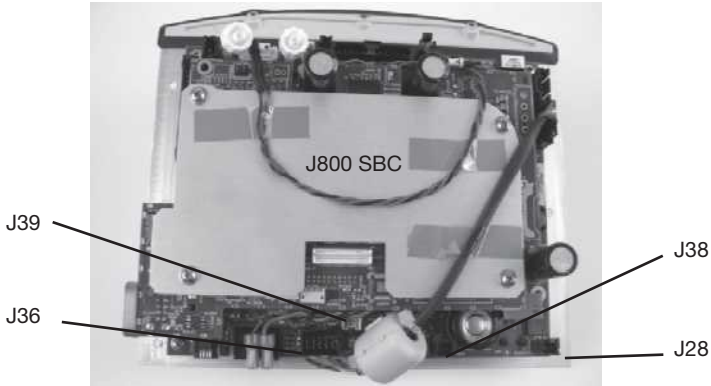
**Figure 5.3 – MCB Connections**

6. Unscrew the luer-lock connectors on the Yellow tube from XD2 and the small Green tube from XD22.



**Figure 5.4 – Pressure Transducers**

7. Disconnect the LCD data cable (GR-CBL3219A) from J800 (push tab release) on the Single Board Computer board (SBC).



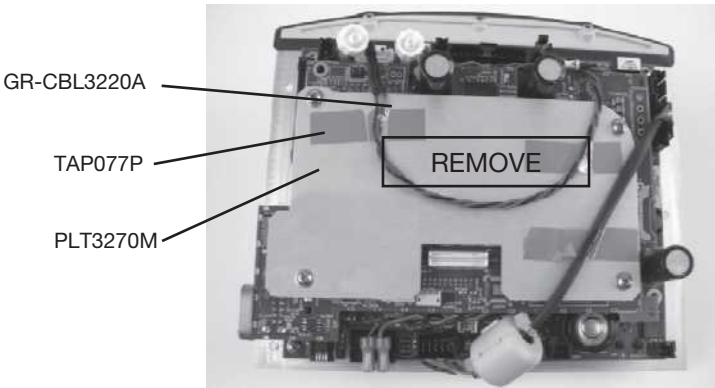
**Figure 5.5 SBC and MCB Connections**

8. Disconnect the remaining cables on the MCB: J28, J36, J38, and J39.
9. Lay the rear panel on the work surface.

### Single Board Computer (GR-PCB3207A) Assembly

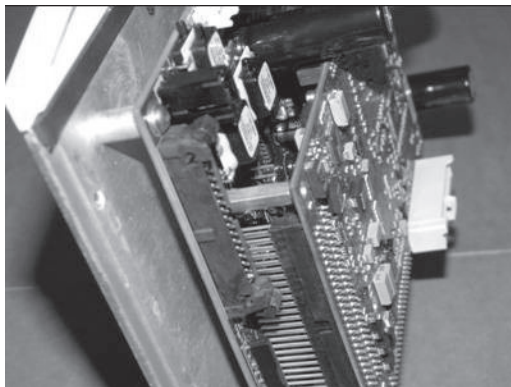
1. Using a #1 Phillips screwdriver, remove the four (4) screws (SCR3274P) and washers (WSH1512P) securing the SBC EMI shield (PLT3270M). Between the EMI shield and the SBC there are 4 spacers (SPC3205P).

**Note:** The routing of the cable GR-CBL3220A (temperature sensor) is very important. If the cable is removed from the EMI . panel, ensure the cable is secured with tape NMI part number TAP077P and is properly routed when installing.



**Figure 5.6 SBC EMI Shield**

2. Carefully disconnect the SBC from the MCB.



**Figure 5.7 SBC and MCB Connection**

3. Reverse the above steps to reinstall.

### **Main Control Board (MCB) Assembly (GR-PCB3252A)**

1. Perform above steps for removal of Single Board Computer (SBC) assembly.
2. When ordering the SBC be sure to order GR-105795 for the Plus Units and GR-105796 for the HT70-2.
3. Disconnect the Power Pac Battery Connector Board cable (GR-PCB3203A) from J5 on the MCB.
4. Disconnect the Power Switch cable (GR-CBL3217A) from SW1 on the MCB.

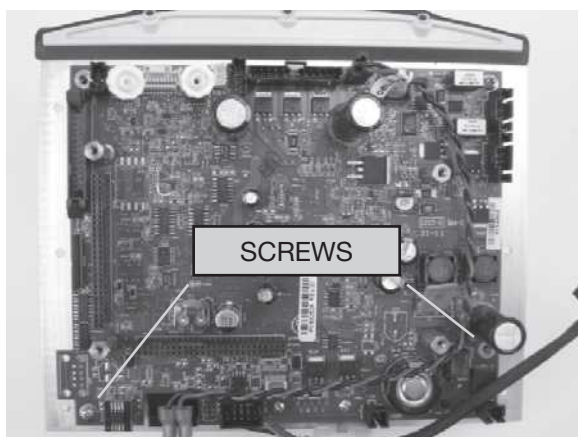


**Figure 5.8 Main Control Board (MCB)**

5. Using a 3/16" nut driver, remove the two (2) hex jack nuts (SCR1855P) that secure the RS232 communication port.

# 5 Service and Repair

6. Using a #1 Phillips screwdriver, remove the four (4) screws (SCR3210P) securing the MCB.



**Figure 5.9 MCB Screws**

7. Reverse the above steps to reinstall.

## **Power Switch Assembly (SWI3204P)**

1. With the rear panel removed, use a small flat screwdriver to carefully press the locking tabs against the sides of the power switch.



**Figure 5.10 – Power Switch**

2. While pressing the locking tabs, press the back of the switch to remove the switch from the back plate.

**Note:** Note the position of the tabs on the switch and be sure to position the new switch in the same configuration. The black wire is

on the left side of the switch and the red wire is on the right side. The ON position faces to the right, looking from the rear of the HT70

3. Reverse the above steps to reinstall.

## **Removal and Replacement of the Top Case Assembly**

Including associated assemblies

### **Inlet Air Filter (HT460300)**

1. Facing the HT70, the inlet air filter is located on the right-hand side.
2. Turn the knurled screws counter-clockwise to remove.
3. Remove the inlet filter cover (CVR3215A) assembly by sliding the cover toward the back of the HT70 and lift out.
4. Remove the Air Inlet Filter Gasket (GKT3201M) by carefully pulling it away from the side of the HT70.
5. Remove the inlet air filter.



***Figure 5.11 – Air Inlet Filter Cover (CVR3215A)***

## 5 Service and Repair



**Figure 5.12 – Air Inlet Filter (HT460300)**



**Figure 5.13 – Air Inlet Filter Gasket (GKT3201M)**

6. Reverse the above steps to reinstall.

**Note:** To reattach Air Inlet Filter Gasket (GKT3201M), use adhesive NMI part number ADH110P. Apply the adhesive to the rear surface of the gasket avoiding the slotted areas. Align the tab in the gasket with the slot on the HT70 and press firmly and allow curing overnight before returning to service.

### **Fan Filter and Guard (FLT3202P)**

1. Facing the HT70, the fan filter is located on the left-hand side.
2. Turn the knurled screw counter-clockwise to remove.
3. Remove the fan filter cover (CVR3216A) assembly by sliding the cover toward the back of the HT70 and lift out.



**Figure 5.14 – Fan Filter Cover**



**Figure 5.15 – Fan Filter and Guard**

4. Using a #1 Phillips screwdriver, remove the four (4) screws (SCR3213P) securing the fan filter and guard (FLT3202P).
5. Discard the filter guard assembly.
6. Install the new filter assembly by reversing the above steps.

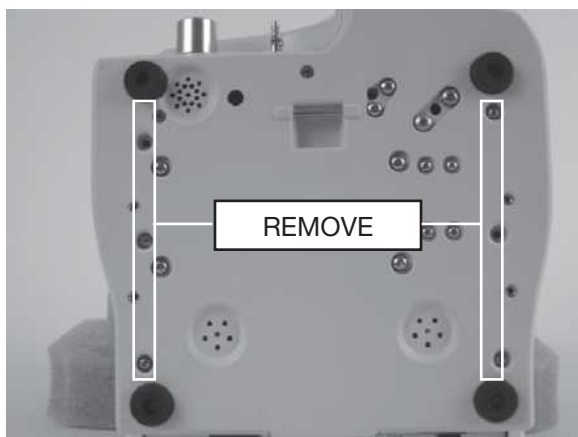
**Note:** The 4 screws (SCR3213P) are secured with a torque setting of 6 – 8 in-lb

#### **Top Case Assembly (CVR3250A)**

1. Refer to the Rear Panel Assembly section to remove the rear panel.
2. Lay the HT70 on the left or right side to expose the screws on the bottom plate.

# 5 Service and Repair

3. Using a #1 Phillips screwdriver, remove the six (6) (SCR3217P) outside screws.



**Figure 5.16 – Top Case Screws**

4. Sit the HT70 upright.
5. Disconnect the inlet tube from the muffler (MUF3210A).
  - a. For the HT70 with O<sub>2</sub> sensor, disconnect the oxygen sensor cable (GR-CBL3200A) from the sensor.



**Figure 5.17 – Muffler w/Oxygen Sensor (optional)**

6. Using a #1 Phillips screwdriver, remove the two (2) screws (SCR3210P) securing the muffler located on the left side when facing the rear of the HT70. Remove the muffler and gasket.
7. Carefully lift the rear portion of the top case Assembly. Ensure the fan clears the pump assembly.

8. Slide the top case assembly forward to clear the Gas Output, Proximal Line, and Exhalation Valve ports.
9. Reverse the above steps to reinstall.

**Note:** When reinstalling the muffler, ensure the gasket (GKT3206M) is between the muffler and the inside of the HT70.

### **Fan Assembly (GR-CBL3208A)**

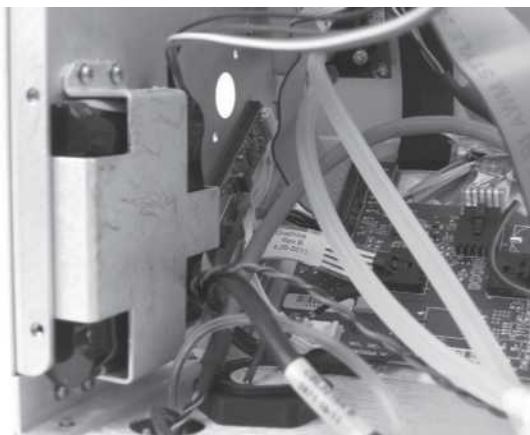
1. Using a short #1 Phillips screwdriver, remove the four (4) screws with captured washers (SCR3218P) securing the fan and four (4) washers (WSH1512P) between the fan and housing
2. Reverse the above steps to reinstall.



**Figure 5.18 – Fan Assembly**

### **Backup Battery (BAT3205A)**

1. Using a #1 Phillips screwdriver, remove the four (4) screws (SCR3215P) securing the bracket (BKT3214M) that holds the backup battery pack.



**Figure 5.19 – Backup Battery**

2. Remove the backup battery.
3. Install the gasket (GKT4408P) on the bottom, opposite end from the wire harness, of the battery pack.

If the gasket on the bracket has come off, replace with a new gasket.

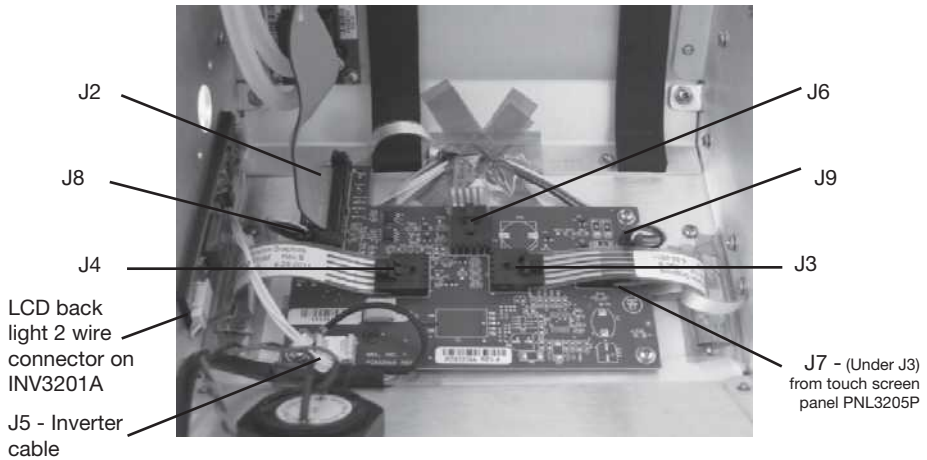
**Note:** The gasket is double-side adhesive. DO NOT remove both protective covers. Only remove one protective cover to attach the gasket to the bottom of the battery or the middle of the bracket.

4. Reverse the above steps to reinstall.

## **Display Board (GR-PCB3206A)**

1. Disconnect the cables from the display board at these connections: J2, J3, J4, J5, J6, J7, J8, and J9.
2. Using a #1 Phillips screwdriver, remove the four (4) screws securing the display board (3 screws (SCR3215P) and 1 screw (SCR3216P))
3. Reverse the above steps to reinstall.

**Note:** During reassembly, ensure the LCD data cable GR-CBL3219A is secured by the yellow cable clamp (CLP2101P). The CLP2101P is secured with SCR3216P and is installed under the GR-PCB3206A. The ground connector for the LCD cable is installed on above the GR-PCB3206A. (Screw→Ground Connector→PCB→CLP2101P →Chassis)



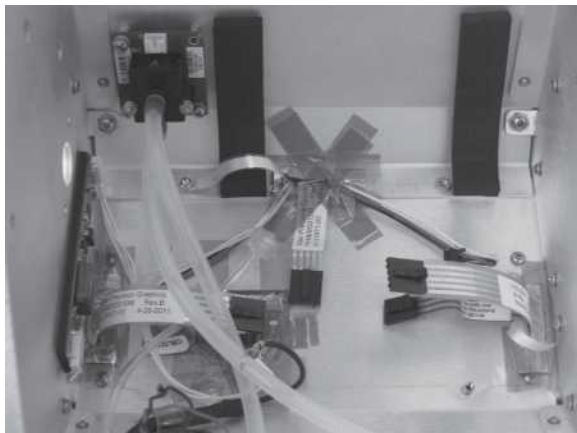
**Figure 5.20 – Display Board**

### **Chassis Assembly (GR-BSE3251A)**

1. Using a #1 Phillips screwdriver, remove the two (2) screws (SCR3215P) securing the ground strips for the left and right switch membranes and use a 3/16" nut driver and remove the nut (NUT2107P) and washer (WSH3207P) securing the top membrane ground strip.
2. Using a #1 Phillips screwdriver, remove the seventeen (17) screws (SCR3215P) securing the top and sides of the chassis.
3. From the inverter board (INV3201A) remove the LCD back light cable from the 2 pin connector.
4. Carefully separate the chassis from the top case.
5. Guide the switch membrane, LCD back light, and touch screen cables through each perspective slot. Take note of each connector to ensure they are not damaged during disassembly.
6. Reverse the above steps to reinstall

**Note:** When feeding the cables back through slots, ensure the EMI shielding (GKT3214M) is intact. If the gasket is not in good condition, replace the gasket. Use tape NMI part number TAP077P to cover all slots as shown in figure 5.21.

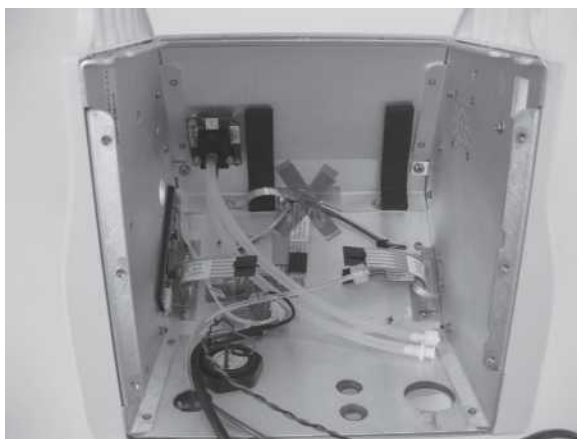
## 5 Service and Repair



**Figure 5.21**

### **Buzzer Assembly (GR-CBL3210A)**

1. Located on the front side of the chassis assembly, turn the securing mounting ring counter-clockwise to loosen and remove the buzzer cable assembly.
2. Be sure to take notice of the position of the silicon o-ring on the buzzer.
3. Reverse the above steps to reinstall.

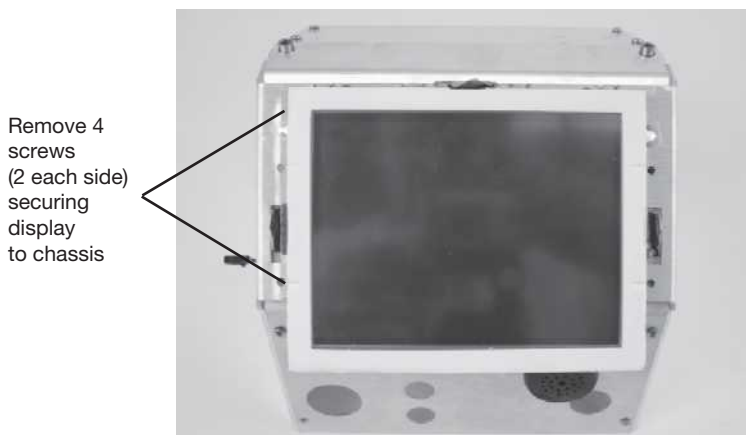


**Figure 5.22 – Buzzer Assembly**

### **LCD/Touch Screen Display (DSP3205A) Assembly**

1. Disconnect the LCD Display cable (GR-CBL3219A) from the back of the display.
2. Using a #1 Phillips screwdriver, remove the four (4) screws P/N SCR2133P securing the LCD/Touch Screen assembly to the chassis.

**Note:** The LCD display and Touch Screen must be replaced as a whole assembly including the LCD gasket. (GKT3208M, PNL3205P and DSP2105P)



**Figure 5.23 LCD Touch Screen Display**

3. Reverse the above steps to reinstall.

#### **Handle Assembly (GR-HDL3215A)**

1. Using a #1 Phillips screwdriver, remove the four (4) screws (SCR3216P) securing the handle to the Top Case assembly.
2. Carefully feed the two (2) sets of LED cables through the feed holes.
3. Reverse the above steps to reinstall.

**Note:** In the event an LED is no longer functioning, the complete handle assembly is replaced.

## 5 Service and Repair



Remove  
4 screws  
(2 each side)  
securing the  
handle to cover  
assembly

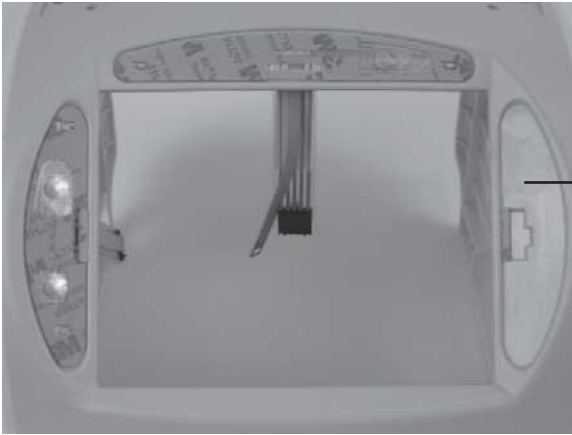
**Figure 5.24 Handle Removal**

### Membrane Switch Assemblies

1. Using a small flat screwdriver, carefully remove the Right (MEM3210M), Top (MEM3212M), or Left (MEM3211M) Membrane Switch assemblies.
2. Feed the cable connection and ground strip through the slot openings.
3. Clean the surface area to remove any adhesive residue.
4. Remove the protective the cover from the new membrane switch.
5. After feeding the cable connection and ground strips through the slot openings, align the membrane switch on the top case and press into place.
6. Verify the membrane switch sits fully inside the recess on the top case.

**Note:** After replacing any of the membrane assemblies, it will be necessary to replace the membrane overlay p/n OVR32XXA (language specific)

OVR3215A – English  
OVR3220A – Symbols  
OVR3221A – Japanese  
OVR3222A – Spanish  
OVR3223A – Portuguese  
OVR3224A – Vietnamese  
OVR3225A – Chinese  
OVR3226A – Russian



Before installing a new membrane, clean the surface of the cover assembly to remove any adhesive.

**Figure 5.25 Membrane removal and replacement**

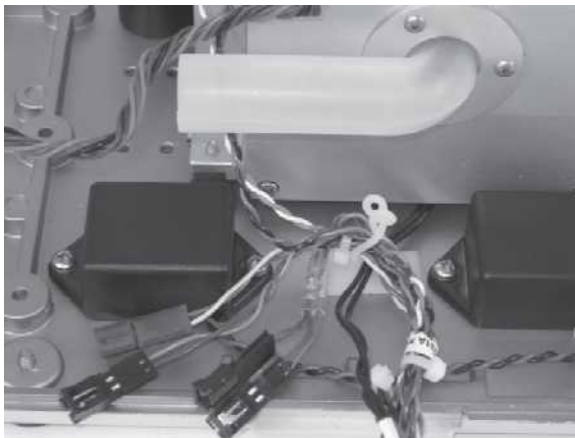
## Removal and Replacement of the Base Assembly

Including associated assemblies

Refer to the Rear Panel Assembly and the Top Case Assembly sections for removal of the rear panel and top case.

### Speaker Assembly (GR-CBL3209A)

1. Using a #1 Phillips screw driver, remove the four (4) screws (SCR3234P) securing the speaker covers.
2. Using a #0.5 Phillips screwdriver, remove the four (4) screws (SCP2025P) securing the speakers to the bottom plate.



**Figure 5.26 – Dual Speaker Assembly**

3. Reverse the above steps to reinstall.

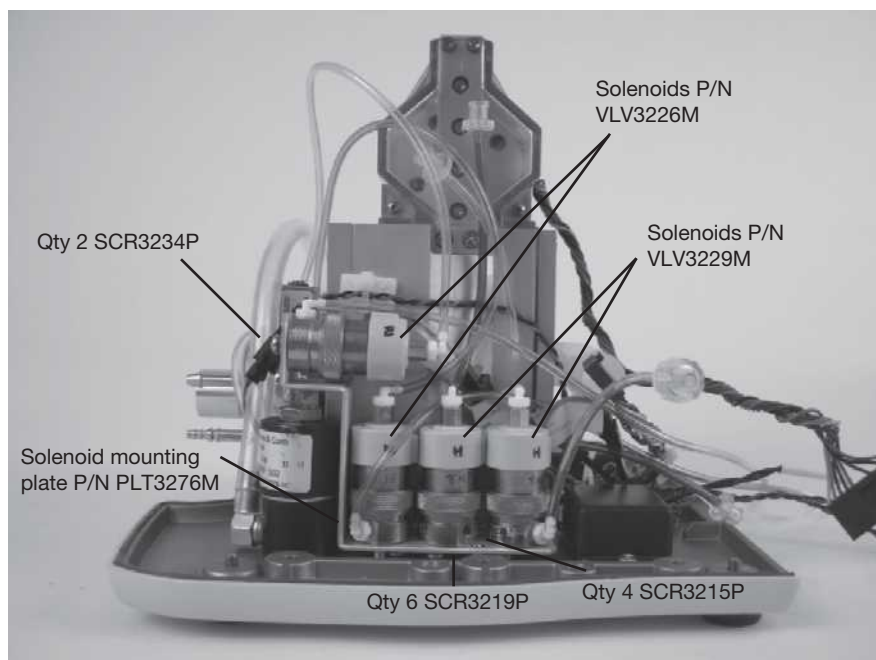
## 5 Service and Repair

**Note:** Under the speakers, there is a gasket (SEL3201M). Before replacing speaker assembly, ensure the gasket is installed.

**Note:** Both speakers must be replaced when replacing either speaker.

### Solenoid Valve Group Assembly (GR-SOL3250A)

1. Refer to Figure 5.27 and 5.28 for the following steps.



**Figure 5.27 Solenoid Valve Group Assembly**



2. Disconnect the Auto-zero Solenoid yellow tube from the 'Y' port connection on the manifold.
3. Disconnect the Auto-zero Solenoid green tube from the 'G' port connection on the manifold.
4. From the bottom of the base assembly, remove the 4 screws p/n SCR3215P securing the solenoid plate p/n PLT3276M to the HT70 base.
5. Remove the plate. To replace any of the four solenoids p/n VLV3226M or VLV3229M, remove the 2 screws securing the solenoid, install the new solenoid and secure with the 2 screws removed.

# 5 Service and Repair

## On / Off Valve (VLV3225M)

1. Remove the clear tube from the top of the on/off valve.
2. From the bottom of the base assembly, remove the 2 screws (SCR3219P) securing the on/off valve. Install the new on/off valve and secure with the 2 screws removed.

**Note:** When reinstalling the On / Off valve, it is important to align the OUT port with the exhaust hole on the bottom of the base plate.

## Safety Valve (VLV3227M)

1. Remove the clear tube from the side of the safety valve.
2. From the bottom of the base assembly, remove the 2 screws (SCR3232P) securing the safety valve. Install the new safety valve and secure with the 2 screws removed.

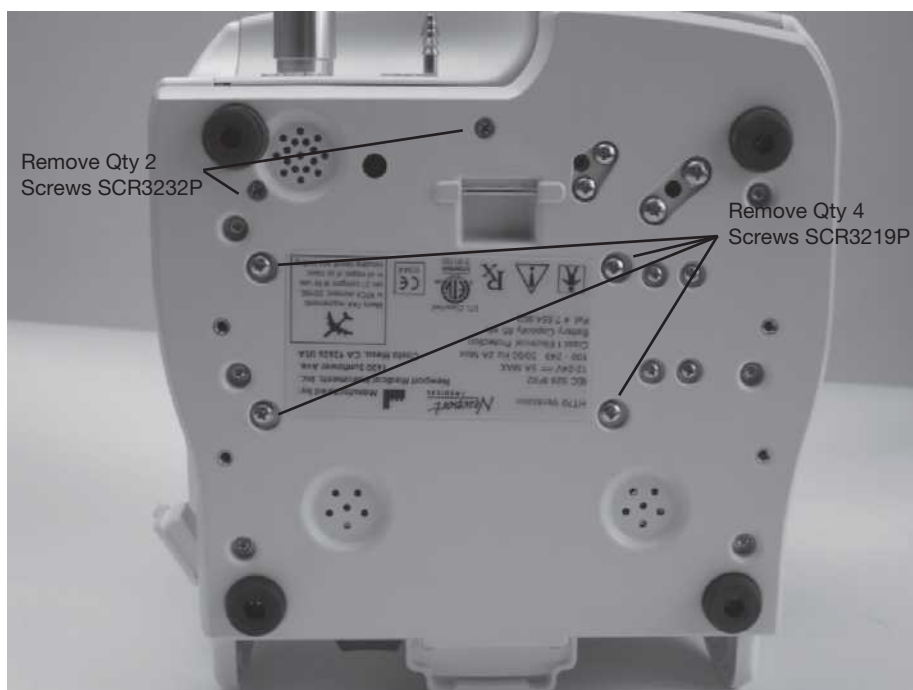
**Note:** When reinstalling the safety valve, route the cable under the pump assembly as shown in figure 5.30.

## Pump Assembly (GR-PMP3230A)

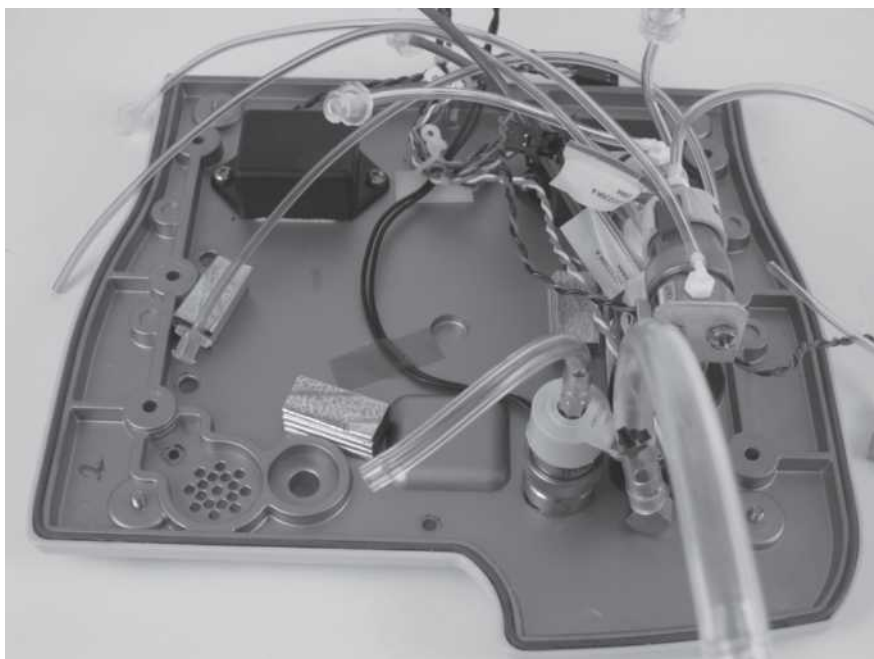
1. Position the base assembly on one side to access the screws securing the pump assembly.
2. Using a Phillips screwdriver, remove the four (4) screws (SCR3219P) and two (2) screws (SCR3232P) securing the pump assembly to the base plate.
3. Reverse the above steps to reinstall.

**Note:** When reinstalling the pump assembly, make sure the 3 stacks of gasket GKT3212P are in place. If you need to replace the gaskets, order Qty 7.5" and cut to 1/2" lengths. See figure 5.30 for location.

When replacing the pump, if you have an older style (PMP3200A, PMP3216A) you will also need to order SPC3206x6 for the difference in weight.



**Figure 5.29 – Pump/Manifold Screws**



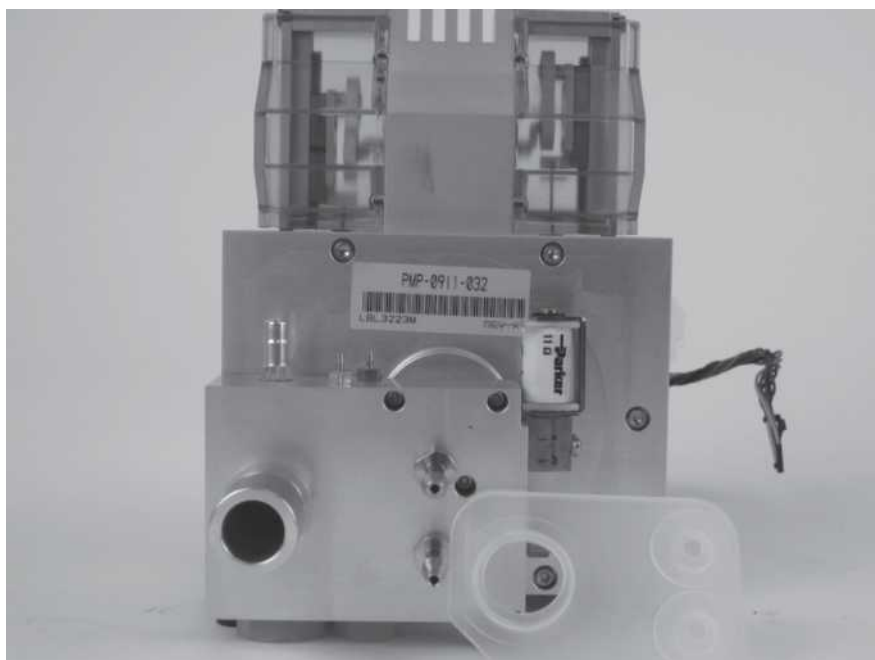
**Figure 5.30 Base Plate**

# 5 Service and Repair

## Manifold Assembly (MNF3200A & GR106764)

1. Refer to the Solenoid Valve Removal section for regarding the disconnection of the pneumatic tubes.
2. Refer to Figure 5.29, Pump Assembly Removal, regarding the removal of the pump assembly from the base plate.
3. Disconnect the black and red wires from the Proportional solenoid.

**Note:** Note the black wire connects to the outside connection on the proportional solenoid. Make sure to reconnect the wires properly.



**Figure 5.31 – Proportional Solenoid Wire Connections**

4. Using a 3/32" hex driver, remove the three (3) screws (SCR3207P) and washers (WSH1250P) securing the manifold to the pump assembly.
5. Using a 1/16" hex driver, remove the two (2) screws (SCR3231P) and washers (WSH3207P) securing the Proportional solenoid valve (VLV3217P) to the manifold.
6. Reverse the above steps to reinstall.

**Note:** When reinstalling the Proportional solenoid valve to the manifold and the manifold to the pump assembly, ensure the o-rings (ORG200P) are fully seated to avoid any possible damage to the o-rings.

**Note:** The gasket (GKT3254M) is installed over the gas outlet and Prox and Exh fittings.

## General Information

This chapter describes how to perform a Quick Check, calibration, and Operational Verification Procedure of the HT70 ventilator. These performance checks ensure the HT70 is in proper operating condition after a service and/or repair has been performed.

**WARNING** Do not use the HT70 ventilator unless it passes the OVP.

## Test Equipment Required

- Certified calibration analyzer PTS2000 or equivalent
- Oxygen Analyzer
- Newport Medical disposable patient breathing circuit (BCD43811P) or equivalent
- Adult 500 mL test lung (LNG800P or LNG600A with an Rp20 parabolic airway restrictor)
- Parabolic Airway Restrictors – Rp20 & Rp50
- Medical grade oxygen source
- Cap (CAP100P) to occlude the patient circuit Y-piece or equivalent (Qty 2)
- 20 cc Locking Syringe (TOL1952P)
- T-connector (TOL200A)
- Oxygen Reservoir (RSV3215A) and Flow Meter (FLW300A)

**Caution** The accuracy of all test equipment (electronic or pneumatic) used for verification or calibration procedures should be certified annually by a testing laboratory before use.

## Quick Check

### Introduction

This procedure is intended to assist qualified operators establish a routine program for verifying proper HT70 operation. Perform the Quick Check procedure before the first patient use, between patient uses, every 6 months.

**HEMOCARE PROVIDERS:** This procedure should be performed prior to delivery of the HT70 to a patient's home and every 6 months.

**NOTE:** If Power Save is On, the screen will go to sleep (go blank) when not used for two minutes. Just touch any button or the screen anywhere to bring it back into view.

# 6 Performance Verification

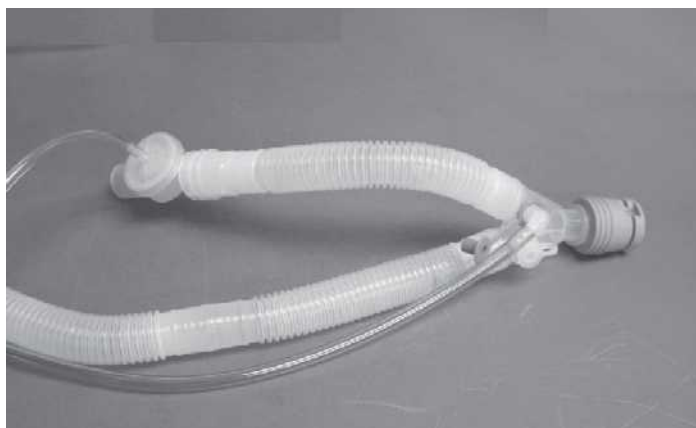
**WARNING** Do not use the HT70 if it fails the Quick Check Procedure.

## Pre-test Inspection

1. Inspect the Air Intake Filter through the filter cover on the right side of HT70. Replace the filter if it is dirty.
2. Examine the test lung and patient circuit to ensure that there are no holes that will cause leaks.
3. Verify that the A.C. power adapter PWR3204P is in good condition.

## Setup and Circuit Check

1. Connect the A.C. power adapter to an A.C. power source.
2. Verify that the External Power LED is lit.
3. Turn the ventilator on and verify that the audible alarm sounds and the LEDs light during the self test.
4. Connect a breathing circuit with exhalation valve
5. Perform the Circuit Check.
  - a. Connect the breathing circuit to the ventilator as it will be used for your patient.
  - b. Touch the *Circuit Check* button at the top of the touch screen.
  - c. For Step 1, occlude the patient connection end of the circuit. (Do not use a test lung.)



**Figure 6.1**

- d. Press the *Accept* button to confirm and start the Circuit Check.
- e. For Step 2, open the end of the patient circuit.
- f. Press the *Accept* button to continue the Circuit Check.
- g. If the test passes, the message “Circuit Check PASSED Press Accept to Confirm” will be displayed.
- h. When the Circuit Check is completed, press the *Accept* button and adjust patient settings as needed and touch the *Start Ventilation* button to begin ventilation.
- i. To cancel the Circuit Check and return to the Startup Screen, press the *Cancel* button.
- j. If the Circuit Check fails:
  - i. The message “Circuit Check FAILED Press Accept to Confirm” will be displayed.
  - ii. Press the *Accept* button to return to Startup Screen.
  - iii. Restart the Circuit Check.
  - iv. Check that the accessories and all tubing connections are properly connected and leak free.
  - v. Then touch the *Circuit Check* button to redo the test.

**WARNING** Do not use the HT70 if the Circuit Check fails, inadequate ventilation may result. Use an alternate method of ventilation. Contact Newport Medical Technical Support.

**NOTE:** The Circuit Check results are logged into Event History and retained after power down.

6. Connect the test lung to the breathing circuit.
7. Press the *Brightness* button and verify that it scrolls through 4 levels of brightness. Set the brightness at desired level.
8. Set the ventilator to the following Standard Test Settings as shown in Table 6.1 and press “Start Ventilation”.

# 6 Performance Verification

## STANDARD TEST SETTINGS (STS)

| Parameter                                    | Patient Setting              |
|----------------------------------------------|------------------------------|
| <b>Advanced/<br/>More Screen</b>             |                              |
| Flow Wave Pattern                            | Square                       |
| NIV                                          | Off                          |
| Slope/Rise                                   | 5                            |
| PS Exp Thresh                                | 25%                          |
| PS Max i-Time                                | 3.0                          |
| <b>Alarm Screen</b>                          |                              |
| ↓P<br>(Low Airway Pressure)                  | 3 cmH <sub>2</sub> O / mbar  |
| ↑P<br>(High Airway Pressure)                 | 99 cmH <sub>2</sub> O / mbar |
| ↑Min Vol<br>(High Inspiratory Minute Volume) | 50 L (maximum setting)       |
| ↓Min Vol<br>(Low Inspiratory Minute Volume)  | 1 L (minimum setting)        |
| ↑RR                                          | 100                          |
| Apnea                                        | 60                           |
| ↑FiO <sub>2</sub>                            | Off                          |
| ↓FiO <sub>2</sub>                            | Off                          |
| Alarm Loudness                               | 1                            |
| <b>Main Screen</b>                           |                              |
| Mode                                         | A/CMV                        |
| Breath Type                                  | VC                           |
| VT                                           | 500 mL                       |
| Flow                                         | 30                           |
| i-Time                                       | 1.0 sec                      |
| RR                                           | 15 b/min                     |
| P <sub>trig</sub>                            | 1 cmH <sub>2</sub> O / mbar  |
| PEEP/CPAP                                    | Off                          |
| PS                                           | 10 cmH <sub>2</sub> O / mbar |
| PC                                           | 15 cmH <sub>2</sub> O / mbar |

**Table 6.1 - Standard Test Settings**

**Quick Check Procedure****1) No External Power Alarm Check**

- a) Disconnect the A.C. power adapter. Verify that there is an audible alarm and the alarm LEDs in the HT70 handle turn on. Verify that the Ext. Power LED turns off, and the Message Area turns yellow and displays the No External Power alarm message. Confirm that HT70 continues to ventilate.
- b) Press the Alarm Silence/Reset button and confirm that its LED lights yellow, the audible alarm turns off and the message area returns to black.
- c) Press the Alarm Silence/Reset button again and confirm that the alarm message clears.
- d) Reconnect the A.C. power adapter. Verify that the Ext. Power LED lights green.

**2) Alarms and Indicators Check****a) High  $\uparrow$ P Alarm**

- i) Set the High Paw alarm limit to 20 cmH<sub>2</sub>O/mbar. Verify that an audible alarm sounds, the RED LED in the handle flashes, the High Paw message displays and that inspiration ends when pressure reaches the high limit. Set the High Paw alarm limit back to 99 cmH<sub>2</sub>O /mbar and verify that the audible alarm stops, the RED LED in the handle is cleared and the alarm message remains. Press the Alarm Silence/Reset button to clear the alarm message.

**b) Low  $\downarrow$ P Alarm**

- i) Disconnect the test lung from the breathing circuit and verify that after two breaths an audible alarm sounds and the Check Circuit or Prox Line/Low Pressure alarm message displays
- ii) Attach the test lung to the breathing circuit and verify that the audible alarm ceases and the alarm message remains. Press Alarm Silence/Reset button to clear the message.

**3) Pressure Gauge / PEEP Check**

- a) Verify that the pressure gauge moves up and down with each breath.

# 6 Performance Verification

- b) Select PEEP and Peak Paw to display in each of two Monitor Data buttons.
- c) Adjust PEEP to 5 cmH<sub>2</sub>O. Verify that the Monitor Data button displays a PEEP value of 4 to 6 cmH<sub>2</sub>O. Reduce PEEP to zero.
- d) Select Pressure Control and set pressure at 20 cmH<sub>2</sub>O. Verify that the Monitor Data button displays a Peak Paw of 17 to 23 cmH<sub>2</sub>O.
- e) Place the ventilator in Volume Control.

## 4) Tidal Volume/Minute Volume/Respiratory Rate Monitor Check

- a) Confirm that Tidal Volume is set to 500.
- b) Select VT, Min Vol and RR Tot to display in each of three Monitor Data buttons. Verify that VT= 450-550, Min Vol = 6-9 and RR Tot = 13-17.

## 5) Integrated Battery System Check

- a) Unplug the A.C. power adapter, clear the alarm with the Alarm Silence/Reset button. Verify that HT70 continues to ventilate and the Power Pac battery gauge (blue icon) reads at least 80%. If battery charge level is not sufficient, plug into external power to fully charge the Integrated Battery System.
- b) Remove the Power Pac battery pack. Verify that HT70 continues to ventilate, the alarm sounds, the alarm LEDs light and the message in the message area indicates that the Backup Battery is in use.
- c) Verify that the battery gauge is now red (for internal backup battery) and reads at least 80%. If the internal backup battery charge level is not sufficient, re-insert the Power Pac battery and plug into external power to fully charge the system.
- d) Replace the Power Pac battery pack and verify that the audible alarm clears but the message remains.
- e) Reconnect the A.C. power adapter into the Power Pac battery pack and confirm that the Ext. Power LED turns green.
- f) Press the Alarm Silence/Reset button repeatedly until all alarm messages are cleared.

**HT70 Ventilator Quick Check Record****Pass / Fail Check-Off Sheet****Preparation for Use Tests****Test Results**

- |                                                          |                                                |
|----------------------------------------------------------|------------------------------------------------|
| 1. No External Power<br>Alarm Check                      | Pass _____ Fail _____                          |
| 2. Alarm & Indicators Check<br>High Paw<br>Low Paw       | Pass _____ Fail _____<br>Pass _____ Fail _____ |
| 3. Pressure Gauge / PEEP Check                           | Pass _____ Fail _____                          |
| 4. Volume / Minute Volume /<br>Respiratory Monitor Check | Pass _____ Fail _____                          |
| 5. Integrated Battery System Check                       | Pass _____ Fail _____                          |

**The ventilator is ready for operation when all tests have passed each test.**

**Note:** Use the space below to document any comments or inspection of the unit, corrective actions taken, or recommendations for further action.

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**Completed by:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Facility:** \_\_\_\_\_

**Serial #:** \_\_\_\_\_

**Unit Hours:** \_\_\_\_\_

**Table 6.2 – Quick Check Record**

## Calibration Procedures

### Introduction

The HT70 calibration procedure should be performed for any repair, scheduled preventive maintenance (every 24 months), 15,000 hour service, or OVP failure. The calibration procedure is not required for routine maintenance except in the event of a failure

### Equipment Setup

- 1) Plug the HT70 into AC power and leave powered off for **30 minutes** prior to testing to ensure the ventilator has sufficient power to complete the calibration testing.



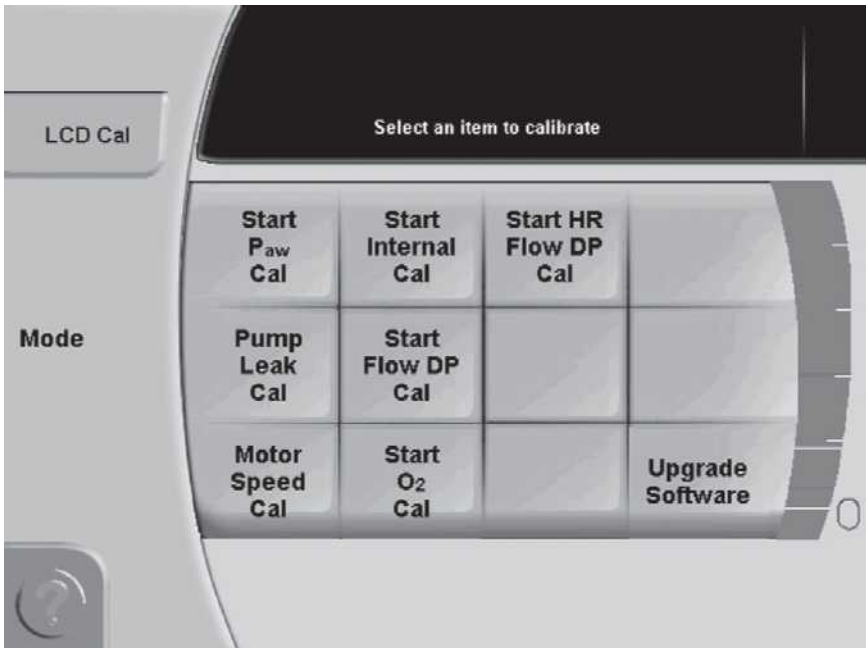
**Figure 6.2 – Equipment Setup**

### Pressure Transducer Calibration

The pressure transducer calibration is based on two reference points: (1) zero pressure (the sensor port open to atmosphere) and (2) reference pressure (an applied pressure of  $60 \pm 0.02$  cmH<sub>2</sub>O to the pressure transducer).

- 1) Remove the Power Pac and, using a #1 Phillips screwdriver, remove the screws securing the Rear Panel Assembly. Refer to section 5 for the detailed instructions.
- 2) Press and hold the manual inflation button and power up the ventilator to activate the Calibration Screen – Figure 6.3.

**Note:** Do not release the manual inflation button until the Start Up window is displayed

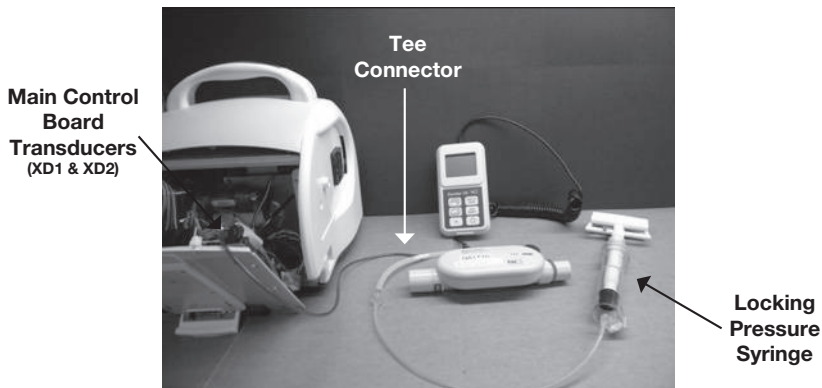


- 3) Disconnect the luer-lock on the Yellow tubing connected to the Airway Pressure Transducer (XD2) mounted to the Main Control Board PCB.
  - a) Select 'Start Paw Cal' to zero the reading.

**Note:** The Message 'Apply zero pressure to the Airway Pressure Transducer. Press zero when ready'
  - b) Verify the sensor port is open to atmosphere and press 'Zero Ready'.

**Note:** The Message 'Apply 60 cmH<sub>2</sub>O pressure to the Airway Pressure Transducer. "Press Pressure Ready" when ready' is displayed.
- 4) Connect the certified test analyzer and locking pressure syringe (TOL1952P) to XD2 using a Tee connector.

## 6 Performance Verification



**Refer to Figure 6.4 – Transducer Calibration**

- 5) Adjust the syringe and monitor the analyzer until a pressure of  $60 \pm 0.02$  cmH<sub>2</sub>O is achieved. When the reading is stable, press the “Pressure Ready”. Avoid over-pressurization above 100 cmH<sub>2</sub>O to prevent damage to the transducers.
- 6) If the calibration is passed, the message “Calibration passed”. Press the ‘Accept Cal’ button to save the cal value; Press the ‘Accept Cal’ button.
  - a) If the test fails, press ‘Cancel Cal’ and repeat the calibration procedure.

**Note:** Any time during the process, the calibration may be canceled by pressing “Cancel Cal”.
- 7) Disconnect the luer-lock on the Green tubing connected to the Internal Pressure Transducer (XD22) mounted to the Main Control Board PCB.
  - a) Select ‘Start Internal Cal’ to zero the reading.

**Note:** The Message “Apply Zero Pressure” to the Internal Pressure Transducer. Press Zero When Ready”
  - b) Verify the sensor port is open to atmosphere and press ‘Zero Ready’.

**Note:** The Message “Apply 60 cmH<sub>2</sub>O” pressure to the Internal Pressure Transducer. Press “Pressure Ready” when ready’ is displayed.
- 8) Connect the certified test analyzer and locking pressure syringe TOL1952P) to XD22 using a Tee connector.
- 9) Adjust the syringe and monitor the analyzer until a pressure of  $60 \pm 0.02$  cmH<sub>2</sub>O is achieved. When the reading is stable, press the “Pressure Ready”. Avoid over-pressurization above 100 cmH<sub>2</sub>O to prevent damage to the transducers.

- 10) If the calibration is passed, the message "Calibration passed. Press the 'Accept Cal' button to save the cal value". Press the 'Accept Cal' button.
  - a) If the test fails, press 'Cancel Cal' and repeat the calibration procedure.
- 11) **Note:** Any time during the process, the calibration may be canceled by pressing "Cancel Cal".
- 12) Reconnect the luer-lock to XD2 and XD22 transducer pressure tubes.
- 13) Power the HT70 off and Reassemble the Rear Panel Assembly and reinstall the Power Pac removed in step 21. Renter calibration screen as performed in step 2.

### Motor Speed Calibration

- 1) Disconnect the circuit from the output port
- 2) On the calibration screen, select "Motor Speed Cal".
- 3) The unit will slowly increase speed motor. Wait approximately 3 minutes for the ventilator to perform and complete this calibration.

**Note:** If the calibration is passed, the message 'Calibration passed. Press the 'Accept Cal' button to save the cal value'.
- 4) Press the 'Accept Cal'.
  - a) If the test fails, press 'Cancel Cal' and repeat the calibration procedure.

### Pump Leak Calibration

- 1) On the Calibration screen, select 'Pump Leak Cal'.

**Note:** The Message "Attach the Rp50 to the patient outlet. Press the button when ready"
- 2) Connect an Rp50 parabolic resistor to the Main Flow Outlet and DO NOT block the open end.
- 3) Select 'Rp50 Ready'.
- 4) The unit will slowly increase speed. Wait for the ventilator to perform and complete this calibration.
- 5) If the calibration is passed, the message 'Calibration passed. Press the 'Accept Cal' button to save the cal value'.
  - a) If the test fails, press 'Cancel Cal' and repeat the calibration procedure.

## 6 Performance Verification

- 6) Remove RP50 from Gas Output and Reconnect breathing circuit to Gas Output.

### FiO2 Sensor Calibration

- 1) Press the 'Start O2 Cal' button.
- 2) Press the 'O2 Ambient Ready' button.

**Note:** During the test the ventilator will display 'Please Wait'. Upon completion, the ventilator will state 'O2 100% Ready'.

- 3) Connect the HT70 Low Flow Oxygen Reservoir to the ventilator.
- 4) Connect the O2 gas source to your flow meter. (note: make sure the flow meter adjustment knob is in the off position.)
- 5) Turn on the oxygen source and turn the flow meter up to 10 L/min to flush the reservoir.
- 6) Press the 'O2 100% Ready' button.

**Note:** During the test the ventilator will display 'Please Wait'.

- 7) After the button is pressed you will need to adjust the flow meter in order to maintain a steady flow of O2 going through the system.
- 8) Once the "Calibration Passed" shows on the message screen you will need to adjust the flow meter down to zero and press Accept Cal button on the main screen.
- 9) Disconnect all accessories from the air inlet port

### LCD Calibration

- 1) Press the 'LCD Cal' button.
- 2) Follow the instructions as indicated on the screen.

### Flow Sensor Transducer Calibration

#### Flow Sensor Board GR-PCB3212A Calibration

**Note:** Flow Sensor Transducer Calibration is only required for PLUS units.

- 1) Select "Clear Tube DP Cal".
- 2) Confirm transducer is open to atmosphere and select "Zero Ready".

- 3) Use a precision pressure regulator to obtain a pressure of 1.0 cmH<sub>2</sub>O.
  - a) Tee a pressure gauge (PTS2000 Certifier FA Plus) to the output of the pressure regulator.
  - b) Apply the pressure to the lower port of the flow sensor connector.
  - c) Adjust the regulator until the output is 1.0 cmH<sub>2</sub>O  $\pm$  0.05.
  - d) Select "Pressure Ready".
- 4) Select "Accept Cal", and then remove the pressure.
- 5) Select "Blue Tube DP Cal".
- 6) Confirm transducer is open to atmosphere and select "Zero Ready".
- 7) Adjust the precision pressure regulator to obtain a pressure of 2.0 cmH<sub>2</sub>O.
  - a) Apply the pressure to the upper port of the flow sensor connector.
  - b) Adjust the regulator until the output is 2.0 cmH<sub>2</sub>O  $\pm$  0.05.
  - c) Select "Pressure Ready".
- 8) Select "Accept Cal" and then remove the pressure.
- 9) Upon completion, press the On/Off Switch and the 'Accept' button to power down the ventilator.

## Operational Verification Procedure (OVP)

**Note:** If the HT70 has been repaired, had a scheduled PM or 15000 hour service performed, or failed during the OVP testing, perform a full system calibration prior to performing the OVP.

## Electrical Safety Testing (EST)

- 1) The electrical safety testing is not required as the HT70 Ventilator is operated on DC power from the AC/DC power supply.

## Equipment Setup

- 1) Plug the HT70 into AC and leave off for **30 minutes** prior to testing to ensure the ventilator has sufficient power to complete the calibration and testing.

# 6 Performance Verification

- 2) Connect a patient circuit and test lung to the HT70 as shown in Figure 6.2.

## Front Panel Test

- 1) Press the On/Off switch once. The ventilator performs a self-test. During the self-test, verify that no error messages are displayed.
- 2) Using the Touch Screen, select More>Utility Settings>Date/Time (For HT70Plus it is time and altitude. Altitude should be set to 0.)
  - a) Using the Touch Screen, touch the year and use ▲Up/▼Down button to select the correct Year.
  - b) Repeat the above step for the Month, Day, Hour, and Minute.
- 3) When finished, select Accept.
- 4) Press all membrane switch buttons and verify that all buttons are functional.

## AC Power Loss and Shutdown Buzzer Alarm Test

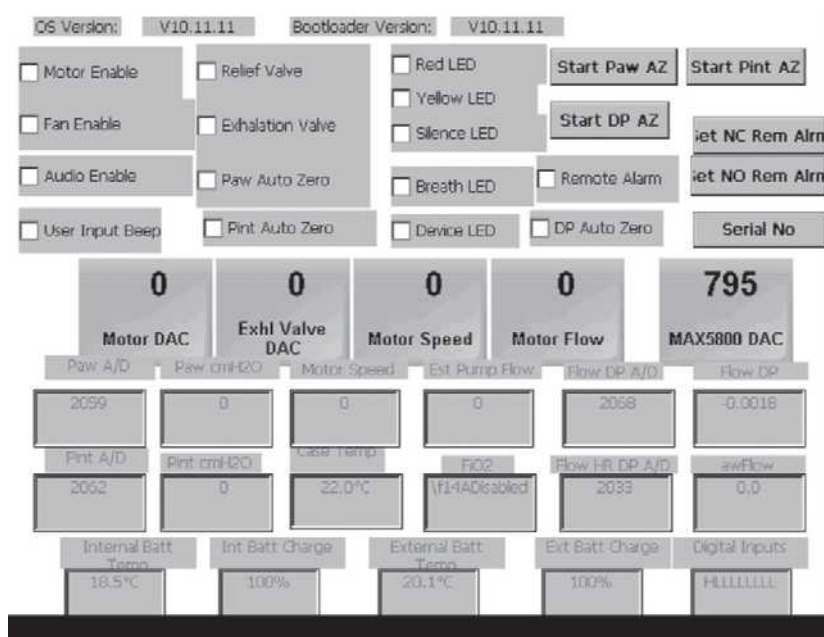
- 1) Disconnect the ventilator from A.C. power. Verify the “Loss of External Power, Switching to battery power” alarm is activated.
- 2) Shutdown the ventilator by pressing the On/Off switch once. Touch the “Accept” button on the touch screen to confirm.
- 3) Verify the audible shutdown alarm activates and allow the alarm to continue for at least two (2) minutes before pressing the Alarm Silence button.
- 4) Connect the ventilator to A.C. power and power up the ventilator.

## Circuit Check

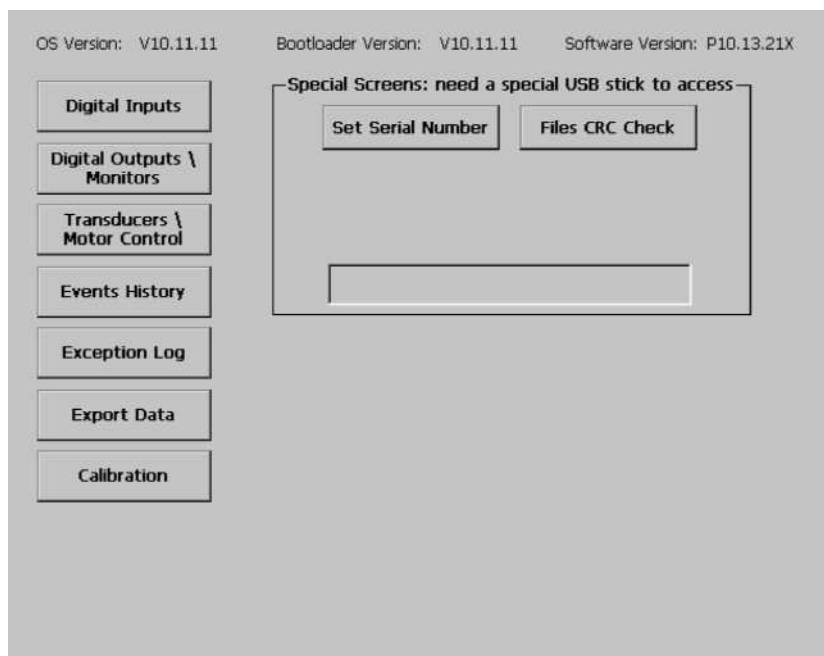
- 1) If the calibration procedures were not required, perform a circuit check as outlined in the Setup and Circuit Check section on page 6-2.

**LED / Solenoid Check**

- 1) Turn on the ventilator and hold “cancel” until the Service screen comes up – Figure 6.5.



**Figure 6.5 – Service Screen for the HT70 Classic**



**Figure 6.6 – Service Screen for HT70 Plus and HT70-2 Models**

# 6 Performance Verification

2) Then select Digital Outputs/Monitors

☐ Red LED      ☐ Fan Enable      ☐ Remote Alarm

☐ Yellow LED      ☐ Audio Enable      **Set NC Rem Alarm**

☐ Silence LED      ☐ User Input Beep      **Set NO Rem Alarm**

☐ Breath LED

☐ Device LED

**Go Back**

Case Temp: 24.6°C      FiO2: Disabled

Internal Batt Temp: 24.2°C      Int Batt Charge: 60%      External Batt Temp: 22.5°C      Ext Batt Charge: 94%

**Figure 6.7 – Digital Inputs & Outputs for HT70 Plus and HT70-2**

3) Then use the touch screen to activate and deactivate each item below to verify operation:

- Red and Yellow LED – The red and yellow indicators activate
- Alarm/Silence LED – The indicator activates
- Breath LED – The indicator activates
- Device Alert LED – The indicator activates
- Fan – Listen for the fan to start

4) To test the remote alarm, use your facility's remote alarm system as applicable.

5) On the screen press (Go Back) to go back to the service screen, then select Transducers/Motor Control.

The screenshot shows a control interface with the following elements:

- Control Checkboxes:**
  - ☐ Motor Enable
  - ☐ Paw Auto Zero
  - ☐ Motor Step Response
  - ☐ Pint Auto Zero
  - ☐ Exhalation Valve
  - ☐ DP AZ
  - ☐ Relief Valve
- Action Buttons:**
  - Start Pint AZ
  - Start Paw AZ
  - Start DP AZ
  - Go Back
- Flow DP:** 0, -12.6386, 0.0
- Flow HR DP:** 0, -1.8022, 0.0
- Paw:** 2050, 0
- Pint:** 2057, 0
- Motor Tics/Sec:** 0
- Motor Tics Delta In ms:** 2
- Zero (max/min) for Flow DP:** 2123 / 2048
- Zero (max/min) for Flow HR DP:** 2844 / 1251
- Zero (max/min) for Paw:** 2385 / 1711
- Zero (max/min) for Pint:** 2385 / 1711
- Span (max/min) for Paw:** 3227 / 2773
- Span (max/min) for Pint:** 3227 / 2773
- Blue Tube Span (max/min):** 2440 / 2388
- Clear Tube Span (max/min):** 999 / 820
- Clear Tube Span (max/min):** 1926 / 1900
- Motor:** 0
- Est Pump Flow:** 0
- Bottom Row Indicators:**
  - Motor DAC: 0
  - Exhl Valve DAC: 0
  - Motor Speed: 0
  - Motor Flow: 0
  - MAX5800 DAC: 940

**Figure 6.8 – Motor Control Screen for HT70 Plus and HT70-2**

- 6) Use the touch screen to activate and deactivate each item below to verify operation:
  - Exhalation Valve – Listen for the solenoid click
  - Relief Valve – Listen for the solenoid to click
  - Paw and Pint Auto Zero - Listen for the solenoid to click for each
  - Start Paw AZ, Start DP AZ and Start Pint AZ – Listen for the solenoid to click for each

### Pressure Relief Valve Test

- 1) Continue in the Service screen for the Pressure Relief Valve Test.
- 2) Select 'Motor Speed' and use ▲Up/▼Down to enter 200 RPM. Press Accept.
- 3) Select 'Relief Valve' to activate the valve.
- 4) Select 'Motor Enable' to activate the motor.
- 5) After the motor starts, use the blue cap (CAP100P) to block the gas outlet.
  - a) Verify the internal pressure increases between 80 - 125 cmH<sub>2</sub>O (displayed on the screen area "Pint cmH<sub>2</sub>O"). Record the value on the Test Record.

## 6 Performance Verification

- b) Verify all air leakage must be from the pressure relief valve.
- 6) Select the 'Enable Motor' to deactivate the motor.
- 7) Remove the blue cap.

### **Pressure Verification Test**

- 1) Continue in the Service screen for the Pressure Verification Test.
- 2) Attach the breathing circuit to the ventilator and connect the T-connector from a calibrated pressure analyzer to the patient connection.
- 3) Use the 2 blue caps to block the end of the breathing circuit and also block the outlet on the exhalation valve to close the system and allow the pressure to build.
- 4) Select 'Motor Enable'.
- 5) Verify the internal (Pint) and airway pressure (Paw) display and the external pressure analyzer are within 10% of each other. Record the values on the Test Record.
- 6) Select the 'Enable Motor' again to deactivate the motor.
- 7) Reset the 'Motor Speed' to 0 and press Accept.
- 8) Remove the blue caps.

### **System Leak Test**

- 1) Continue in the Service screen for the System Leak Test.
- 2) Connect the flow monitor to the Main Flow Outlet
- 3) Set the Motor Flow to 30 Lpm and press Accept.
- 4) Remove the air inlet filter cover and filter.
- 5) Completely block the inlet air for more than 3 seconds and measure the flow.
- 6) Select 'Motor Enable'.
- 7) Verify the flow is less than 1 Lpm. Record the results on the Test Record.

- 8) Select the 'Enable Motor' again to deactivate the motor.
- 9) Reinstall the air inlet filter and cover.

### Flow Measurement

- 1) Continue in the Service screen for the Flow Measurement Test.
- 2) Connect the flow monitor between the Main Flow Outlet and the breathing circuit.
- 3) Select 'Relief Valve' to activate the valve. A check mark will be displayed to verify activation.
- 4) Select 'Motor Enable'.
  - a) Verify the measured flow is between 25.5 and 34.5 Lpm.
  - b) Record the value on the Test Record.
- 5) Set Motor Flow to 60 Lpm and verify the measure flow is between 51 to 69 Lpm.
- 6) Select the 'Motor Enable' again to deactivate the motor.
- 7) Exit the Service Screen.
  - a) Press and hold the On/Off Switch until the system shuts down (approximately 20 seconds).
  - b) Press the Alarm Silence button to deactivate the alarm.

### Emergency Intake Valve Test

- 1) With the ventilator off, attach a test lung (LNG600P) to the Main Flow Outlet.

**Note:** Test lung needs to be deflated and squeezed before connecting to the HT70.

- 2) Verify air can be pulled through the emergency intake valve by gently squeezing the test lung and creating a negative effort (vacuum) in the HT70.
- 3) Remove the test lung

### Patient Effort Indicator Test

- 1) Power up the ventilator with Standard Test Settings and attach breathing circuit and test lung (LNG600P) with an Rp20.

## 6 Performance Verification

- 2) Touch the “Start Ventilation” on the touch screen.
- 3) Set RR to 1 b/min and trigger breaths by squeezing the test lung hard enough to create a negative pressure equal to or greater than 1 cmH<sub>2</sub>O/mbar.  
**Note:** Negative efforts are displayed by a yellow bar on the graph.
- 4) Verify the Patient Effort Indicator (on touch screen) blinks and the ventilator delivers a mandatory breath every successful squeeze of the test lung.
- 5) Set RR to 10 b/min, PEEP to Off, and P<sub>trig</sub> to 0.2 cmH<sub>2</sub>O/mbar.
  - a) Verify that there is no auto-triggering by looking at RR monitor and Patient Effort indicator.
- 6) Change the settings to SPONT mode with Pressure support set at 10 and P<sub>trig</sub> to 1 cmH<sub>2</sub>O.
- 7) Trigger a breath by squeezing the test lung, and verify the ventilator delivers pressure support breaths.

### Pressure Control and PEEP Test

- 1) Set the ventilator to Standard Test Settings (STS) except for the following: RR 10, i-Time 3.0, and P<sub>trig</sub> 5.0.
- 2) Attach the calibration analyzer inline with the patient circuit and the test lung (LNG600P) with airway restrictor (Rp20).
- 3) Set PC 15 and PEEP 3.
  - a) Verify the ventilator monitored Peak pressure is within  $\pm 2$  cmH<sub>2</sub>O and the PEEP is within  $\pm 1$  cmH<sub>2</sub>O of the measured pressures.
- 4) Set PC 15 and PEEP 5.
  - a) Verify the ventilator monitored Peak pressure is within  $\pm 2$  cmH<sub>2</sub>O and the PEEP is within  $\pm 2$  cmH<sub>2</sub>O of the measured pressures.
- 5) Set PC 30 and PEEP 15.
  - a) Verify the ventilator monitored Peak pressure is within  $\pm 3$  cmH<sub>2</sub>O and the PEEP is within  $\pm 2$  cmH<sub>2</sub>O of the measured pressures.

**Note:** The inspiratory pressure shall be maintained during entire inspiration time. To verify this watch the bar graph on the monitor.

- 6) Record the results on the Test Record.

### **FiO<sub>2</sub> Verification Test**

- 1) Set the ventilator to Standard Test Settings, with PEEP set to 2 cmH<sub>2</sub>O, and connect the test lung (LNG600P) with an Rp20.
- 2) Attach the oxygen analyzer to the Main Flow Outlet on the HT70 with a T-Adapter in-line with your breathing circuit.
- 3) Attach the HT70 Low Flow Reservoir to the Fresh Gas Inlet port and the flow meter hose to the oxygen source.
- 4) Start ventilation and allow the several breaths to be delivered at each to ensure stable readings. Record the Displayed and Measured values on the Test Record
  - a) Set flow meter to 3 LPM, Verify both the HT70 and Calibrated O<sub>2</sub> analyzer are within 3% of each other.
  - b) Set flow meter to 5 LPM, Verify both the HT70 and Calibrated O<sub>2</sub> analyzer are within 3% of each other.
- 5) Turn off the oxygen source, disconnect the Low Flow Reservoir from the HT70.

### **Volume Control Test**

- 1) Set the ventilator to Standard Test Settings and connect the test lung (LNG600P) with an Rp20.
- 2) Place the PTS2000 Certifier FA analyzer at the gas outlet in order to measure the tidal volume. Wait for measurements to stabilize.

**Note:** Be sure the analyzer is set to measure using the ATP (Ambient Temperature and Pressure) parameters.

- 3) Set the following parameters and record the values on the Test Record.
  - a) VT 200 mL / Flow 30 / iTime 0.4 s / RR 10 bpm. Spec: 180 – 220 mL.

## 6 Performance Verification

b) VT 500 mL / Flow 50 / iTime 0.6 s / RR 10 bpm. Spec: 450 – 550 mL.

4) Connect the expiratory flow sensor to the circuit and record the values on the Test Record (only for HT70 Plus Units)

a) VT 200mL / Flow 30 / iTime 0.4s / RR 10 bpm. Spec: 180 – 220 mL.

b) VT 500mL / Flow 50 / iTime 0.6s / RR 10 bpm. Spec: 450 – 550 mL.

**Note:** For best results when testing with flow sensor, please ensure that a 6 inch extension is used between the flow sensor and the test lung.

### Manual Inflation Test

1) Set ventilator to the Standard Test Settings. Set RR to 1. Press, hold and release the Manual inflation button.

2) Verify that a manual inflation occurs each time the manual inflation button is pressed. Verify inflation is terminated when the button is released. Set RR to 15.

3) Record the results on the Test Record.

### Pressure Alarm Test

**Note:** All activated alarms trigger an audible tone, visual LED indicators, and a displayed message.

1) Set the ventilator per Standard Test Settings (STS).

2) Set high pressure alarm 10 cmH<sub>2</sub>O.

a) Verify the High Pressure alarm is activated.

b) Verify the Red LED indicators are activated.

c) Verify the message 'High Pressure' is displayed.

d) Reset the High Pressure Alarm setting to 99 cmH<sub>2</sub>O.

3) Set the low pressure alarm 2 cmH<sub>2</sub>O higher than the displayed Peak Paw.

- a) Verify the Low Pressure alarm is activated.
- b) Verify the Red LED indicators are activated.
- c) Verify the message 'Low Pressure' is displayed.
- 4) Return the alarm settings to the STS.
- 5) Disconnect the individual limbs of the patient circuit and verify the appropriate alarm is activated.
  - a) Remove the Proximal Line tube and verify the 'Check Circuit' or 'Prox Line' alarm is activated.
    - i) Reconnect the tube and verify the alarm is cleared.
  - b) Remove the Exhalation Valve tube and verify the 'Low Pressure' is activated.
    - i) Reconnect the tube and verify the alarm is cleared.
  - c) Remove the Gas Output tube and verify the 'Check Circuit' or 'Prox Line' alarm is activated.
    - i) Reconnect the tube and verify the alarm is cleared.
- 6) Block the exhalation valve completely to trigger an 'Occlusion' Alarm.
  - a) Verify the ventilator audible alarm is activated.
  - b) Verify the Red LED indicators are activated.
  - c) Verify the message 'Circuit Occlusion' is displayed and the pressure does not rise.
  - d) Continue to block the exhalation valve and verify the message 'Sustained Occlusion' is displayed and the relief valve opens in approximately 10 seconds.
  - e) Unblock the exhalation valve and verify the alarm is cleared.
- 7) Press the 'Alarm/Silence Reset' button multiple times to clear the displayed alarm history.

# 6 Performance Verification

## Integrated Battery System Check

- 1) Unplug the A.C. power supply.
  - a) Verify the HT70 continues to ventilate.
  - b) Verify the audible alarm and handle alarm indicators activate.
  - c) Verify the HT70 displays 'No Ext Power / Switching to Battery Power'.
  - d) Verify the battery level indicator has changed to blue and the power level indicates at least 80%.
  - e) Press the Alarm Silence/Reset button to clear the alarm.
- 2) Remove the Power Pac battery.
  - a) Verify the HT70 continues to ventilate.
  - b) Verify the audible alarm activates and handle alarm indicators flash.
  - c) Verify the HT70 displays 'Switching to Backup Battery'.
  - d) Verify the battery level indicator changes to red and power level indicates at least 80%.

**Note:** If battery power level indicates less than 80%, connect the HT70 to external power to fully charge the Integrated Battery System.
- 3) Reinstall the Power Pac battery.
  - a) Verify the alarm clears, but the message remains.
  - b) Press the Alarm Silence/Reset button to clear the alarm message.
- 4) Reconnect the A.C. power supply.
  - a) Verify the External Power indicator activates.
- 5) If any alarm messages are displayed, press the Alarm Silence Reset button until all alarm messages are cleared.

## General Information

This troubleshooting guide is intended to assist a Newport Medical factory-trained and authorized technician to troubleshoot and repair the HT70 Ventilator. Although the table will be advantageous during the evaluation of the problem, all issues may not be addressed.

The guide includes Mechanical, Pneumatic and Electronic troubleshooting.

Contact the Newport Medical Technical Services Department for additional assistance.

### WARNING

Hazardous voltages are present inside the ventilator. Disconnect electrical power and oxygen sources before attempting any disassembly. Failure to do so could result in injury to service personnel and/or equipment.

**Table 7.1 TROUBLESHOOTING GUIDE**

| Problem                                             | Potential Cause                         | Corrective Action                                                                              |
|-----------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------|
| Ventilator fails the circuit check                  | Leaking patient circuit                 | Check breathing circuit for leaks and/or proper connections<br><br>Change the patient circuit  |
|                                                     | Safety valve does not close             | Replace the Pump Assembly                                                                      |
|                                                     | Manifold leaks                          | Replace the Pump Assembly                                                                      |
| Low proximal pressure (Paw) with normal output flow | Faulty patient circuit                  | Check breathing circuit for leaks and/or proper connections                                    |
|                                                     | External humidifier chamber leaking     | Bypass humidifier and retest. If problem is solved, inspect and adjust the external humidifier |
|                                                     | Pressure transducers out of calibration | Perform a complete calibration of the ventilator. Refer to section 6, Calibrations             |
|                                                     | Faulty exhalation valve                 | Replace valve                                                                                  |
|                                                     | <i>cont.</i>                            | <i>cont.</i>                                                                                   |

# 7 Troubleshooting

| Problem                                                                        | Potential Cause                                                         | Corrective Action                                                                                                                                                        |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low proximal pressure ( $P_{aw}$ ) with normal output flow<br><br><i>cont.</i> | Pressure relief valve leaking                                           | Perform the Pressure Relief Valve Check in the OVP section. If the failure persists, contact Newport Medical Technical Support for further assistance.                   |
|                                                                                | Emergency intake valve leaking                                          | Perform the Emergency Intake Valve Check in the OVP section. If the failure persists, contact Newport Medical Technical Support for further assistance.                  |
| Low proximal pressure ( $P_{aw}$ ) with low output flow                        | Air inlet filter blocked                                                | Replace air filter (ensure that it is checked / changed weekly).                                                                                                         |
|                                                                                | Internal leak (i.e. ruptured pump diaphragm, leaking check valve, etc.) | Inspect check valves. Faulty or damaged check valves require the replacement of the pump assembly. Contact the Newport Medical Technical Support for further assistance. |
|                                                                                | Pressure relief valve leaking                                           | Perform the Pressure Relief Valve Check in the OVP section. If the failure persists, contact Newport Medical Technical Support for further assistance.                   |
| Proximal pressure increasing slowly (Device Alert alarm may or may not occur)  | Emergency intake valve leaking                                          | Perform the Emergency Intake Valve Check in the OVP section. If the failure persists, contact Newport Medical Technical Support for further assistance.                  |
|                                                                                | Blocked air inlet                                                       | Replace air inlet filter (ensure that it is checked / changed weekly).                                                                                                   |
|                                                                                | High $FiO_2$ setting with a depleted oxygen supply                      | Inspect oxygen source and replace if necessary                                                                                                                           |
|                                                                                |                                                                         | If problem persists, remove the air/oxygen mixer. If the problem is resolved, replace the mixer                                                                          |
|                                                                                |                                                                         | If the problem persists, replace the pump/manifold assembly.                                                                                                             |

| Problem                                  | Potential Cause                                                          | Corrective Action                                                                                                                                                                                                   |
|------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unstable baseline                        | Patient circuit leak                                                     | <p>Check patient circuit for leaks</p> <p>Check patient for proper connection and compliance</p> <p><b>Note:</b> If using a star lumen tubing, make sure the tubing is not connected directly to the HT70</p>       |
|                                          | Internal solenoid valve or valve assembly is not functioning properly    | <p>Check the internal tubes and verify all are securely connected</p> <p>If problem persists, perform a full system calibration</p> <p>If the problem persists, replace the valve that is leaking or inoperable</p> |
|                                          | Main Control Board transducers (XD1 and XD2) are not properly calibrated | <p>Perform the Pressure Transducer Calibration</p>                                                                                                                                                                  |
| Baseline shows PEEP with no PEEP setting | Water inside exhalation valve                                            | <p>Disconnect exhalation drive line tubing and drain any trapped water.</p> <p>If problem persists, replace the exhalation valve</p>                                                                                |
|                                          | Kinked or occluded exhalation tube                                       | <p>Check and replace if necessary</p>                                                                                                                                                                               |
|                                          | Kinked or occluded proximal pressure tube                                | <p>Check and replace if necessary</p>                                                                                                                                                                               |
|                                          | Dirty or occluded proximal pressure inline filter                        | <p>Replace the inline filter</p>                                                                                                                                                                                    |
|                                          | Contaminated check valves                                                | <p>Replace pump assembly</p>                                                                                                                                                                                        |
|                                          | Malfunctioning Proportional solenoid                                     | <p>Replace Proportional solenoid</p>                                                                                                                                                                                |

# 7 Troubleshooting

| Problem                                       | Potential Cause                                  | Corrective Action                                                                                                                                                 |
|-----------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proximal pressure has slow return to baseline | Kinked or occluded proximal pressure line        | Check and replace the inline proximal filter if necessary<br><br>Heat Moisture Exchanger (HME) is causing resistance during exhalation . Remove or change the HME |
| Pump louder than normal                       | Large leak in the patient circuit                | Check and replace patient circuit if necessary                                                                                                                    |
|                                               | Internal muffler disconnected                    | Remove the air inlet filter and check the screws securing the muffler<br><br>Remove the rear panel assembly to inspect the muffler and replace if necessary       |
|                                               | Pump assembly malfunction                        | Replace the pump assembly                                                                                                                                         |
| Vibration noise (i.e. bezel)                  | Improper assembly                                | Inspect and tighten all screws securing the rear panel, case, and base assemblies                                                                                 |
|                                               | Inlet air filter cover or fan filter cover loose | Inspect and tighten the screws securing the covers                                                                                                                |
| Unit auto-triggers                            | Leak in patient circuit                          | Check and replace patient circuit if necessary                                                                                                                    |
|                                               | Leak at patient interface                        | Check patient for any leaks at the cuff or mask                                                                                                                   |
|                                               | Out of Calibration                               | Perform transducer calibration                                                                                                                                    |
| Does not operate on backup battery            | Backup battery discharged                        | Connect to AC and charge for at least 3 hours                                                                                                                     |
|                                               | Backup battery will not charge                   | Replace the backup battery                                                                                                                                        |
|                                               | Backup battery older than two (2) years          | Replace the backup battery                                                                                                                                        |
|                                               | Main Control Board malfunction                   | Replace Main Control Board                                                                                                                                        |

| Problem                                           | Potential Cause                                  | Corrective Action                                                                                                       |
|---------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Does not operate on Power Pac                     | Power Pac discharged                             | Connect to AC and charge for at least 3 hours                                                                           |
|                                                   | Power Pac will not charge                        | Replace the Power Pac battery                                                                                           |
|                                                   | Replace the Power Pac battery                    | Power Pac older than two (2) years                                                                                      |
|                                                   | Main Control Board malfunction                   | Replace the Main Control Board                                                                                          |
| Does not operate on AC power                      | Malfunctioning or damaged power supply           | Replace power supply                                                                                                    |
|                                                   | Main Control Board malfunction                   | Replace Main Control Board                                                                                              |
|                                                   | Power Pac malfunction                            | Replace Power Pac                                                                                                       |
| Dim display<br>No display<br>Intermittent display | Display brightness set to low                    | Press the 'Brightness' button to adjust the display                                                                     |
|                                                   | Display backlight non-functional                 | Replace the LCD display assembly                                                                                        |
|                                                   | Display board non-functional                     | Replace the display board                                                                                               |
|                                                   | SBC board non-functional                         | Replace the SBC board                                                                                                   |
|                                                   | LCD display cable loose                          | Remove rear panel and case assemblies and the display board. Inspect the LCD cable for proper connection to the display |
|                                                   | LCD display cable has damaged wires or connector | Replace the LCD display cable                                                                                           |

# 7 Troubleshooting

| Problem                                                                                     | Potential Cause                         | Corrective Action                                                                                            |
|---------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Touch Screen unresponsive                                                                   | Loose connection                        | Remove the rear panel and case assemblies and inspect the cables for proper connections on the display board |
|                                                                                             | Damaged cables                          | Replaced the LCD display assembly                                                                            |
|                                                                                             | Main Control board malfunction          | Replace the Main Control board                                                                               |
|                                                                                             | Single Board Computer board malfunction | Replace SBC                                                                                                  |
| Indicators do not activate:<br>Alarm/Silence<br>Device Alert<br>External Power<br>Handlebar | Loose cable connection                  | Remove the rear panel and case assemblies and inspect the cables for proper connections on the display board |
|                                                                                             | Switch Membrane malfunction             | Replace the Switch Membrane                                                                                  |
|                                                                                             | Switch Membrane damaged                 |                                                                                                              |
|                                                                                             | Broken wires                            | Replace handle assembly                                                                                      |
| Buttons unresponsive                                                                        | Loose cable connection                  | Remove the rear panel and case assemblies and inspect the cables for proper connections on the display board |
|                                                                                             | Switch Membrane malfunction             | Replace the Switch Membrane                                                                                  |
|                                                                                             | Switch Membrane damaged                 |                                                                                                              |
|                                                                                             | Display board malfunction               | Replace the display board                                                                                    |
| ON / OFF Switch unresponsive                                                                | Loose cable connection                  | Remove the rear panel and inspect the switch cable for proper connections on the Main Control board          |
|                                                                                             | Switch malfunction                      | Replace the Switch                                                                                           |
|                                                                                             | Main Control Board malfunction          | Replace Main Control Board                                                                                   |

| Problem                | Potential Cause                                                                                                                                                                                                                                                                                                                                                                                        | Corrective Action                                                                                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Alert<br>LED on | Motor Fault error<br><br>Internal Pressure (Pint) error<br><br>System Error                                                                                                                                                                                                                                                                                                                            | Refer to the displayed message window or Event History log for additional information                                                                                                     |
| Motor Fault            | Pump assembly malfunction<br><br>Main Control board malfunction                                                                                                                                                                                                                                                                                                                                        | Replace Pump assembly<br><br>Replace Main Control board                                                                                                                                   |
| Device Alert           | Internal Pressure (Pint) error                                                                                                                                                                                                                                                                                                                                                                         | Review the Event History log. If Pint failure is indicated, contact Newport Medical Technical Support for further assistance                                                              |
| System Error           | Speed table calibration error<br><br>Airway pressure zero calibration error<br><br>Airway Pressure calibration<br><br>Internal pressure zero calibration error<br><br>Internal pressure calibration error<br><br>Pump leak calibration error<br><br>Airway (Paw) auto zero error<br><br>Internal (Pint) auto zero error<br><br>FiO2 Ambient (21%) calibration error<br><br>FiO2 100% calibration error | Review Event History log for specific error<br><br>Perform the complete system calibration<br><br>If the error persists, contact Newport Medical Technical Support for further assistance |

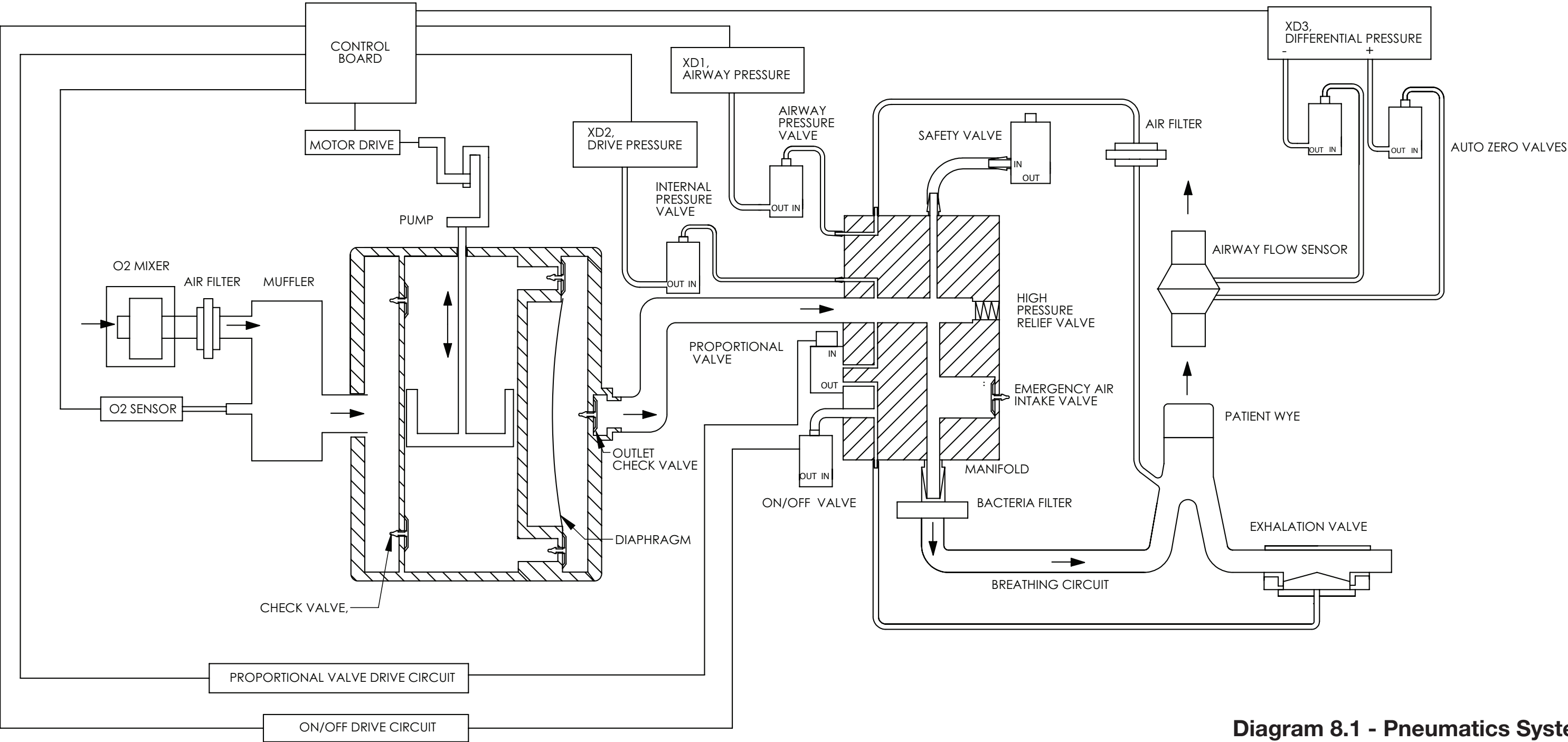
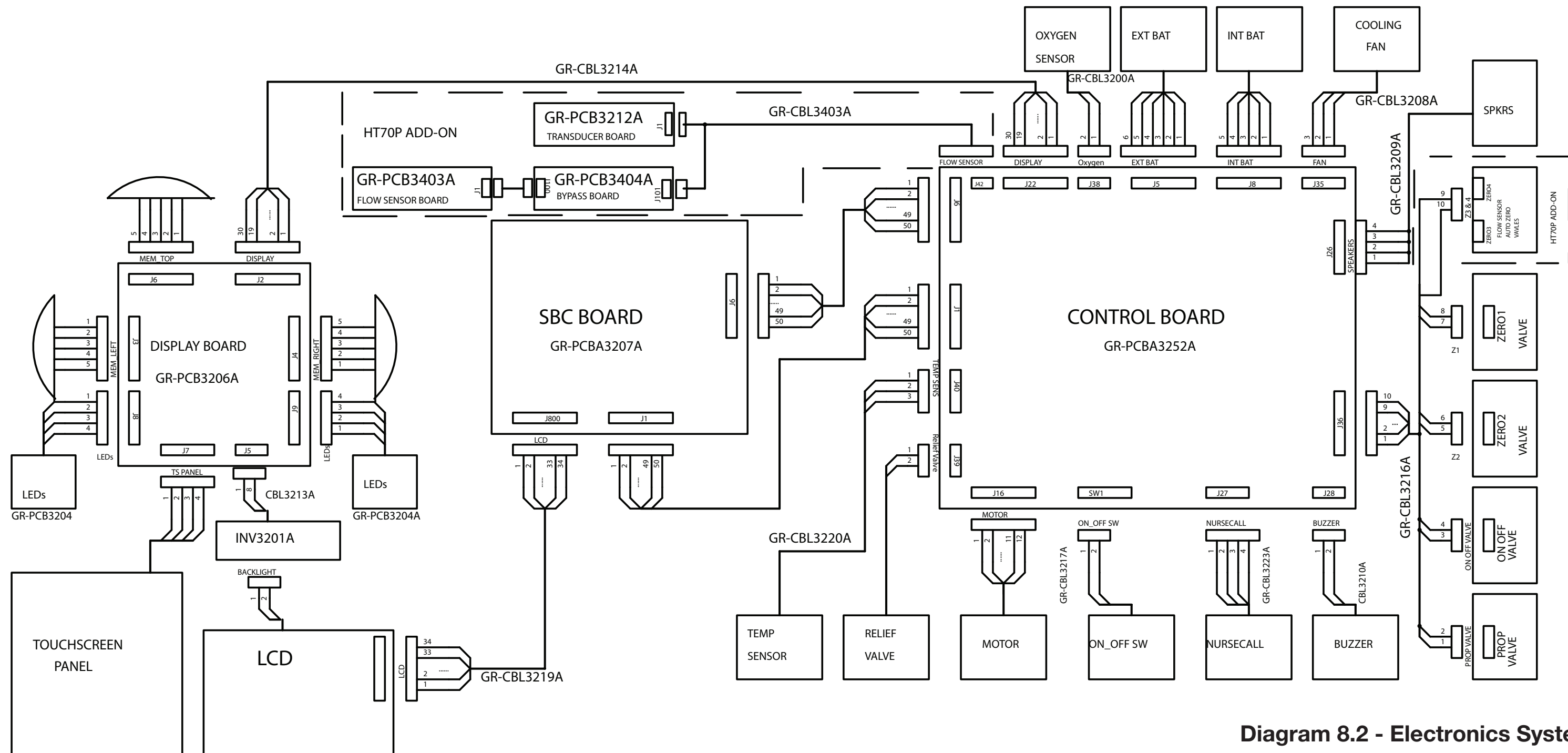
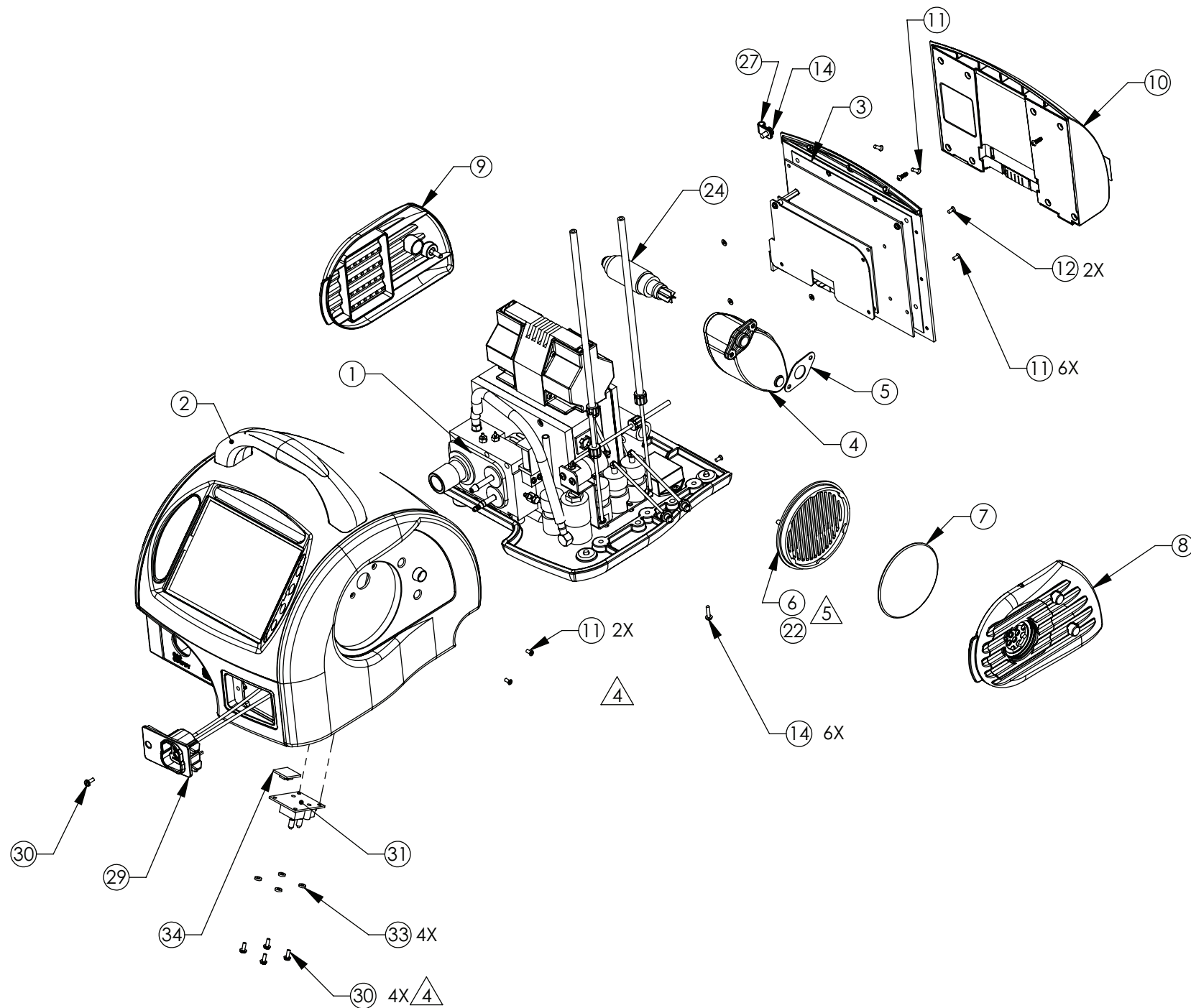


Diagram 8.1 - Pneumatics System



**Diagram 8.2 - Electronics System**



**Diagram 8.3 - HT70 Plus Exploded View**

| Item | Qty. | Description                                       | Part No.    |
|------|------|---------------------------------------------------|-------------|
| 1    | 1    | HT790 BASE ASSEMBLY W/ FLOW SENSOR, NEXT GEN      | PLT3219M    |
| 2    | 1    | HT70 CASE ASSEMBLY, NEXT GEN                      | GR-CVR3250A |
| 3    | 1    | HT7 REAR PANEL ASSEMBLY, NEXT GEN                 | GR-PNL3250A |
| 4    | 1    | ASSEMBLY, INLET MUFFLER                           | MUF3210A    |
| 5    | 1    | SILICONE MUFFLER GASKET                           | GKT3206M    |
| 6    | 1    | GASKET, FILTER                                    | GKT3201M    |
| 7    | 1    | FILTER, INLET, 80MM DIAMETER                      | HT460300-1  |
| 8    | 1    | COVER INLET FILTER ASSEMBLY                       | CVR3215A    |
| 9    | 1    | COVER FAN FILTER ASSEMBLY                         | CVR3216A    |
| 10   | 1    | BATTERY ASSEMBLY W/ LOCKING DC CONNECTOR          | BAT3271A    |
| 11   | 9    | FLAT HEAD PHILLIPS, 300SS, 4-40 X 1/4: LG, 100DEG | SCR3210P    |
| 12   | 2    | SCREW, 4-40 X 3/8" FLAT HEAD PHILLIPS, SST        | SCR3211P    |
| 14   | 7    | SCREW, 4-40 X 1/2" PH PAN SQ CONE SEM SS          | SCR3217P    |
| 22   | AR   | ADHESIVE, RTV, MEDICAL GRADE                      | ADH110P     |
| 24   | 1    | OXYGEN SENSOR                                     | SEN2103P    |
| 27   | 1    | NYLON LOOP CLAMP FOR 3/16" OD BLACK               | CLP3202P    |
| 29   | 1    | RECEPTACLE, FLOW SENSOR ASSEMBLY                  | GR-SEN3421A |
| 30   | 5    | SCREW 4-40 X 5/16" PH PAN SQ CONE SEMS SS         | SCR3216P    |
| 31   | 1    | TRANSDUCER DAUGHTER BOARD                         | GR-PCB321A  |
| 33   | 4    | WASHER, FLAT, .115 ID X .250 OD X .062 THK, NYLON | WSH1020P    |
| 34   | 1    | CABLE TIE HOLDER, 3/4" L X 3/4" W X 3/16" H       | CTH100P     |

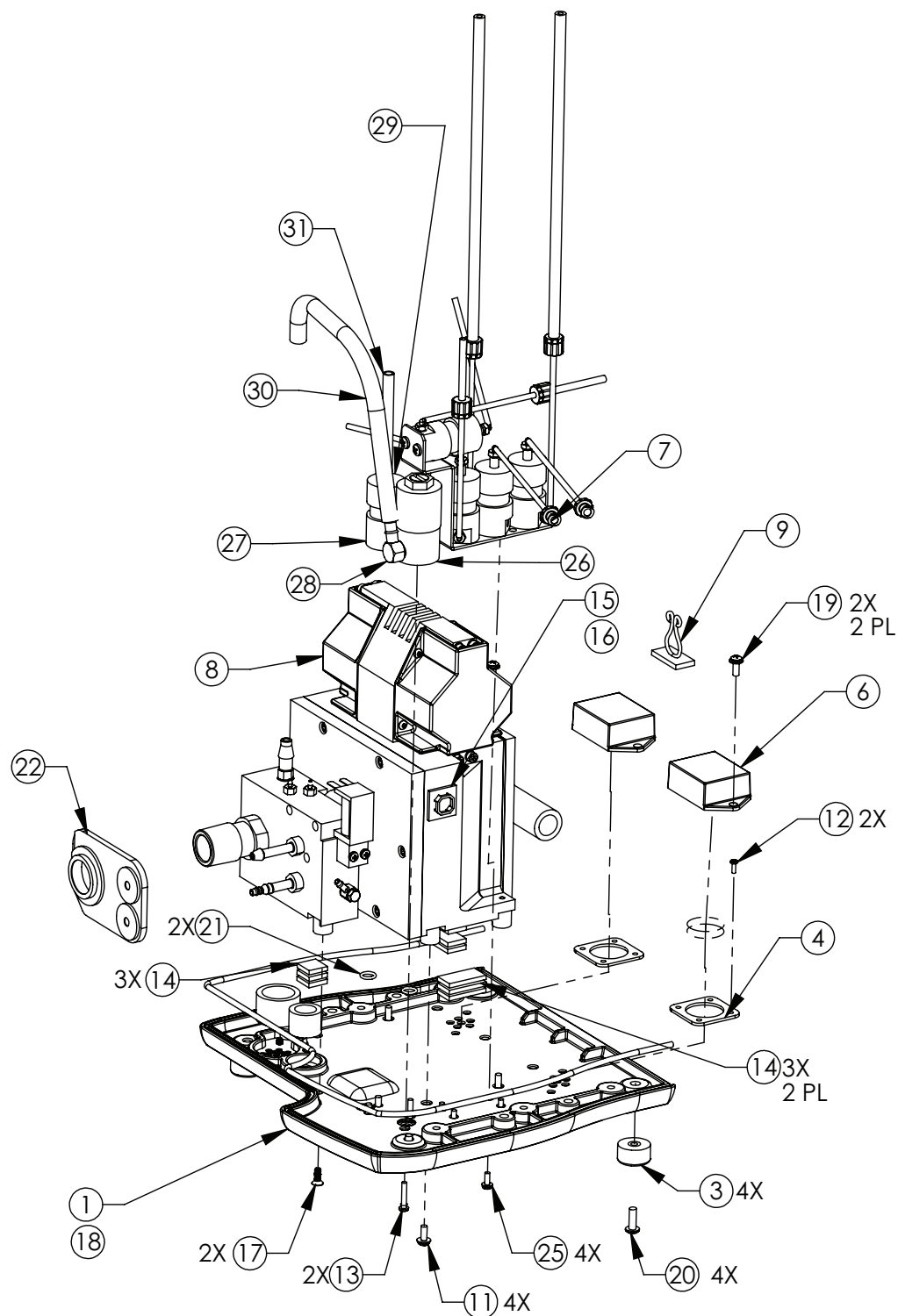
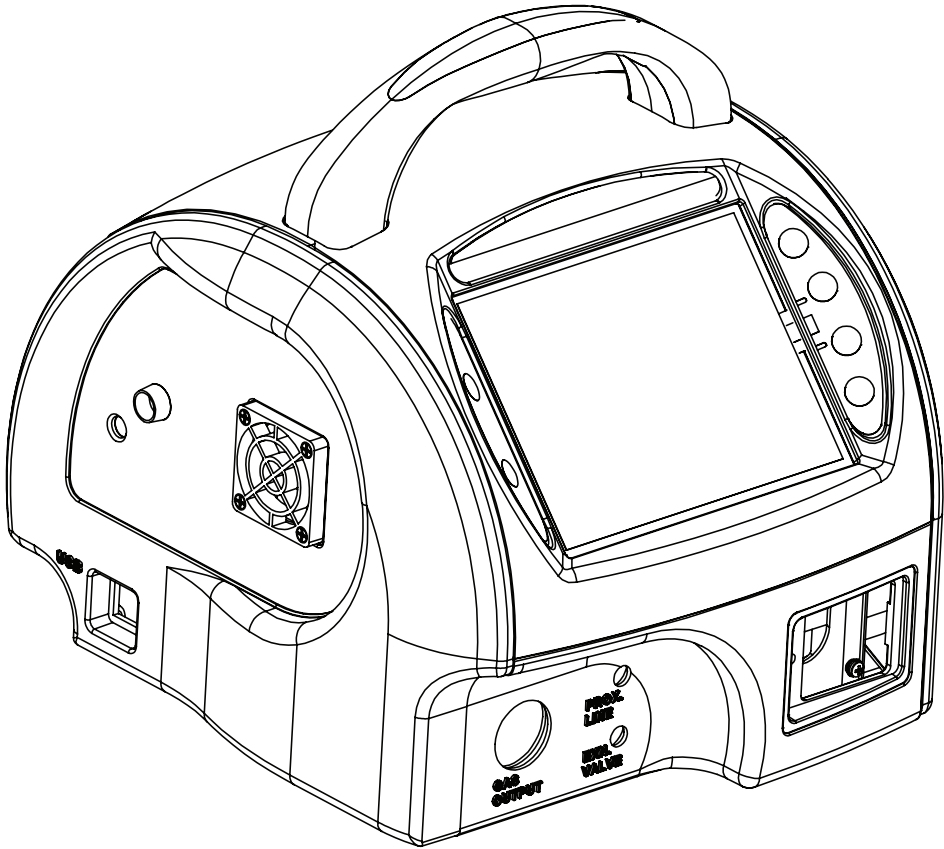


Diagram 8.4 - Base Assembly

| Item | Qty. | Description                                                                       | Part No.    |
|------|------|-----------------------------------------------------------------------------------|-------------|
| 1    | 1    | MOLDED BOTTOM PLATE                                                               | PLT3219M    |
| 2    | 25"  | GASKET CORD., .125" OD, SILICONE SPRONGE                                          | GKT3250P    |
| 3    | 4    | RUBBER BUMPER W/ UNTHREADED THROUGH HOLE WITH WASHER 3/4" BASE DIA. 13/32" HEIGHT | BPR3201P    |
| 4    | 2    | SEAL, FOAM                                                                        | SEL3201M    |
| 5    | 1    | CABLE ASSEMBLY SPEAKER CABLE                                                      | GR-CBL3209A |
| 6    | 1    | SOLENOID ASSEMBLY, HT70 PLUS                                                      | SOL3250A    |
| 7    | 1    | PUMP AND MANIFOLD ASSEMBLY, NEXT GEN                                              | GR-PMP3230A |
| 8    | 1    | NYLON CABLE TIE                                                                   | TYR3201P    |
| 9    | 1    | CABLE, ASSY VALVE, HT70 PLUS                                                      | GR-CBL3231A |
| 10   | 6    | SCREW, 6-32 X 3/8" PH PAN SQ CONE SEM SS                                          | SCR3219P    |
| 12   | 4    | SCREW 2-56 X .250, PAN HEAD PHIL, SST                                             | SCP2025P    |
| 13   | 2    | SCREW, 8-32 X 1/2 PH PAN SQ CONE SEM SS                                           | SCR3232P    |
| 14   | 7.5" | SOFT SHIELD (EMI FOAM CORE GASKET) 4002, 2.7MMTHICK, 12.7M WIDE                   | GKT3212P    |
| 15   | 1    | MOUNT, TIE WRAP                                                                   | CTH100P     |
| 16   | 1    | TYWRAP, SMALL                                                                     | TYR100P     |
| 17   | 2    | SCREW, 6-32 X 3/8" SELF-LOCK FLAT HEAD PHIL, SS                                   | SCR3233P    |
| 18   | A/R  | ADHESIVE SEALANT                                                                  | ADH108P     |
| 19   | 4    | SCREW, 6-32 X 5/16" PH W/ ATTACHED LOCK WASHER                                    | SCR3234P    |
| 20   | 4    | SCREW, 8-32 X 1/2" PAN HD PHILLIPS, SST, NYLON                                    | SCR3244P    |
| 21   | 2    | O-RING, BUNA-N, 70A DUROMETER                                                     | ORG05307P   |
| 22   | 1    | GASKET, PUMP MANIFOLD OUTLET                                                      | GKT3254M    |
| 23   | 1    | GASKET SLEEVE, 1.00" OD, 0.75" ID, SILICONE                                       | GKT3251M    |
| 24   | 1    | GASKET SLEEVE, 0.75" OD, 0.50" ID, SILICONE                                       | GKT3252M    |
| 25   | 4    | SCREW, 4-40 X 5/16 PH PAN SQ CONE SEM SS                                          | SCR3216P    |
| 26   | 1    | SOLENOID VALVE                                                                    | VLV3227M    |
| 27   | 1    | SOLENOID VALVE, PROX.                                                             | VLV3225M    |
| 28   | 1    | CONNECTOR 1/4" TUBE X 1/8 NPT BRASS ELBOW                                         | CON3205P    |
| 29   | 1    | CONNECTOR, 1/8T X 10-32M                                                          | CON200P     |
| 30   | 8"   | TUBING, 1/4 ID X 3/8 OD CLEAR TYGON                                               | TUB3210P    |
| 31   | 3.5" | TUBING, 1/8 OD X 1/4 OD                                                           | TUB300P     |
| 32   | A/R  | PIPE PASTE - LOX8                                                                 | PPS1000P    |



| Item | Qty. | Description                                                   | Part No.    |
|------|------|---------------------------------------------------------------|-------------|
| 1    | 1    | CHASSIS                                                       | BSE3201M    |
| 2    | 1    | COVER FRONT CHASSIS HT70+                                     | CVR3255M    |
| 3    | 1    | BUZZER CABLE                                                  | CBL3210A    |
| 4    | 1    | COVER REAR                                                    | CVR3210M    |
| 5    | 1    | DISPLAY, LCD                                                  | DSP2105P    |
| 6    | 1    | TOUCHSCREEN                                                   | PNL3205P    |
| 7    | AR   | TAPE, DOUBLE COATED 1/4 WIDE                                  | TAP078P     |
| 8    | 1    | HOUSING, CENTER, CHASSIS                                      | HSG3217M    |
| 9    | 1    | HOUSING, CENTER BOTTOM, LOWER PROFILE                         | HSG3216M    |
| 10   | 1    | HANDLE ASSEMBLY                                               | GR-HDL3215A |
| 11   | 1    | TOP MEMBRANE SWITCH                                           | MEM3212M    |
| 12   | 1    | LEFT MEMBRANE SWITCH                                          | MEM3211M    |
| 13   | 1    | RIGHT MEMBRANE SWITCH                                         | MEM3210M    |
| 14   | 1    | HOUSING, AIR INTAKE SIDE                                      | HSG3223M    |
| 15   | 1    | FAN INTAKE FILTER KIT                                         | FLT3202P    |
| 16   | 1    | CABLE ASSY, FAN                                               | GR-CBL3208A |
| 17   | 1    | HOUSING, ACCESSORY SIDE                                       | HSG3222M    |
| 18   | 1    | BATTERY, INTERNAL                                             | BAT3205A    |
| 19   | 4.5" | GASKET, EMI SHIELDING                                         | GKT3214M    |
| 20   | 4    | GASKET, CLOSED CELL, DOUBLE SIDED, 3/4" W, 0.125" THICK, E150 | GKT4408P    |
| 21   | 1    | BRACKET, BATTERY                                              | BKT3214M    |
| 22   | 1    | CABLE, LCD DATA                                               | GR-CBL3219A |
| 23   | 1    | "E" CLAMP, 1/4 HOLDING DIAMETER                               | CLP2102P    |
| 24   | 1    | HT70 DISPLAY BOARD WITH EXTERNAL INVERTER                     | GR-PCB3206A |
| 25   | 1    | CABLE ASSY, DISPLAY TO CONTROL PCB, RIBBON                    | GR-CBL3214A |
| 26   | 4    | SCREW, M2 X 0.4 X 3MM, PAN HD PH, SST                         | SCR2133P    |
| 27   | 4    | FLAT HD PHILLIP, T8-8 SS, 4-40, 7/16" LG                      | SCR3213P    |
| 28   | 23   | SCREW 4-40 X 5/16" PH PAN SQ CONE SEMS SS                     | SCR3216P    |
| 29   | 17   | SCREW 4-40 X 1/4" PH PAN SQ CONE SEMS SS                      | SCR3215P    |
| 30   | 2    | SCREW, 4-40 X 3/4" PAN HD, PATH                               | SCR3277P    |

| Item | Qty. | Description                                           | Part No. |
|------|------|-------------------------------------------------------|----------|
| 31   | 4    | WASHER, ID .115 TO .130, OD .240 TO .290, SST         | WSH1512P |
| 32   | 1    | GASKET, LCD                                           | GKT3208M |
| 33   | AR   | ADHESIVE, CLEAR TWO PART                              | ADH111P  |
| 34   | 1    | O-RING, SILICONE 1 1/4 ID X 1 7/16 OD                 | ORG3216P |
| 35   | 1    | SCREW, 4-40 X 1/2 PH PAN SQ CONE SEM SST              | SCR3217P |
| 36   | 11"  | TAPE, KAPTON POLYIMIDE, 1/2'                          | TAP077P  |
| 37   | 3    | WASHER, SPRING LOCK #4, 18-8 SST                      | WSH3207P |
| 38   | 1    | HEX NUT, 4-40 THD, 3/16" W, 1/16" H, SST              | NUT2107P |
| 39   | 4    | SCREW, 4-40 X 11/16" PH PAN, 18-8 SST                 | SCR3218P |
| 40   | 1    | SCREW, 4-40 X .25", 100 DEG FLAT HEAD PHILLIPS, 300 S | SCR3210P |
| 41   | 6"   | GASKET, 1/4 X 3/4 ADHESIVE BACKED FOAM                | GKT3209P |
| 42   | 1    | MICROSEMI BACKLIGHT INVERTER                          | INV3201A |
| 43   | 1    | CABLE BACKLIGHT INVERTER                              | CBL3213A |
| 44   | 1    | GASKET, HOUSING CENTER BOTTOM                         | GKT3253M |
| 45   | 1    | COVER, USB CUTOUT                                     | CVR3256M |
| 48   | AR   | ADHESIVE, RTV SEALANT                                 | ADH108P  |
| 49   | 3"   | GASKET CORD, .125 OD SILICONE SPONGE                  | GKT3250P |

Diagram 8.5 - HT70 Plus Case Assembly (1 of 7)

| Item | Qty. | Description                                            | Part No. |
|------|------|--------------------------------------------------------|----------|
| 1    | 1    | CHASSIS                                                | BSE3201M |
| 2    | 1    | COVER FRONT CHASSIS HT70+                              | CVR3255M |
| 3    | 1    | BUZZER CABLE                                           | CBL3210A |
| 4    | 1    | COVER REAR                                             | CVR3210M |
| 5    | 1    | DISPLAY, LCD                                           | DSP2105P |
| 6    | 1    | TOUCH SCREEN                                           | PNL3205P |
| 7    | 1    | TAPE, DOUBLE COATED 1/4 WIDE                           | TAP078P  |
| 26   | 4    | SCREW, M2 X 0.4 X 3MM, PAN HD PH, SST                  | SCR2133P |
| 28   | 1    | SCREW 4-40 X 5/16", PH PAN SQ CONE SEMS SS             | SCR3216P |
| 29   | 7    | SCREW 4-40 X 1/4", PH PAN SQ CONE SEMS SS              | SCR3215P |
| 32   | 1    | GASKET, LCD                                            | GKT3208M |
| 34   | 1    | O-RING, SILICONE 1 1/4 ID X 1 7/16 OD                  | ORG3216P |
| 35   | 1    | SCREW, 4-40 X 1/2 PH PAN SQ CONE SEM, SST              | SCR3217P |
| 36   | 2 IN | TAPE, KAPTON, POLYIMIDE, 1/2"                          | TAP077P  |
| 37   | 1    | WASHER, SPRING LOCK #4, S.S.                           | WSH3207P |
| 38   | 1    | NUT, HEX 4-40 THD, S.S.                                | NUT2107P |
| 40   | 1    | SCREW, 4-40 X .25", 100 DEG FLAT HEAD PHILLIPS, 300 SS | SCR3210P |

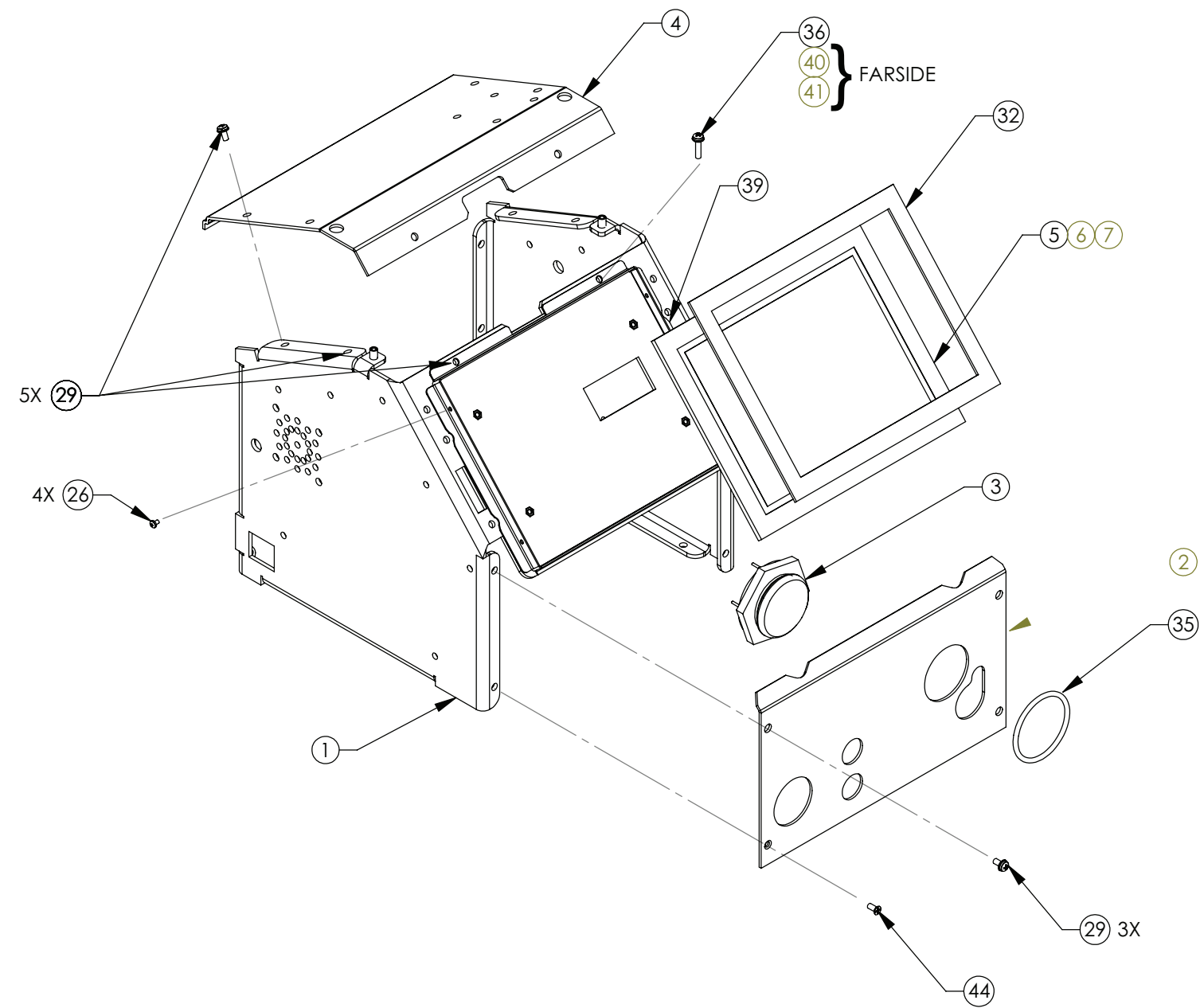
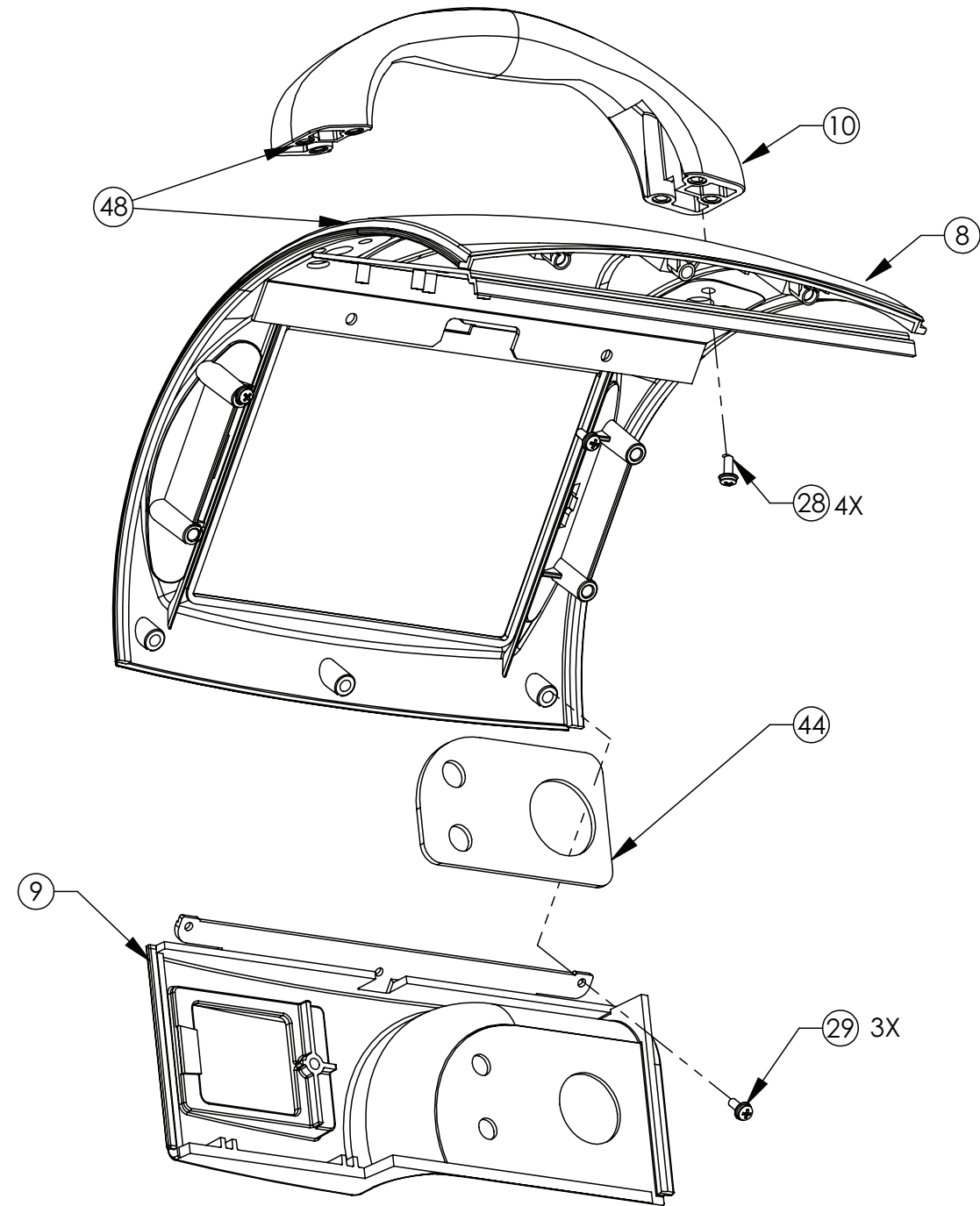
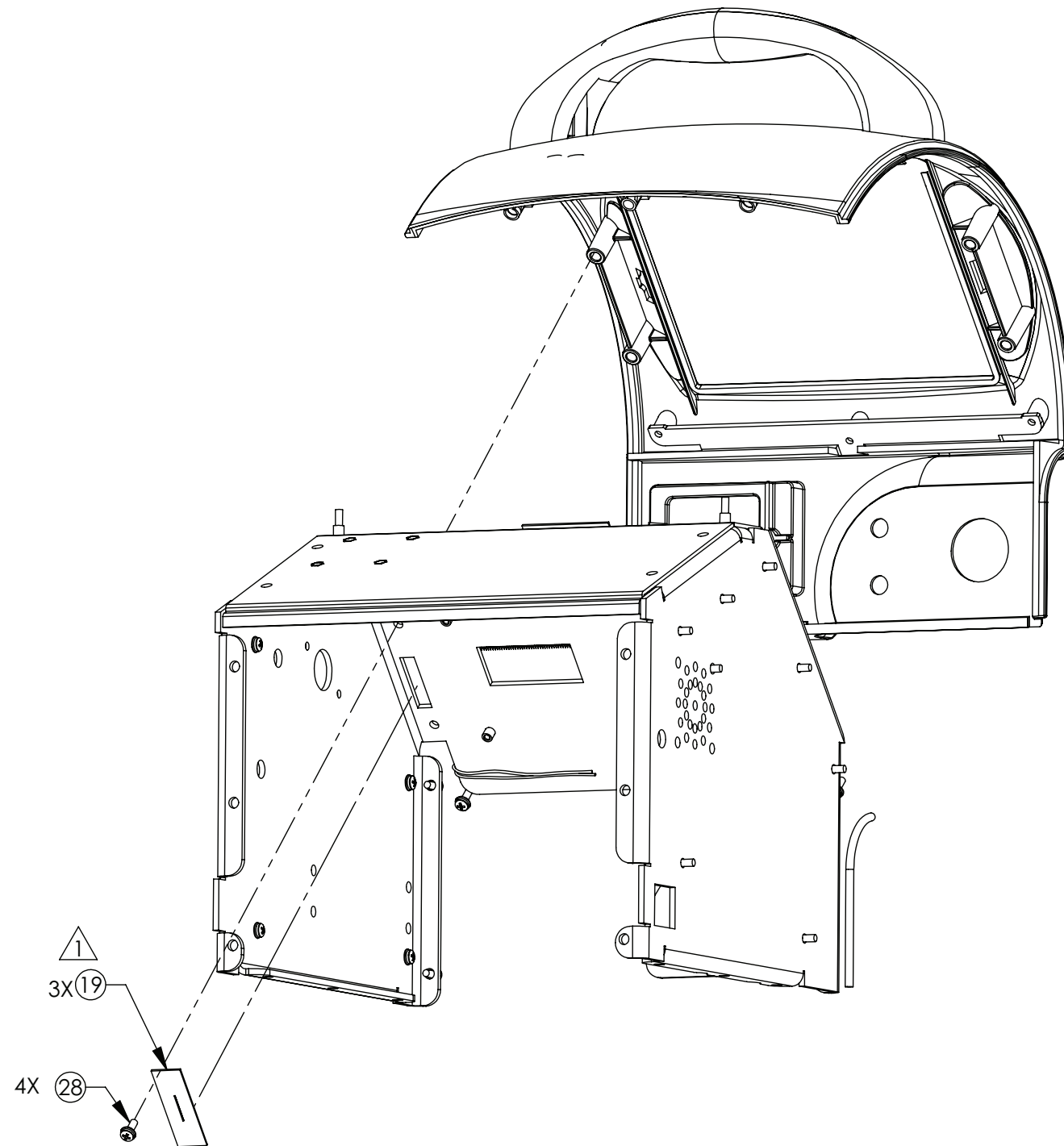


Diagram 8.5 - HT70 Plus Case Assembly (2 of 7)

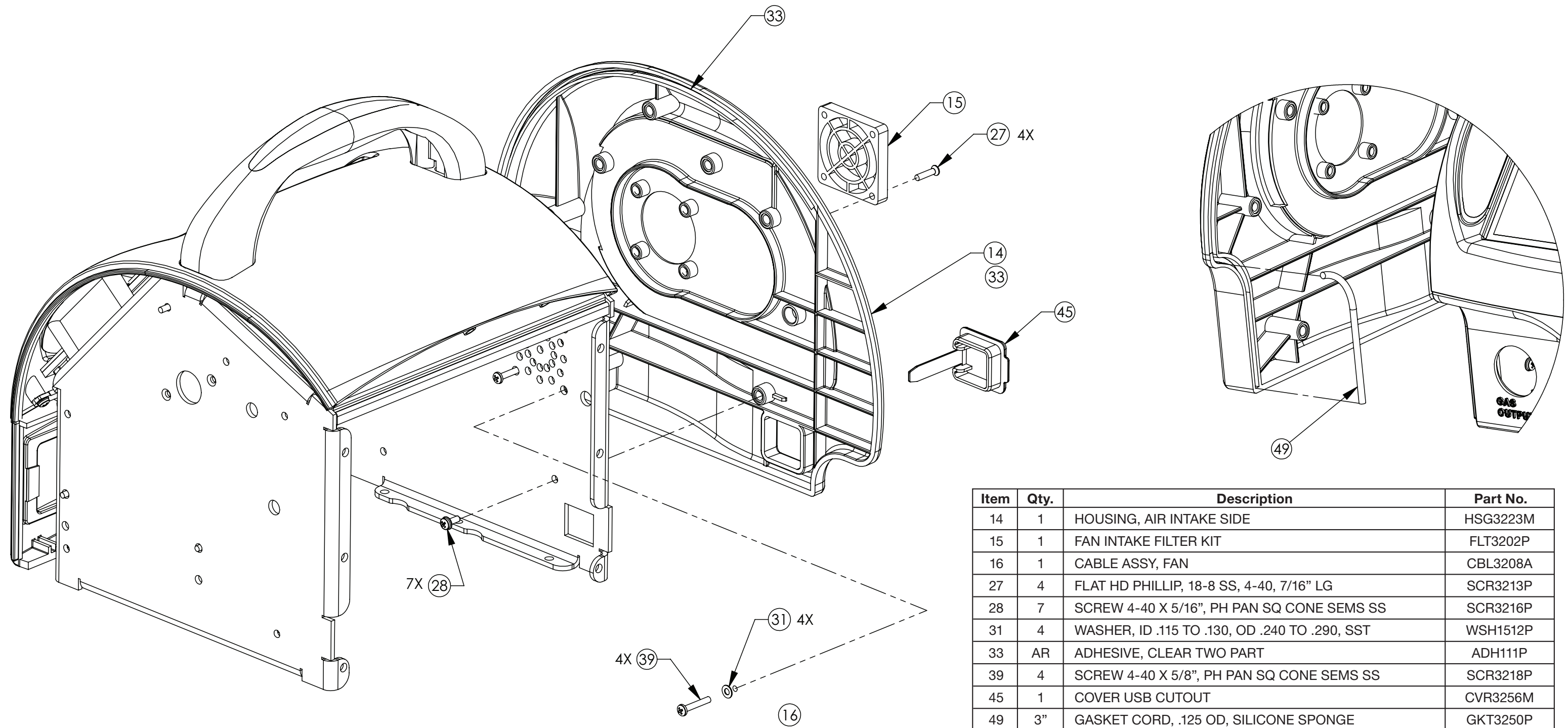


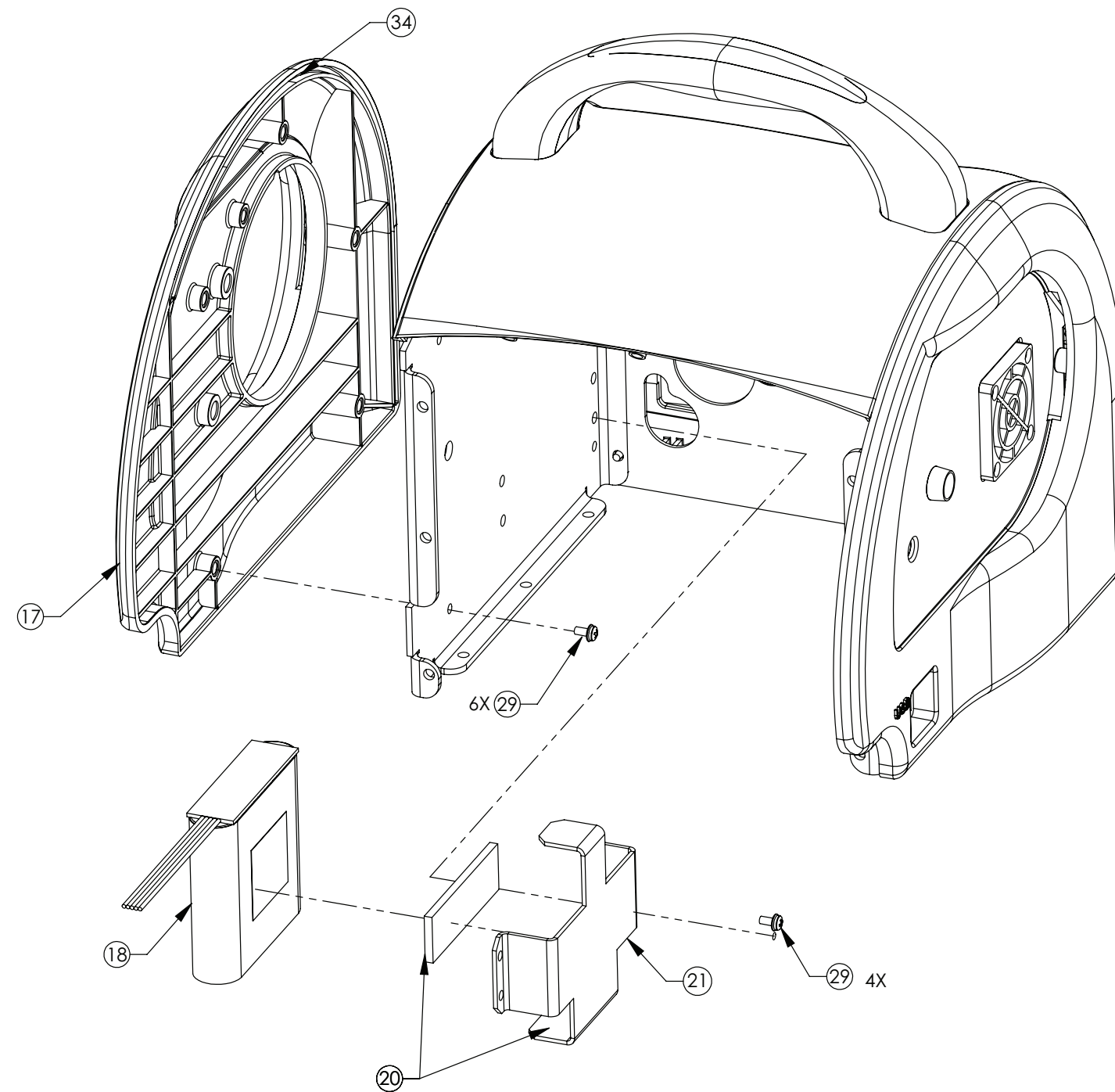
| Item | Qty. | Description                               | Part No. |
|------|------|-------------------------------------------|----------|
| 8    | 1    | HOUSING CENTER, CHASSIS                   | HSG3217M |
| 9    | 1    | HOUSING, CENTER BOTTOM, LOWER PROFILE     | HSG3216M |
| 10   | 1    | HANDLE ASSEMBLY                           | HDL3215A |
| 28   | 4    | SCREW 4-40 X 5/16" PH PAN SQ CONE SEMS SS | SCR3216P |
| 29   | 3    | SCREW 4-40 X 1/4" PH PAN SQ CONE SEMS SS  | SCR3215P |
| 33   | AR   | ADHESIVE, CLEAR TWO PART                  | ADH111P  |
| 44   | 1    | GASKET, HOUSING CENTER BOTTOM             | GKT3253M |
| 48   | AR   | ADHESIVE, RTV SEALANT                     | ADH108P  |

Diagram 8.5 - HT70 Plus Case Assembly (3 of 7)



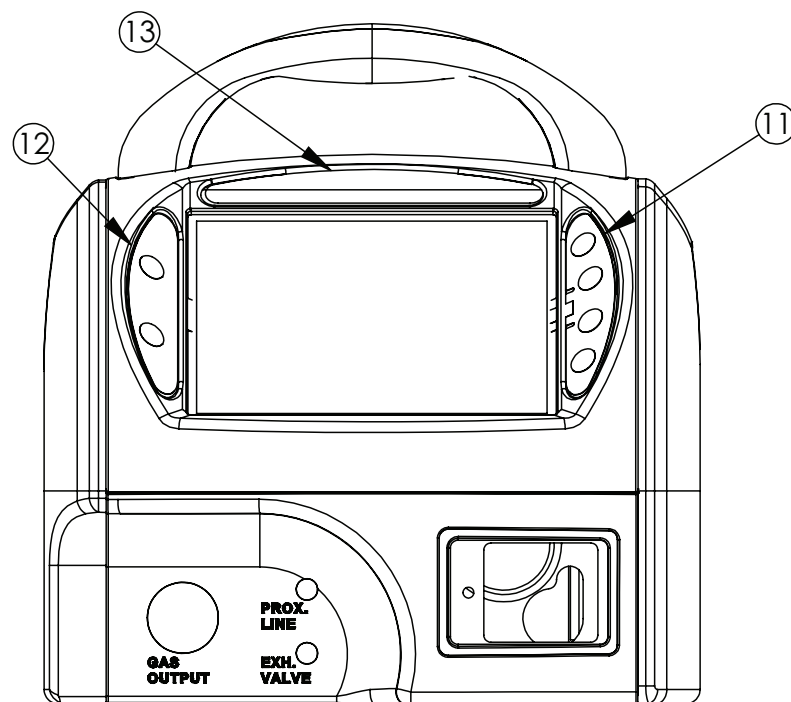
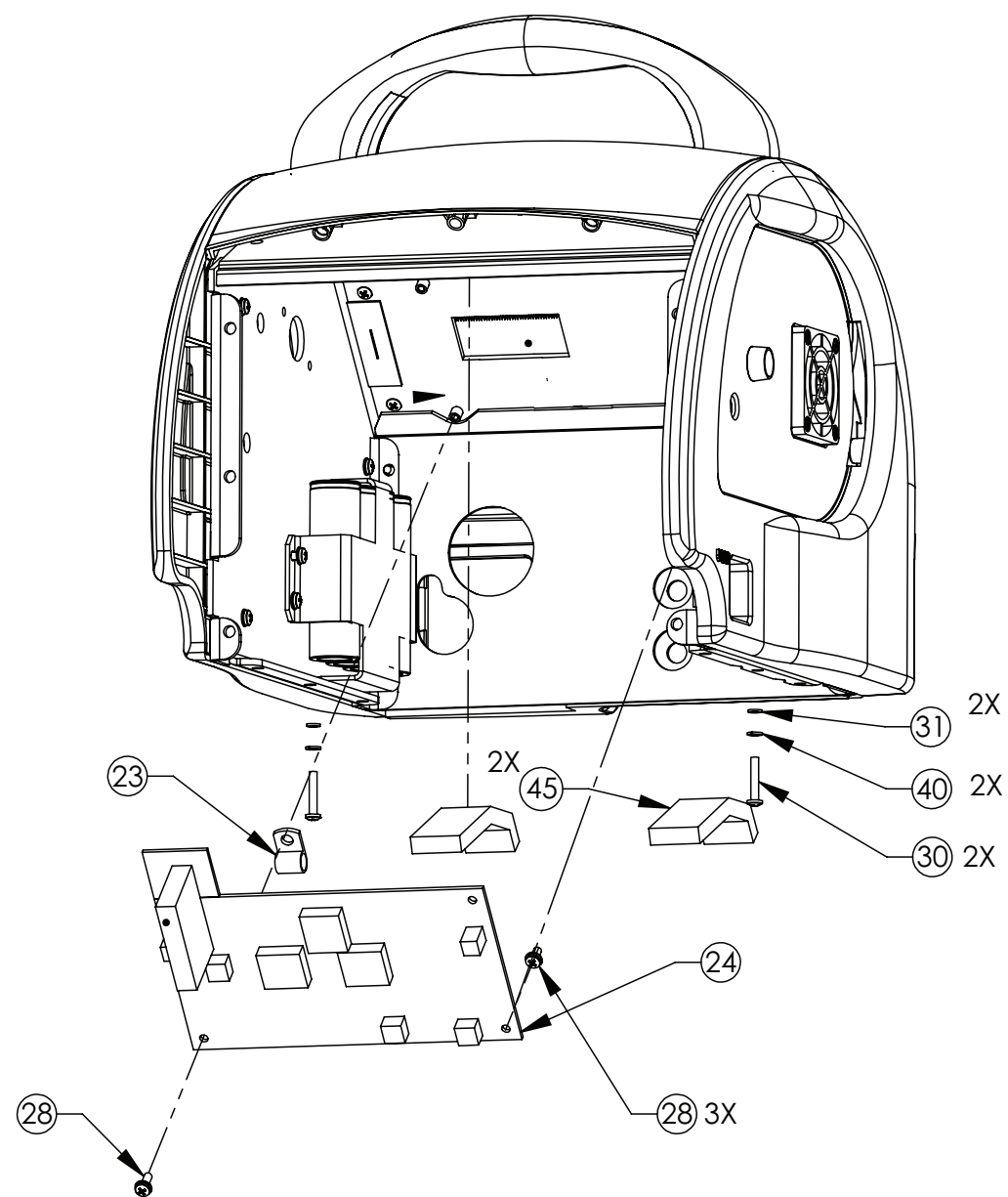
| Item | Qty. | Description                              | Part No. |
|------|------|------------------------------------------|----------|
| 19   | 4.5" | GASKET, EMI SHIELDING                    | GKT3214M |
| 28   | 4    | SCREW 4-40 X 5/16" PH PAN SQ CONE SEM SS | SCR3216P |
| 36   | 9 IN | TAPE, KAPTON POLYIMIDE, 1/2"             | TAP077P  |





| Item | Qty. | Description                                                  | Part No. |
|------|------|--------------------------------------------------------------|----------|
| 17   | 1    | HOUSING, ACCESSORY SIDE                                      | HSG3222M |
| 18   | 1    | BATTERY, INTERNAL                                            | BAT3205A |
| 20   | 3    | GASKET, CLOSED CELL, DOUBLE SIDED, 3/4" W, 0.125 THICK, E150 | GJT4408P |
| 21   | 1    | BRACKET, BATTERY                                             | BKT3214M |
| 28   | 6    | SCREW 4-40 X 5/16" PH PAN SQ CONE SS                         | SCR3216P |
| 29   | 4    | SCREW 4-40 X 1/4" PH PAN SQ CONE SS                          | SCR3215P |
| 33   | AR   | ADHESIVE, CLEAR TWO PART                                     | ADH111P  |

**Diagram 8.5 - HT70 Plus Case Assembly (6 of 7)**



| Item | Qty. | Description                                 | Part No. |
|------|------|---------------------------------------------|----------|
| 11   | 1    | TOP MEMBRANE SWITCH                         | MEM3212M |
| 12   | 1    | LEFT MEMBRANE SWITCH                        | MEM3211M |
| 13   | 1    | RIGHT MEMBRANE SWITCH                       | MEM3210M |
| 22   | 1    | CABLE, LCD DATA                             | CBL3219A |
| 23   | 1    | "E" CLAMP, 1/4 HOLDING DIAMETER             | CLP2102P |
| 24   | 1    | HT70 DISPLAY BOARD WITH EXTERNAL INVERTER   | PCB3206A |
| 25   | 1    | CABLE ASSY, DISPLAY TO CONTROL PCB, RIBBON  | CBL3214A |
| 28   | 1    | SCREW 4-40 5/16" PH PAN SQ CONE SEMS SS     | SCR3216P |
| 29   | 3    | SCREW 4-40 1/4" PH PAN SQ CONE SEMS SS      | SCR3215P |
| 30   | 2    | SCREW 4-40 X 3/4 PAN HD, PATCH              | SCR3277P |
| 31   | 2    | WASHER, ID .115 TO .130 OD .240 TO .290 SST | WSH1512P |
| 37   | 3    | WASHER, SPRING LOCK #4, 18-8 SST            | WSH3207P |
| 38   | 1    | HEX NUT, 4-40 THD, 3/16" W, 1/16" H, SST    | NUT3207P |
| 41   | 6"   | GASKET, 1/4 X 3/4 ADHESIVE BACKED FOAM      | GKT3209P |

Diagram 8.5 - HT70 Plus Case Assembly (7 of 7)